CLINICAL CONDITIONS

Neurology Review Articles

9 physiology topics and 20 conditions, from stroke to movement disorders

For educational use only. This is a study aid, not a clinical reference.

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PHYSIOLOGY

Neurons and action potentials: structure, channels, and conduction

Neurons receive input on dendrites, integrate it at the axon hillock, and conduct all-or-none action potentials to their terminals. The resting potential of about −70 mV reflects potassium leak and the Na/K-ATPase, and voltage-gated sodium and potassium channels drive depolarization and repolarization. Refractory periods limit firing rate and keep conduction one-way, and myelin allows fast saltatory conduction. Demyelination, sodium channel mutations, and sodium channel toxins disrupt this process, and sodium channel blockers exploit it. Here, we review neuron structure, membrane channels, the action potential, and their failures.

Key points

- The resting potential is about −70 mV, close to the potassium equilibrium potential, because potassium leak channels dominate at rest.
- At threshold, about −55 mV, voltage-gated sodium channels open, producing an all-or-none spike.
- Sodium channel inactivation and potassium efflux repolarize the membrane, and inactivation causes the absolute refractory period.
- Myelin from oligodendrocytes and Schwann cells allows saltatory conduction between nodes of Ranvier.
- Lidocaine and phenytoin block voltage-gated sodium channels.

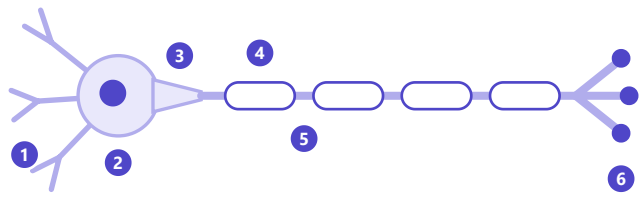


Fig. 1 | A neuron: dendrites, 1, receive input; the soma, 2, holds the nucleus; the axon hillock, 3, starts the action potential; myelin, 4, insulates the axon; nodes of Ranvier, 5, regenerate the signal; and axon terminals, 6, release neurotransmitter. 1, Dendrites; 2, Soma; 3, Axon hillock; 4, Myelin; 5, Nodes of Ranvier; 6, Axon terminals.

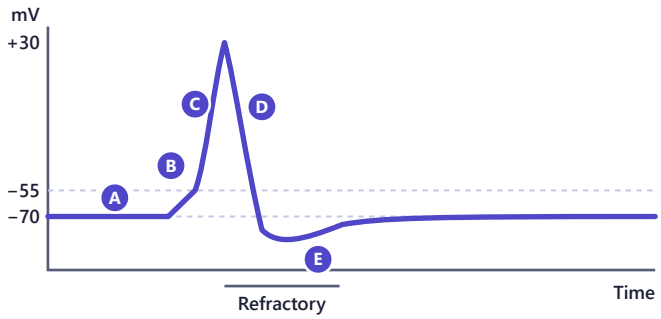


Fig. 2 | A neuronal action potential plotted as voltage over time: A, resting at minus 70 millivolts; B, a stimulus reaches threshold at minus 55; C, depolarization as sodium enters; D, repolarization as potassium leaves; E, hyperpolarization before the return to rest. The refractory period spans the spike and hyperpolarization. A, Resting potential; B, Threshold; C, Depolarization; D, Repolarization; E, Hyperpolarization.

Structure

Dendrites receive afferent signals, the soma holds the nucleus, and the axon hillock initiates the action potential, which the axon carries to terminals that release neurotransmitter. Myelin, from oligodendrocytes in the CNS and Schwann cells in the periphery, insulates the axon, leaving nodes of Ranvier packed with voltage-gated sodium channels. Most neurons, including motor neurons, are multipolar; some are bipolar; and sensory neurons of the dorsal root ganglia are pseudounipolar.

Glial cells support neurons. Astrocytes buffer potassium, take up neurotransmitter, and with their foot processes help form the BBB; microglia, of monocyte origin, are the immune cells of the CNS; oligodendrocytes myelinate many CNS axons each, whereas each Schwann cell myelinates one PNS axon; and ependymal cells line the ventricles, with choroid epithelium regulating CSF. The BBB consists of capillary endothelial cells joined by tight junctions, supported by pericytes, astrocyte foot processes, and the basal lamina.

Channels and the action potential

Potassium leak channels make the resting membrane most permeable to potassium, so the resting potential of about −70 mV lies near the potassium equilibrium

potential, and the Na/K-ATPase pumps 3 sodium out and 2 potassium in to maintain the gradients. When summed input reaches threshold, about −55 mV, voltage-gated sodium channels open and sodium rushes in, an all-or-none depolarization. Sodium channels then inactivate and voltage-gated potassium channels open, repolarizing the membrane, and slow potassium channel closure causes hyperpolarization. At the terminal, voltage-gated calcium channels trigger transmitter release, and ligand-gated channels generate postsynaptic potentials.

During the absolute refractory period, inactivated sodium channels prevent another spike; during the relative refractory period, a stronger stimulus is needed. These periods set the maximum firing rate and keep conduction one-way. In unmyelinated axons, depolarization spreads to adjacent membrane slowly, whereas in myelinated axons current jumps from node to node, saltatory conduction, more than 10 times faster.

When it fails

Multiple sclerosis destroys oligodendrocytes, MOGAD targets myelin, and NMOSD destroys astrocytes through antibodies against AQP4, with secondary demyelination, slowing or blocking conduction. Tetrodotoxin blocks sodium channels, causing numbness and weakness, and sodium channel mutations such as SCN1A raise excitability and cause epilepsy.

Pharmacology

Local anesthetics such as lidocaine block voltage-gated sodium channels in nerves, so pain signals fail to conduct. Antiseizure drugs such as phenytoin, carbamazepine, and lamotrigine reduce sodium influx in neurons, lowering excitability.

Relationships to other topics

Synapses and neurotransmitters. The action potential ends at the terminal, where calcium entry converts the electrical signal into chemical transmission.

PHYSIOLOGY

Synapses and neurotransmitters: release, receptors, and clearance

At chemical synapses, an action potential opens voltage-gated calcium channels, and calcium-activated SNARE proteins fuse vesicles to release neurotransmitter. Transmitters act on fast ionotropic or slower metabotropic receptors, producing excitatory or inhibitory postsynaptic potentials that sum at the axon hillock, and are cleared by reuptake or enzymes. Glutamate is the main excitatory and GABA the main inhibitory transmitter, with acetylcholine, dopamine, norepinephrine, and serotonin modulating specific systems. Here, we review synaptic transmission, the major transmitters, their failures, and drugs that target them.

Key points

- Calcium entry activates SNARE proteins, which fuse vesicles and release transmitter.
- Ionotropic receptors are fast ion channels; metabotropic receptors act through G proteins.
- Glutamate is the main excitatory transmitter, GABA the main inhibitory transmitter in the brain, and glycine inhibitory in the cord.
- Acetylcholine is cleared by AChE; small-molecule transmitters are cleared by reuptake.
- Benzodiazepines increase GABA-A channel opening, and cholinesterase inhibitors raise acetylcholine.

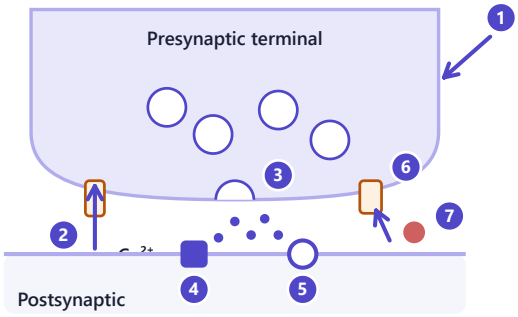


Fig. 1 | A chemical synapse: 1, an action potential reaches the terminal; 2, voltage-gated calcium channels open; 3, vesicles fuse and release neurotransmitter; 4, it binds ionotropic receptors and, 5, metabotropic receptors; it is cleared by, 6, reuptake or, 7, enzymes. 1, Action potential; 2, Calcium channels; 3, Vesicle fusion; 4, Ionotropic receptors; 5, Metabotropic receptors; 6, Reuptake; 7, Enzymes.

Synaptic transmission

An action potential depolarizes the terminal, opening voltage-gated calcium channels, and calcium activates SNARE proteins that fuse vesicles with the membrane. Transmitter binds ionotropic receptors, ion channels that act quickly, or metabotropic receptors, G protein-coupled receptors with slower, longer effects. Excitatory postsynaptic potentials depolarize the next neuron and inhibitory potentials hyperpolarize it, as chloride enters through GABA-A receptors, and inputs sum at the axon hillock. Transmitter is cleared by reuptake into the terminal or glia, for small molecules only, or by enzymes such as AChE. Electrical synapses pass current directly through gap junctions with almost no delay.

Neurotransmitters

Glutamate, acting on AMPA and NMDA receptors, is the main excitatory transmitter and underlies plasticity and memory. GABA, acting on GABA-A chloride channels and GABA-B G protein-coupled receptors, is the main inhibitory transmitter in the brain, and glycine is inhibitory mainly in the cord. Acetylcholine acts on nicotinic ion channels and muscarinic M1–M5 receptors at muscle, autonomic synapses, and the basal forebrain. Dopamine from the SNc and ventral tegmental area acts on D1 and D2 families, norepinephrine from the locus coeruleus drives arousal, and serotonin from the raphe nuclei modulates mood, sleep, and gut motility.

When it fails

Loss of cholinergic neurons contributes to memory decline in Alzheimer disease, and death of SNc dopamine neurons leaves too little dopamine in Parkinson disease. Glutamate excitotoxicity contributes to both, seizures arise when excitation outweighs GABA inhibition, and in hepatic encephalopathy, ammonia enhances GABA-A chloride flow.

At the neuromuscular junction, myasthenia gravis targets postsynaptic nicotinic AChR or MuSK, causing fatigable weakness that worsens through the day, with ptosis and diplopia in 85% and thymoma in 10%; pyridostigmine, a cholinesterase inhibitor, is followed by

immunosuppression. Lambert-Eaton syndrome targets presynaptic voltage-gated calcium channels, reducing ACh release, and causes proximal weakness, autonomic dysfunction, and reduced reflexes that improve with repeated contraction, often with small cell lung cancer. In Huntington disease, striatal GABA neurons die.

Pharmacology

Benzodiazepines increase how often GABA-A chloride channels open, causing sedation and seizure control. Cholinesterase inhibitors such as donepezil block acetylcholine breakdown, botulinum toxin blocks acetylcholine release at the neuromuscular junction, and SSRIs block serotonin reuptake.

Relationships to other topics

Neurons and action potentials. The action potential arriving at the terminal opens the calcium channels that start release.

Basal ganglia pathways. Dopamine acts on D1 and D2 striatal neurons to balance the direct and indirect pathways.

PHYSIOLOGY

Ascending tracts: dorsal columns, spinothalamic tracts, and spinocerebellar tracts

Ascending tracts carry sensation from the body to the thalamus and cortex or to the cerebellum. The dorsal column-medial lemniscus carries fine touch, vibration, and conscious proprioception, ascends on the same side, and crosses in the medulla. The spinothalamic tract carries pain, temperature, and crude touch and crosses in the cord within 1–2 segments of entry. The spinocerebellar tracts carry unconscious proprioception to the cerebellum on the same side. Where each crosses determines the side of deficits after a lesion. Here, we review their routes and failures.

Key points

- The dorsal columns ascend on the same side and cross in the medulla as the medial lemniscus.
- Spinothalamic fibers cross in the anterior white commissure 1–2 segments above entry.
- A cord hemisection causes same-side dorsal column loss and opposite-side pain and temperature loss.
- Spinocerebellar tracts end in the cerebellum on the same side, so their lesions cause same-side ataxia.

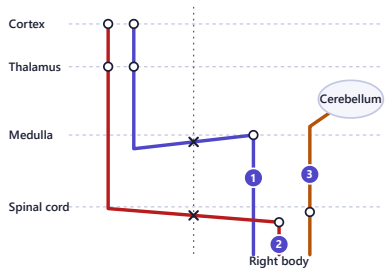


Fig. 1 | Ascending tracts from the right side of the body. 1, the dorsal column-medial lemniscus ascends on the same side and crosses in the medulla. 2, the spinothalamic tract crosses in the cord, one to two segments above entry. Both reach the left thalamus and cortex. 3, the dorsal spinocerebellar tract stays on the same side and enters the cerebellum. 1, Dorsal column-medial lemniscus; 2, Spinothalamic tract; 3, Spinocerebellar tracts.

Routes

In the dorsal column-medial lemniscus, large DRG fibers ascend on the same side, gracilis from below T6 and cuneatus from T6 up, to the nucleus gracilis and cuneatus in the medulla. Second-order internal arcuate fibers cross and ascend as the medial lemniscus to the VPL thalamus, which projects to S1. The pathway carries fine touch, vibration, conscious proprioception, and two-point discrimination.

In the spinothalamic tract, thin Aδ and C fibers synapse in the dorsal horn, and second-order fibers cross in the anterior white commissure 1–2 segments above entry. The lateral tract carries sharp pain and temperature, with sacral fibers most lateral, and the anterior tract carries crude touch; both reach the VPL thalamus and S1, insula, and cingulate cortex.

The dorsal spinocerebellar tract arises in the Clarke nucleus (C8–L3) and ascends uncrossed through the inferior cerebellar peduncle, carrying leg and trunk information. The ventral tract arises from spinal border cells, crosses twice, and enters through the superior cerebellar peduncle, so both end on the same side, carrying unconscious proprioception from muscle spindles and tendon organs.

Dermatome landmarks localize sensory levels: C6 supplies the thumb, C7 the middle finger, T4 the nipple, T10 the umbilicus, L4 the anterior knee and medial malleolus, and S1 the lateral malleolus.

When it fails

A cord hemisection, Brown-Séquard syndrome, causes same-side dorsal column loss and weakness with opposite-side pain and temperature loss. Posterior cord lesions and dorsal column loss from B12 deficiency or tabes dorsalis cause vibration and position loss with a positive Romberg test, and syringomyelia interrupts crossing spinothalamic fibers, causing cape-like pain and temperature loss. Medial medullary syndrome damages the medial lemniscus, causing opposite-side vibration and position loss, and spinocerebellar loss, as in Friedreich ataxia, causes same-side ataxia.

Relationships to other topics

Descending tracts. Lesions often hit both: in a hemisected cord, the corticospinal tract adds same-side weakness to the sensory pattern.

PHYSIOLOGY

Descending tracts: corticospinal, corticobulbar, and brainstem motor tracts

Descending tracts carry motor commands to cranial nerve nuclei and the spinal cord. The lateral corticospinal tract crosses at the pyramids and drives skilled distal movement, and the corticobulbar tract reaches cranial nerve nuclei mostly bilaterally. Brainstem tracts set posture and tone: the rubrospinal tract favors upper limb flexion, and the vestibulospinal and pontine reticulospinal tracts favor extension, whereas the medullary reticulospinal tract inhibits extensors. Lesions above or below the pyramidal decussation and the red nucleus produce predictable weakness and posturing. Here, we review these tracts and their failures.

Key points

- About 90% of corticospinal fibers cross at the pyramidal decussation, so lesions above it cause opposite-side weakness.
- Corticobulbar input is mostly bilateral, and the upper face receives input from both hemispheres.
- The rubrospinal tract crosses in the midbrain and favors upper limb flexion.
- Lesions above the red nucleus cause decorticate posturing; lesions below it cause decerebrate posturing.

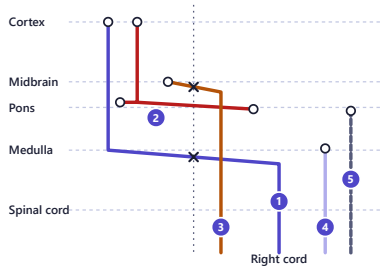


Fig. 1 | Descending tracts from the left cortex and brainstem. 1, the lateral corticospinal tract crosses at the pyramids to the right cord. 2, the corticobulbar tract reaches cranial nerve nuclei on both sides. 3, the rubrospinal tract crosses in the midbrain. 4, the vestibulospinal and, 5, reticulospinal tracts descend on their own side. 1, Lateral corticospinal; 2, Corticobulbar; 3, Rubrospinal; 4, Lateral vestibulospinal; 5, Reticulospinal.

Corticospinal and corticobulbar tracts

The lateral corticospinal tract arises from M1, the premotor cortex and SMA, and S1, and descends through the posterior limb of the internal capsule, cerebral peduncle, basis pontis, and pyramid; 90% of fibers cross at the pyramidal decussation to reach ventral horn neurons for distal limbs, driving skilled movement. The anterior corticospinal tract, about 10%, stays uncrossed until its segment and serves axial and proximal muscles on both sides. The corticobulbar tract arises from the face area of the motor cortex, passes through the genu of the internal capsule, and reaches the motor nuclei of CN V, VII, and IX–XII, mostly bilaterally; because the upper face receives input from both hemispheres, a cortical lesion spares the forehead.

Brainstem tracts

The rubrospinal tract arises from the red nucleus, which receives cortical and cerebellar input, crosses in the midbrain, and favors upper limb flexion. The lateral vestibulospinal tract, from the lateral vestibular nucleus, and the pontine reticulospinal tract descend uncrossed and excite antigravity extensors, whereas the medullary reticulospinal tract, mainly uncrossed, inhibits extensors, facilitates flexors, and contributes to autonomic and breathing control. A lesion above the red nucleus causes decorticate posturing, with flexed arms and extended legs, and a lesion below it causes decerebrate posturing, with extension of all limbs.

When it fails

Capsular or brainstem lesions above the decussation cause opposite-side spastic hemiparesis, and cord lesions below it cause same-side spasticity, hyperreflexia, and a Babinski sign. Corticobulbar lesions cause supranuclear facial palsy, sparing the forehead, and bilateral lesions cause pseudobulbar palsy, with dysarthria, dysphagia, and a brisk jaw jerk.

Relationships to other topics

Ascending tracts. Combined lesions localize precisely: a cord hemisection adds same-side corticospinal weakness to the sensory pattern.

Basal ganglia pathways. The basal ganglia shape movement by acting on the thalamus and cortex, which then act through the corticospinal tract.

PHYSIOLOGY

Basal ganglia pathways: direct, indirect, and the role of dopamine

The basal ganglia select and scale movement through two opposing striatal output pathways. In the direct pathway, D1 medium spiny neurons inhibit the GPi and SNr, releasing the thalamus to drive movement; in the indirect pathway, D2 neurons inhibit the GPe, freeing the STN to excite the GPi and restrain movement. Dopamine from the SNc excites D1 and inhibits D2 neurons, so both actions favor movement. Dopamine loss causes parkinsonism, whereas indirect-pathway loss and STN lesions cause chorea and ballismus. Here, we review the circuit, its failures, and drugs that act on it.

Key points

- The GPi and SNr tonically inhibit the VA and VL thalamus.
- The direct pathway (D1) reduces that inhibition; the indirect pathway (D2, GPe, STN) increases it.
- Dopamine excites D1 and inhibits D2 neurons, favoring movement through both pathways.
- Dopamine loss in Parkinson disease causes bradykinesia; indirect-pathway loss in Huntington disease and STN lesions cause hyperkinesia.

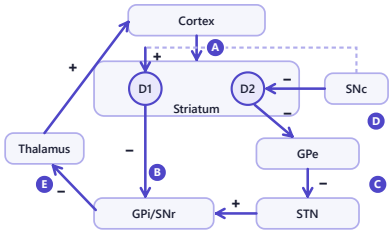


Fig. 1 | The basal ganglia loop. A, the cortex excites the striatum. B, in the direct pathway, D1 neurons inhibit the GPi and SNr, releasing the thalamus. C, in the indirect pathway, D2 neurons inhibit the GPe, freeing the STN to excite the GPi, which inhibits the thalamus. D, dopamine from the SNc excites D1 and inhibits D2 neurons, so both changes favor movement. E, the thalamus excites the cortex. A, Cortex; B, Direct pathway; C, Indirect pathway; D, Dopamine; E, Thalamus.

Circuit

The striatum, made of the caudate and putamen, contains GABA medium spiny neurons that receive cortical and dopamine input. The GPi and SNr form the output and tonically inhibit the VA and VL thalamus, which relays to the motor cortex. In the direct pathway, D1 neurons inhibit the GPi and SNr, releasing the thalamus and allowing movement. In the indirect pathway, D2 neurons inhibit the GPe, which then inhibits the STN less, and the STN, the loop's only glutamate nucleus, excites the GPi, restraining movement. Dopamine from the SNc excites D1 and inhibits D2 neurons, so both actions favor movement.

When it fails

In Parkinson disease, loss of SNc dopamine weakens the direct pathway and strengthens the indirect pathway, increasing GPi inhibition and causing bradykinesia, and dopamine-blocking drugs cause parkinsonism by blocking D2 receptors. In Huntington disease, indirect-pathway neurons die first, reducing GPi inhibition and causing chorea, and an STN lesion removes the GPi's excitation, releasing the thalamus and causing hemiballismus.

Pharmacology

Levodopa crosses the BBB and becomes dopamine, with carbidopa blocking its peripheral conversion, and dopamine agonists such as pramipexole stimulate receptors directly. MAO-B and COMT inhibitors slow dopamine and levodopa breakdown. VMAT2 inhibitors such as tetrabenazine block vesicular loading of dopamine, reducing chorea, and D2 antagonists such as haloperidol reduce hyperkinesia but can cause parkinsonism.

Relationships to other topics

Synapses and neurotransmitters. Dopamine acts through D1 and D2 receptors, members of the metabotropic receptor family.

Descending tracts. The basal ganglia act through the thalamus and cortex, which drive movement through the corticospinal tract.

PHYSIOLOGY

Olfactory, optic, and vestibulocochlear nerves: the special senses

Cranial nerves I, II, and VIII carry smell, sight, hearing, and balance. Olfactory receptor neurons pass through the cribriform plate to the olfactory bulb and reach cortex without a thalamic relay. The optic pathway crosses only its nasal fibers at the chiasm, so lesions produce characteristic field defects, and CN II forms the afferent limb of the pupillary light reflex. CN VIII carries hearing, which becomes bilateral above the cochlear nuclei, and vestibular input to the eye, spinal, and cerebellar pathways. Here, we review these nerves and their failures.

Key points

- Smell reaches cortex without a thalamic relay, and olfactory receptor neurons regenerate.
- Only nasal fibers cross at the chiasm, so chiasmal lesions cause bitemporal hemianopia and tract lesions homonymous hemianopia.
- CN II is the afferent and CN III the efferent limb of the pupillary light reflex.
- Hearing becomes bilateral above the cochlear nuclei, so one-sided deafness localizes to the ear, nerve, or nucleus.

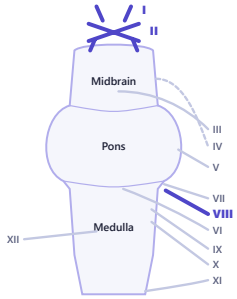


Fig. 1 | The brainstem from the front with the olfactory, optic, and vestibulocochlear nerves highlighted. Brainstem from the front; IV leaves from the back

Olfactory and optic nerves

Olfactory receptor neurons, which regenerate for life, pass through the cribriform plate to synapse on mitral cells in the olfactory bulb, which project to the piriform cortex, amygdala, and entorhinal cortex without a thalamic relay. Ammonia tests the trigeminal nerve rather than smell. Retinal ganglion cells form the optic nerve, a CNS tract myelinated by oligodendrocytes, which passes through the optic canal to the chiasm, where nasal fibers cross. Each optic tract carries the opposite visual field to the LGN, and radiations reach V1 in the calcarine cortex, the lower field through Meyer loop in the temporal lobe. Optic nerve lesions blind one eye, chiasmal lesions cause bitemporal hemianopia, tract and V1 lesions cause homonymous hemianopia, and Meyer loop lesions cause an upper quadrantanopia. Pupil fibers leave before the LGN for the pretectum, and CN III constricts both pupils, so a damaged optic nerve produces an RAPD.

Argyll Robertson pupils constrict to a near target but not to light and mark neurosyphilis.

Vestibulocochlear nerve

Hair cells of the cochlea and vestibular labyrinth synapse on bipolar neurons of the spiral and Scarpa ganglia, whose axons pass through the internal acoustic meatus with CN VII to the cochlear and vestibular nuclei. Hearing becomes bilateral at the trapezoid body and ascends through the lateral lemniscus, inferior colliculus, and MGN to the Heschl gyrus, so one-sided deafness localizes to the ear, nerve, or cochlear nucleus. Vestibular nuclei project through the MLF for the VOR, to the vestibulospinal tracts, and to the cerebellum.

When it fails

Smell is lost early in Parkinson disease and after head injury. Optic neuritis demyelinates CN II, causing painful vision loss with an RAPD, and strokes behind the chiasm cause homonymous hemianopia. Vestibular schwannoma causes one-sided hearing loss and tinnitus, and vestibular neuritis causes vertigo with an abnormal head impulse.

Relationships to other topics

Oculomotor, trochlear, and abducens nerves. CN III forms the efferent limb of the pupillary light reflex,

and the vestibular nuclei drive eye movements through the MLF.

PHYSIOLOGY

Oculomotor, trochlear, and abducens nerves: moving the eyes

Cranial nerves III, IV, and VI move the eyes. CN III supplies four extraocular muscles and the levator, and its parasympathetic fibers constrict the pupil; CN IV supplies the superior oblique and is the only nerve to leave the back of the brainstem; CN VI supplies the lateral rectus. Their palsies produce characteristic deviations, and the pattern of pupil involvement in CN III palsy separates compression from ischemia. The MLF links the abducens and oculomotor nuclei for conjugate gaze. Here, we review these nerves and their failures.

Key points

- CN III palsy turns the eye down and out with ptosis and, if compressive, a dilated pupil.
- A PComm aneurysm dilates the pupil early, whereas a diabetic infarct spares it.
- CN IV palsy causes hypertropia and extorsion, with a head tilt away from the weak side.
- CN VI palsy causes esotropia and is a false localizing sign of raised ICP.
- An MLF lesion causes INO, as in MS.

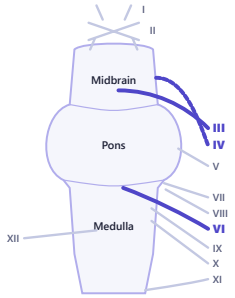


Fig. 1 | The brainstem from the front with the oculomotor, trochlear, and abducens nerves highlighted; the trochlear nerve leaves from the back. Brainstem from the front; IV leaves from the back

Eye muscles

The lateral rectus (VI) abducts, the medial rectus (III) adducts, the superior rectus (III) elevates, and the inferior rectus (III) depresses. The superior oblique (IV) depresses and intorts, especially when adducted, the inferior oblique (III) elevates and extorts when adducted, and the levator (III) lifts the lid.

Sympathetic fibers reach the eye through 3 neurons: from the hypothalamus to the ciliospinal center at C8–T2, out at T1 to the superior cervical ganglion, and along the internal carotid artery to the iris dilator and Müller muscle. A lesion anywhere along the path, as in lateral medullary syndrome, Pancoast tumors, or carotid dissection, causes Horner syndrome: ptosis, miosis, and anhidrosis.

The three nerves

CN III arises with the Edinger-Westphal nucleus at the superior colliculus level, leaves through the interpeduncular fossa beside the PComm, runs in the lateral wall of the cavernous sinus, and enters the orbit through the superior orbital fissure, its parasympathetic fibers synapsing in the ciliary ganglion. Its palsy turns the eye down and out with ptosis and a dilated pupil; because pupil fibers ride on the surface, a PComm aneurysm dilates the pupil early, whereas a diabetic infarct spares it. Every CN IV fiber crosses before leaving the back of the midbrain, and its palsy causes hypertropia and extorsion with a head tilt away from the weak side. CN VI arises under the facial colliculus, climbs the clivus, and runs free in the cavernous sinus; its palsy causes esotropia and can be a false localizing sign of raised ICP, and its nucleus is the horizontal gaze center. The MLF links the VI nucleus to the opposite III nucleus, and its lesion causes INO.

When it fails

A PComm aneurysm compresses CN III and blows the pupil, and Weber syndrome combines a same-side CN III palsy with opposite hemiparesis. MS demyelinates the MLF, causing INO, and progressive supranuclear palsy causes a supranuclear vertical gaze palsy with intact nerves.

Relationships to other topics

Olfactory, optic, and vestibulocochlear nerves. CN II and III form the pupillary light reflex, and vestibular input drives the eyes through the MLF.

PHYSIOLOGY

Trigeminal and facial nerves: face sensation, mastication, and expression

The trigeminal nerve carries sensation from the face through three divisions and supplies the muscles of mastication, with four nuclei extending from the midbrain to C2. The facial nerve supplies facial expression and the stapedius, carries parasympathetic fibers to the lacrimal and salivary glands, and carries taste from the front of the tongue. Together they form the corneal reflex. Upper and lower motor neuron facial palsies differ by forehead involvement, and trigeminal nuclear lesions cause dissociated facial sensory loss. Here, we review these nerves and their failures.

Key points

- CN V has mesencephalic, principal sensory, spinal, and motor nuclei; the spinal nucleus carries facial pain and temperature to C2.
- V1, V2, and V3 leave through the superior orbital fissure, foramen rotundum, and foramen ovale.
- The corneal reflex runs V1 in and VII out.
- The facial nerve passes through the parotid without supplying it.
- A UMN facial palsy spares the forehead; an LMN palsy does not.

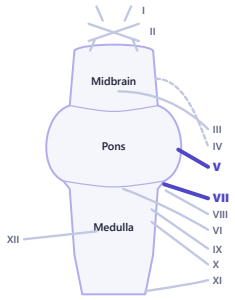


Fig. 1 | The brainstem from the front with the trigeminal and facial nerves highlighted. Brainstem from the front; IV leaves from the back

Trigeminal nerve

CN V has a mesencephalic nucleus for jaw proprioception, a principal sensory nucleus for touch, a spinal nucleus for pain and temperature extending to C2, and a motor nucleus. Its ophthalmic division, through the superior orbital fissure, supplies the forehead and cornea; its maxillary division, through the foramen rotundum, the cheek and upper jaw; and its mandibular division, through the foramen ovale, the lower jaw and front of the tongue, plus the masseter, temporalis, and pterygoids. The angle of the jaw is supplied by C2–C3, and the jaw deviates toward a weak V3.

Trigeminal neuralgia causes brief, electric shock-like pain on one side, usually in V2 or V3, from vascular compression of the root, often by the superior cerebellar artery, and is treated first with carbamazepine or oxcarbazepine.

Facial nerve

The facial nucleus lies in the pons, and its fibers loop over the VI nucleus before leaving at the cerebellopontine angle with CN VIII. At the geniculate ganglion, the greater petrosal nerve leaves for the lacrimal gland; in the facial canal, branches supply the stapedius and form the chorda tympani, which carries taste from the front two-thirds of the tongue and fibers to the submandibular and sublingual glands. Only motor fibers leave the stylomastoid foramen, and the nerve divides into 5 branches within the parotid without supplying it. The corneal reflex runs V1 in and VII out, and loss of the stapedius causes hyperacusis.

Ramsay Hunt syndrome, varicella zoster of the facial nerve with a vesicular rash, causes about 6% of facial palsy.

When it fails

A UMN lesion weakens the lower face and spares the forehead, whereas Bell palsy, an LMN lesion, weakens the whole half of the face with hyperacusis and taste loss. Lateral medullary syndrome damages the spinal trigeminal nucleus, causing same-side facial pain and temperature loss, and vestibular schwannoma can stretch CN VII at the cerebellopontine angle.

Relationships to other topics

Descending tracts. Corticobulbar input to the upper facial nucleus is bilateral, which is why UMN lesions

spare the forehead.

PHYSIOLOGY

Glossopharyngeal, vagus, accessory, and hypoglossal nerves: the lower cranial nerves

The lower cranial nerves leave the medulla and upper cervical cord. CN IX supplies the stylopharyngeus, the parotid, the carotid sinus and body, and the back of the tongue, forming the afferent limb of the gag reflex. CN X supplies the palate, pharynx, and larynx and carries parasympathetic fibers to the heart, lungs, and gut. CN XI supplies the sternocleidomastoid and trapezius, and CN XII the tongue. Their lesions cause dysphagia, hoarseness, weakness of head turning, and tongue deviation, and they localize medullary strokes. Here, we review these nerves and their failures.

Key points

- CN IX is the afferent and CN X the efferent limb of the gag reflex.
- The left recurrent laryngeal nerve hooks under the aortic arch.
- The uvula deviates away from a vagal lesion, and the tongue toward a hypoglossal lesion.
- The right sternocleidomastoid turns the head to the left.
- Lateral medullary syndrome affects IX and X; medial medullary syndrome affects XII.

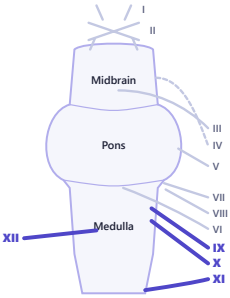


Fig. 1 | The brainstem from the front with the glossopharyngeal, vagus, accessory, and hypoglossal nerves highlighted. Brainstem from the front; IV leaves from the back

The four nerves

CN IX supplies its only muscle, the stylopharyngeus, sends parasympathetic fibers to the parotid through the otic ganglion, carries pressure and oxygen signals from the carotid sinus and body that slow the heart, and carries taste and touch from the back third of the tongue, forming the afferent limb of the gag reflex. CN X, from the nucleus ambiguus, supplies the palate, pharynx, and larynx and forms the efferent limb of the gag, and from the dorsal motor nucleus sends parasympathetic fibers to the heart, lungs, and gut to the splenic flexure; the left recurrent laryngeal nerve hooks under the aortic arch, and the uvula deviates away from a vagal lesion. CN XI arises from C1–C5, rises through the foramen magnum, and leaves through the jugular foramen to supply the sternocleidomastoid and trapezius, and can be injured in posterior triangle node biopsy. CN XII arises in the floor of the fourth ventricle, leaves between the pyramid and olive through the hypoglossal canal, and supplies all tongue muscles except palatoglossus; its lesion deviates the tongue toward the lesion, with wasting and fasciculations.

When it fails

Lateral medullary syndrome damages the nucleus ambiguus, causing dysphagia, hoarseness, and a reduced gag, and medial medullary syndrome damages CN XII, deviating the tongue toward the lesion with opposite hemiparesis. Bilateral UMN lesions cause pseudobulbar palsy, with dysarthria and dysphagia but no wasting, and recurrent laryngeal nerve injury causes hoarseness.

Relationships to other topics

Descending tracts. Corticobulbar fibers reach these motor nuclei, so bilateral UMN lesions cause pseudobulbar palsy.

STROKES AND LESIONS

Stroke syndromes: localizing the lesion from the cortex to the spinal cord

Stroke symptoms follow the anatomy of the lesion rather than its cause. Cerebral syndromes reflect the ACA, MCA, PCA, and watershed territories, brainstem syndromes produce crossed signs, and spinal cord lesions separate motor, pain, and proprioceptive pathways. Hemorrhagic strokes add features of raised ICP and meningeal irritation. Here, we review these syndromes and their complications, and consider how to distinguish those that resemble each other.

Key points

- Brainstem strokes are crossed: cranial nerve signs are ipsilateral, and limb signs are contralateral.
- ACA favors the leg, whereas MCA favors the face and arm.
- Dominant MCA lesions cause aphasia; non-dominant lesions cause neglect.
- Thunderclap headache with meningeal signs suggests SAH.

Cerebral ischemic syndromes

ACA syndrome causes contralateral leg weakness, greater than arm, contralateral leg sensory loss, urinary incontinence, abulia, and a grasp reflex. MCA syndrome causes contralateral face and arm weakness, greater than leg, gaze toward the lesion, contralateral face and arm sensory loss, and contralateral homonymous hemianopia; dominant lesions add aphasia and Gerstmann syndrome, and non-dominant lesions add extinction, hemispatial neglect, and anosognosia. PCA syndrome usually causes no weakness, with contralateral sensory loss if the thalamus is involved, contralateral homonymous hemianopia with macular sparing, alexia without agraphia (dominant), and prosopagnosia (non-dominant). MCA/ACA watershed syndrome, from low perfusion or ICA stenosis, causes proximal arm and shoulder weakness sparing the hand and face ("man in a barrel" if bilateral), mild proximal arm sensory loss, and transcortical motor aphasia (dominant). MCA/PCA watershed syndrome usually causes no weakness and mild sensory change, with visual field loss, transcortical sensory aphasia (dominant), neglect (non-dominant), and Balint syndrome if bilateral.

Dominant-hemisphere lesions cause characteristic aphasias. Broca aphasia, from the inferior frontal gyrus, is nonfluent with intact comprehension and impaired repetition. Wernicke aphasia, from the posterior superior temporal gyrus, is fluent but paraphasic, with impaired comprehension and repetition. Conduction aphasia, from the arcuate fasciculus, is fluent with relatively intact comprehension but impaired repetition, and global aphasia affects all three. Transcortical motor and sensory aphasias resemble Broca and Wernicke aphasia but preserve repetition.

Lacunar infarcts follow lipohyalinosis or microatheroma of small penetrating arteries, mainly with hypertension and diabetes, and lack cortical signs. Pure motor stroke, the most common at 45%, involves the posterior limb of the internal capsule, corona radiata, or ventral pons; pure sensory stroke involves the thalamus; sensorimotor stroke involves the thalamus and internal capsule; ataxic hemiparesis involves the internal capsule, pons, or corona radiata; and dysarthria-clumsy hand syndrome involves the pons or internal capsule. Amaurosis fugax, a painless, curtain-like loss of vision in one eye that resolves, usually follows emboli from the carotid bifurcation and warns of stroke.

Brainstem strokes

In the midbrain, supplied by P1 paramedian branches and the top of the basilar, Weber syndrome (ventromedial) causes ipsilateral CN III palsy with a dilated pupil and contralateral hemiparesis. Benedikt syndrome (paramedian) causes ipsilateral CN III palsy, contralateral tremor and ataxia, with or without contralateral hemiparesis, and contralateral vibration and proprioception loss, combining Weber and Claude features. Claude syndrome causes ipsilateral CN III palsy and contralateral ataxia, with no weakness. In the pons, ventromedial pontine syndrome, from basilar paramedian branches, causes contralateral hemiparesis, dysarthria, and ataxic hemiparesis, and its caudal form (Raymond syndrome) adds ipsilateral CN VI palsy. Ventral pontine (Millard-Gubler) syndrome causes contralateral hemiplegia, ipsilateral CN VI and CN VII palsy (forehead included), and horizontal diplopia. Inferior medial pontine (Foville) syndrome causes ipsilateral horizontal gaze palsy and CN VII palsy (forehead included), contralateral hemiparesis, contralateral vibration and proprioception loss, and eyes

that deviate toward the weak side; both are supplied by basilar paramedian and short circumferential branches. Lateral pontine (AICA) syndrome causes ipsilateral CN VII palsy (forehead included) and ataxia with preserved strength, crossed pain and temperature loss (ipsilateral face and contralateral body), and ipsilateral hearing loss, vertigo, and Horner syndrome. Marie-Foix syndrome, a lateral pontine syndrome from basilar perforators and the AICA, adds contralateral hemiparesis and contralateral vibration and proprioception loss to ipsilateral limb ataxia, CN VII palsy, hearing loss, vertigo, and nystagmus. Locked-in syndrome, from bilateral ventral pons injury via the basilar artery, causes quadriplegia and anarthria, usually with preserved sensation, consciousness, vertical eye movements, and blinking. In the medulla, lateral medullary (Wallenberg) syndrome, from the vertebral artery or PICA, causes ipsilateral palate, pharynx, and vocal cord weakness and ataxia with preserved strength, crossed pain and temperature loss, and ipsilateral Horner syndrome, vertigo, dysphagia, and hiccups. Medial medullary (Dejerine) syndrome, from the anterior spinal artery, causes contralateral hemiparesis sparing the face, ipsilateral tongue weakness with deviation toward the lesion, and contralateral vibration and proprioception loss.

Spinal cord lesions

Central cord syndrome, from hyperextension injury and syringomyelia, causes arm weakness greater than leg, with the hands worst, cape-like pain and temperature loss, and sacral sparing. Anterior cord syndrome, from the anterior spinal artery, causes bilateral weakness and pain and temperature loss below the lesion, preserved vibration and proprioception, and bladder and bowel dysfunction. Posterior cord syndrome, from the posterior spinal artery, tabes dorsalis, and B12 deficiency, causes sensory ataxia with preserved strength, bilateral vibration and proprioception loss, and a positive Romberg sign. Brown-Séquard syndrome (cord hemisection), from penetrating trauma and compression, causes ipsilateral weakness and vibration and proprioception loss below the lesion, contralateral pain and temperature loss, and ipsilateral Horner syndrome if at or above T1.

Hemorrhagic strokes

Intracerebral hemorrhage causes headache and cortical focal neurologic deficits. Subarachnoid hemorrhage causes thunderclap headache and meningeal signs: photophobia, neck stiffness, and positive Kernig and Brudzinski signs. Both cause raised ICP, with nausea and vomiting, posturing, and Cushing triad, along with loss of consciousness in intracerebral hemorrhage and CN deficits in subarachnoid hemorrhage. Subarachnoid hemorrhage is complicated by rebleeding, highest early before the aneurysm is secured; vasospasm and delayed cerebral ischemia on days 3–14; hydrocephalus; hyponatremia from SIADH or cerebral salt wasting; and seizures.

Complications

Stroke complications are cerebral edema (malignant MCA infarct, with herniation in the first several days), hemorrhagic transformation, aspiration pneumonia from dysphagia, seizures, DVT and PE from immobility, spasticity and contractures, post-stroke depression, and central post-stroke pain, often from a thalamic infarct but possible after any sensory pathway lesion.

Relationships to other conditions

Ischemia and hemorrhage. Both can cause focal deficits, and non-contrast CT head separates them.

Lateral pontine vs lateral medullary. Both cause crossed pain and temperature loss, ataxia, vertigo, and Horner syndrome with preserved strength. CN VII palsy and hearing loss favor the pons; dysphagia, hiccups, and palate weakness favor the medulla.

Midbrain syndromes. All three share CN III palsy. Contralateral hemiparesis means Weber, contralateral ataxia means Claude, and tremor with sensory loss means Benedikt.

Medial medullary vs ventral pontine. Both cause contralateral hemiparesis. Ipsilateral tongue weakness points to the medulla, and CN VI and CN VII palsy to the pons.

Anterior vs posterior cord. Anterior cord lesions lose strength and pain and temperature but preserve vibration and proprioception, whereas posterior cord lesions lose vibration and proprioception but preserve strength.

STROKES AND LESIONS

Acute ischemic stroke: embolic, thrombotic, and low-flow injury and the race to reperfuse

Blocked or inadequate arterial blood flow leads to cerebral ischemia, causing an infarct core surrounded by a salvageable penumbra. Embolus is the most common mechanism; the others are thrombus, global hypoperfusion, vasculitis, and dissection. Management is a race to reperfuse within fixed time windows, followed by prevention tailored to the cause. Here, we review its causes, imaging, acute treatment, and recurrence prevention, and consider how it relates to hemorrhagic stroke.

Key points

- Glucose and non-contrast CT head come first, to rule out hypoglycemia and hemorrhage.
- IV alteplase or tenecteplase is given within 4.5 hours of onset.
- Thrombectomy for large vessel occlusion extends up to 24 hours with penumbra.
- Prevention of recurrence is chosen by cause.

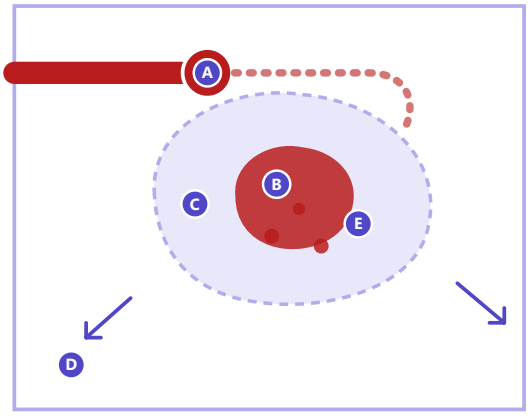


Fig. 1 | An artery blocked by an embolus, with a core of infarcted brain surrounded by a penumbra downstream. Arrows show swelling, and small spots show bleeding into the infarct. A, Arterial occlusion; B, Infarct core; C, Penumbra; D, Cerebral edema; E, Hemorrhagic transformation.

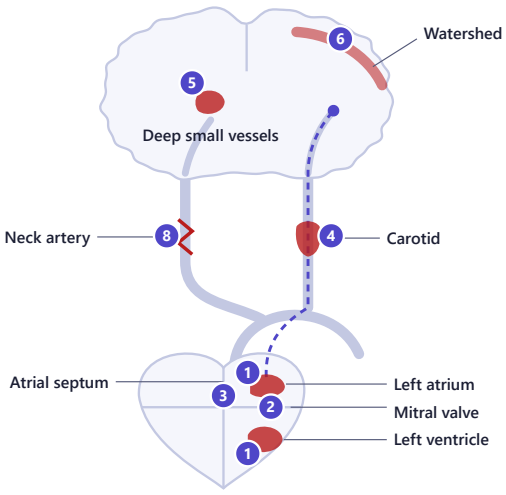


Fig. 2 | Diagram of the heart, aortic arch, neck arteries, and brain showing where each cause of ischemic stroke starts. 1, Cardiac embolus; 2, Infective endocarditis; 3, Paradoxical embolus; 4, Large vessel thrombus; 5, Small vessel thrombus; 6, Global hypoperfusion; 8, Dissection.

Pathophysiology

Blocked or inadequate flow causes cerebral ischemia with decreased oxygen delivery. An embolus or thrombus stops flow downstream. In the infarct core, flow is too low for cells to survive, so injury is irreversible, whereas in the penumbra, flow stops function but not survival, so reperfusion can reverse deficits. Failing ion pumps swell cells, causing cerebral edema, with malignant MCA infarct and herniation in the first several days, and reperfusion of injured vessels can cause hemorrhagic transformation, with blood within the infarct.

Etiology and mechanisms

Cardiac embolus, an important cause, arises from a left atrial thrombus in A-fib or flutter, mechanical valve, or RHD, or from a left ventricular thrombus in LV aneurysm from MI, anterior or lateral MI, or HFrEF. Infective endocarditis is a separate source of cardiac embolus and a contraindication to thrombolysis. Paradoxical embolus occurs with ASD, patent foramen ovale, and atrial septal aneurysm, where a right-to-left shunt lets a venous thrombus reach arterial circulation. Global hypoperfusion from cardiac arrest, acute respiratory failure, or ICA stenosis with low BP decreases cerebral oxygen delivery, causing watershed infarcts. Large vessel thrombus is atherosclerotic thrombus in the ICA, MCA, and vertebrobasilar arteries. Small vessel thrombus causes lacunar infarcts from the lenticulostriate arteries or small pontine branches. Vasculitis from autoimmune disease, infections, or primary CNS vasculitis causes arterial wall inflammation and can affect any artery. Dissection from trauma, iatrogenic injury, or connective tissue diseases causes an intimal tear with thrombus and embolus.

Diagnosis

Glucose and non-contrast CT head rule out hypoglycemia and hemorrhage. If there is no hemorrhage, CT angiogram of the head and neck looks for large vessel occlusion, and CT perfusion or MRI looks for penumbra if onset is 4.5–24 hours ago or unknown. Cardiac embolus is sought with ECG, telemetry, and echocardiogram, looking for A-fib and thrombus. Paradoxical embolus is sought with bubble echocardiogram, looking for PFO. Large vessel thrombus is assessed by CT angiogram or carotid duplex for ICA stenosis, small vessel thrombus by MRI for lacunar infarct, vasculitis by ESR, CRP, autoimmune serology, and CSF, dissection by CT or MR angiogram of the neck, and infective endocarditis by blood cultures and echocardiogram. Vascular risk factors are assessed with lipids and HbA1c. NIHSS grades stroke severity.

Stroke mimics

Stroke mimics must be recognized without delaying reperfusion. Hypoglycemia causes deficits that resolve with glucose. A seizure with Todd paralysis is favored by a witnessed seizure, tongue biting, and a post-ictal state. Migraine with aura spreads over minutes, with positive symptoms and past similar attacks. Functional neurologic disorder shows inconsistent signs, a positive Hoover sign, and normal imaging. Brain tumor or abscess has a gradual onset, with fever in an abscess, and is seen on CT or MRI with contrast. Bell palsy causes isolated facial weakness that includes the forehead, peripheral vertigo causes isolated vertigo with a reassuring HINTS examination, and systemic illness such as sepsis, hyponatremia, or drugs can unmask an old deficit.

Management

Reperfusion uses IV alteplase or tenecteplase within 4.5 hours of onset, and mechanical thrombectomy for large vessel occlusion up to 24 hours with penumbra. Aspirin is given within 24–48 hours, waiting 24 hours after thrombolysis. Supportive care comprises a swallow screen before oral intake, treating fever and glucose, and DVT prophylaxis. BP is kept below 185/110 for thrombolysis, and otherwise allowed up to 220/120. Thrombolysis contraindications are hemorrhage on CT, head trauma within 3 months, intracranial or spinal surgery within 3 months, active bleeding, platelets under 100000/μL, INR over 1.7, infective endocarditis, aPTT over 40 seconds, therapeutic LMWH within 24 hours, recent DOAC use with possible ongoing effect, suspected aortic dissection, and an intracranial neoplasm with high bleeding risk. Prior ischemic stroke is weighed case by case rather than excluding thrombolysis. After a minor non-cardioembolic stroke (NIHSS 3 or less) or high-risk TIA not treated with thrombolysis, aspirin plus clopidogrel is started within 24 hours and continued for 21 days before changing to one antiplatelet.

Prevention of recurrence

Cardiac embolus is treated with anticoagulation: DOAC, or warfarin for a mechanical valve. Infective endocarditis is treated with ABX. Paradoxical embolus is treated with antiplatelet therapy, with PFO closure if under 60 with no other cause. Large vessel thrombus is treated with antiplatelet therapy and high-intensity statin, with carotid endarterectomy for symptomatic ICA stenosis of 70–99%. Small vessel thrombus is treated with antiplatelet therapy, statin, and BP control. Global hypoperfusion is treated by treating the cause of low perfusion. Vasculitis is treated with immunosuppression and treating any infection. Dissection is treated with antiplatelet therapy or anticoagulation.

In A-fib, the CHA_2DS_2-VASc score estimates stroke risk to guide anticoagulation, with 1 point each for congestive heart failure, hypertension, diabetes, vascular disease, age 65–74, and female sex, and 2 points each for age 75 or older and prior stroke or TIA; a score of 2 or more marks high risk.

Relationships to other conditions

Intracerebral hemorrhage. Non-contrast CT separates them, and BP goals are opposite: ischemic stroke tolerates up to 220/120, whereas hemorrhage is actively lowered. Hemorrhagic transformation links the two.

Subarachnoid hemorrhage. Thunderclap headache with meningeal signs favors SAH, and hemorrhage on CT contraindicates thrombolysis.

Stroke syndromes. Deficits localize the vessel, and watershed syndromes follow low perfusion or ICA stenosis.

Infective endocarditis. It causes ischemic stroke by embolism and contraindicates thrombolysis, and it causes hemorrhage through mycotic aneurysms.

Embolic disease elsewhere. A-fib also embolizes to the SMA, causing acute mesenteric ischemia.

Intracerebral hemorrhage: causes by location, and limiting hematoma growth

Vessel rupture bleeds into the brain parenchyma, forming a hematoma that compresses tissue and raises ICP. Hypertension is the most common cause, and cerebral amyloid angiopathy is a common cause of lobar bleeds in older adults, with location helping to separate them. Management aims to limit hematoma growth, reverse coagulopathy, and control ICP. Here, we review its causes, workup, and treatment, and consider how it relates to ischemic stroke and subarachnoid hemorrhage.

Key points

- Hypertension bleeds in deep structures, especially the basal ganglia; amyloid angiopathy bleeds in the cortex.
- If systolic is 150–220, lower it to a target of 140.
- Reversal is matched to the drug.
- Cerebral venous sinus thrombosis is treated with anticoagulation, even with hemorrhage.

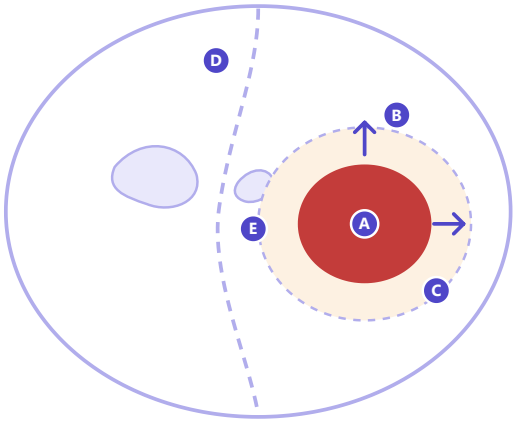


Fig. 1 | A brain in coronal section with a hematoma in one hemisphere. Swelling surrounds the hematoma, arrows show it growing, the midline is pushed across, and the nearby ventricle is compressed. A, Tissue compression; B, Hematoma growth; C, Surrounding edema; D, Raised ICP; E, Hydrocephalus.

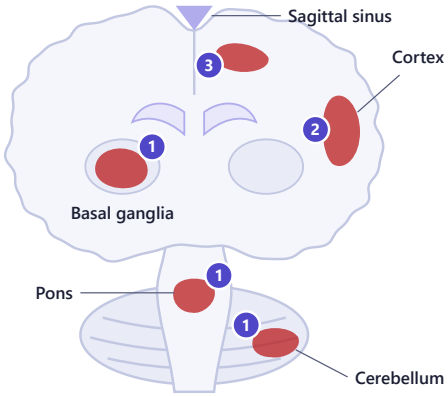


Fig. 2 | Coronal brain diagram: hypertension bleeds in the basal ganglia, pons, and cerebellum; amyloid angiopathy bleeds in the cortex; venous sinus thrombosis bleeds near the cortex under the sagittal sinus. 1, Hypertension; 2, Cerebral amyloid angiopathy; 3, Cerebral venous sinus thrombosis.

Pathophysiology

A weakened vessel ruptures into the brain parenchyma, forming a hematoma. The hematoma presses on brain tissue, causing cortical focal neurologic deficits, and bleeding can continue, causing worsening deficits, which is why BP control and reversal limit it. Injured tissue swells around the hematoma, adding to its mass. This added mass raises ICP, causing headache, nausea and vomiting, loss of consciousness, posturing, and Cushing triad, and blood or mass effect can block CSF flow, causing hydrocephalus that may need an EVD.

Etiology and mechanisms

Hypertension, the most common cause, bleeds in the basal ganglia, thalamus, pons, and cerebellum, and less commonly the cortex. Cerebral amyloid angiopathy, a common cause of lobar bleeds in older adults, bleeds in

cortical structures and may present with dementia. Coagulopathy follows warfarin, heparin, DOACs, tPA, aspirin, clopidogrel, and liver disease. Hemorrhagic transformation follows ischemic stroke, through conversion from tPA or reperfusion injury. Arteriovenous malformation is a vascular abnormality of young patients. Mycotic aneurysm follows infective endocarditis vegetation deposits. Cerebral venous sinus thrombosis follows factor V Leiden, decreased AT-III, decreased protein C or S, APS, OCP, and pregnancy, with obstructed venous drainage causing venous infarction or hemorrhage near the cortex. Malignancy, including melanoma, RCC, choriocarcinoma, and thyroid carcinoma, disrupts the BBB. Causes from coagulopathy onward can bleed at any site. Primary ICH, from hypertension or amyloid angiopathy, accounts for 85% of cases and is a diagnosis of exclusion, whereas secondary ICH has an identifiable cause, such as a vascular malformation, tumor, or bleeding diathesis. Chronic hypertension causes lipohyalinosis and Charcot-Bouchard aneurysms in penetrating arterioles.

Clinical presentation

Patients have headache and cortical focal neurologic deficits, with raised ICP causing nausea and vomiting, loss of consciousness, posturing, and Cushing triad.

Diagnosis

Non-contrast CT head shows hyperdense blood in the parenchyma, bloods comprise FBC, PT/INR, aPTT, and LFTs, and CT angiogram is obtained if the patient is young, the bleed is lobar, or there is no hypertension. Hypertension shows a deep location on CT. Cerebral amyloid angiopathy is evaluated by MRI with SWI, showing lobar microbleeds. Coagulopathy is evaluated by those bloods. Hemorrhagic transformation is seen on CT head after stroke or thrombolysis as blood within the infarct. Cerebral venous sinus thrombosis is evaluated by CT or MR venography, with the empty delta sign. Malignancy is evaluated by MRI with contrast, showing an enhancing mass. Arteriovenous malformation is evaluated by CT angiogram, with catheter angiography to confirm. Mycotic aneurysm is evaluated by blood cultures, echocardiogram, and CT angiogram. The ICH score estimates 30-day mortality. Mortality reaches 50% at 30 days, with most deaths in the first 24 hours, and depends on hematoma size and location, GCS on admission, and intraventricular extension or hydrocephalus.

The ICH score predicts 30-day mortality from five items: GCS 3–4 (2 points) or 5–12 (1 point), age 80 or older, infratentorial origin, volume of 30 mL or more, estimated as A × B × C / 2 on CT, and IVH, each 1 point.

Management

Hematoma growth is limited with IV nicardipine, clevidipine, or labetalol, and reversal of any coagulopathy; if systolic is 150–220, it is lowered to a target of 140. ICP is lowered with head of bed at 30°, hypertonic saline or mannitol, and EVD for hydrocephalus. Surgical evacuation is used for a cerebellar bleed over 3 cm with brainstem compression, decreased consciousness, or hydrocephalus. Supportive care comprises ICU admission, treating fever and hyperglycemia, antiseizure medication only for seizures, and intermittent pneumatic compression. Warfarin is reversed with 4-factor PCC and IV vitamin K to an INR below 1.4, with PCC acting faster than FFP, heparin with protamine, dabigatran with idarucizumab, apixaban or rivaroxaban with andexanet alfa or 4-factor PCC, and tPA with cryoprecipitate. For aspirin or clopidogrel, the offending drug is stopped. Liver disease is treated with vitamin K and FFP. By cause, hypertension needs long-term control below 130/80; amyloid angiopathy needs antithrombotics avoided where possible; hemorrhagic transformation needs antithrombotics stopped; venous sinus thrombosis needs anticoagulation, even with hemorrhage; malignancy needs surgery, radiation, or systemic therapy; arteriovenous malformation needs surgical resection, embolization, or radiosurgery; and mycotic aneurysm needs IV ABX for infective endocarditis, with endovascular or surgical repair if ruptured.

Relationships to other conditions

Ischemic stroke. Both cause focal deficits and CT separates them. Thrombolysis and antithrombotics that treat ischemia can cause hemorrhage, and BP goals differ.

Subarachnoid hemorrhage. Blood lies in the parenchyma here and in the subarachnoid space there. AVM and mycotic aneurysm cause both, and thunderclap headache with meningeal signs favors SAH.

Dementia. Cerebral amyloid angiopathy can present with dementia as well as lobar bleeds.

Stroke complications. Hemorrhagic transformation is both a complication of ischemic stroke and a cause of intracerebral hemorrhage.

Subarachnoid hemorrhage: aneurysmal rupture and the threats that follow

Vessel rupture, from trauma or a weakened wall such as an aneurysm, bleeds into the subarachnoid space, causing meningeal irritation and raised ICP. Patients describe a thunderclap headache, and triggers include exercise, straining, and anger. Aneurysms favor the AComm, then the PComm. Severity is graded by Hunt and Hess, and management centers on securing the aneurysm and preventing delayed complications. Here, we review its causes, workup, and treatment, and consider how it relates to other stroke types.

Key points

- Thunderclap headache with meningeal signs is the classic presentation.
- If CT is negative and suspicion remains, lumbar puncture looks for xanthochromia.
- Securing the aneurysm early prevents rebleeding.
- Nimodipine prevents ischemia, and vasospasm occurs on days 3–14.

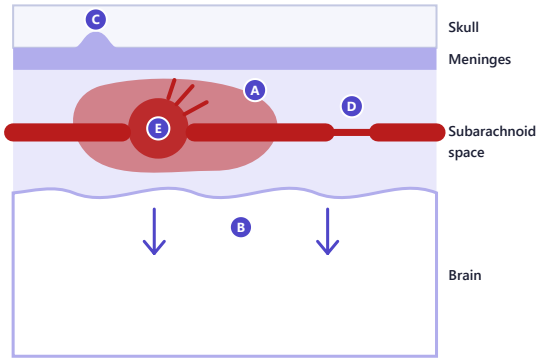


Fig. 1 | A cross-section from skull to brain. An artery in the subarachnoid space has a ruptured aneurysm, and blood spreads through the space. Lettered sites mark irritated meninges, raised pressure on the brain, a blocked arachnoid granulation, a narrowed artery, and the aneurysm that can bleed again. A, Meningeal irritation; B, Raised ICP; C, Hydrocephalus; D, Vasospasm; E, Rebleeding.

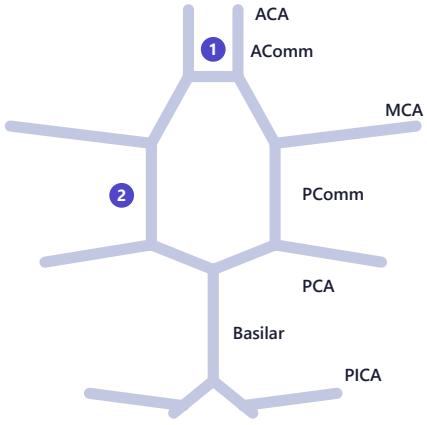


Fig. 2 | Circle of Willis with the two most common aneurysm sites marked: the anterior communicating artery and the posterior communicating artery. About 85% of aneurysms arise in the anterior circulation. 1, AComm; 2, PComm.

Pathophysiology

Trauma or a weakened wall such as an aneurysm leads to vessel rupture, triggered by exercise, straining, or anger, and blood spreads through the CSF of the subarachnoid space. Blood irritates the meninges, causing thunderclap headache, photophobia, and neck stiffness, and added blood volume raises ICP, causing nausea and vomiting, CN deficits, posturing, and Cushing triad. Blood blocks CSF absorption at the arachnoid granulations, causing hydrocephalus, and blood around arteries narrows them on days 3–14, causing vasospasm and delayed cerebral ischemia. The unsecured aneurysm can rupture again, with the highest risk early.

Etiology and mechanisms

Traumatic SAH follows blunt force or penetrating injury. Non-traumatic causes are aneurysms (saccular or berry, giant, fusiform, dissection pseudoaneurysm, and

mycotic); hypertension, smoking, and sympathomimetics, with alcohol, OCP, and pregnancy less certain, which cause hemodynamic stress and inflammation at bifurcations and leave a thin or absent tunica media; AVM; and inherited conditions (Ehlers-Danlos syndrome, PKD, fibromuscular dysplasia, coarctation of the aorta, and Marfan syndrome). About 85% of aneurysms arise in the anterior circulation, most often at the AComm, with the PComm also common. Triggers are coffee or cola, physical exercise, sneezing, straining, sexual intercourse, anger, and being startled.

Clinical presentation and complications

Patients have a thunderclap headache, meningeal signs (photophobia, neck stiffness, and positive Kernig and Brudzinski signs), and raised ICP (nausea and vomiting, CN deficits, posturing, and Cushing triad). Complications are rebleeding, highest early before the aneurysm is secured; vasospasm and delayed cerebral ischemia on days 3–14; hydrocephalus; hyponatremia from SIADH or cerebral salt wasting; and seizures.

Diagnosis

Non-contrast CT head shows blood in the subarachnoid space. If CT is negative and suspicion remains, lumbar puncture looks for xanthochromia. CT angiogram finds the source, with catheter angiography if negative. The Hunt and Hess scale grades severity: grade 1 is asymptomatic, or minimal headache and slight neck stiffness; grade 2 is moderate to severe headache and neck stiffness, with no deficit except CN palsy; grade 3 is drowsiness, confusion, or mild focal deficit; grade 4 is stupor, moderate to severe hemiparesis, and possibly early decerebrate rigidity; and grade 5 is deep coma, decerebrate posturing, and a moribund appearance. A higher grade means a worse prognosis. By the Ottawa SAH rule, alert patients over 15 years with a new severe headache peaking within 1 hour are investigated if they have neck pain or stiffness, are aged 40 or older, had a witnessed LOC, developed the headache during exertion, describe a thunderclap headache, or have limited neck flexion. CT within 6 hours of onset has a sensitivity near 100%, so a normal scan can rule out SAH without a lumbar puncture; after 6 hours, sensitivity falls, and a lumbar puncture follows a normal scan.

Management

The aneurysm is secured early by endovascular coiling or surgical clipping. BP is controlled with IV nicardipine or labetalol until then, keeping systolic below 160. Nimodipine prevents ischemia. ICP is lowered with head of bed at 30°, hypertonic saline or mannitol, and EVD for hydrocephalus. Supportive care comprises ICU admission, pain relief, and reversal of any coagulopathy. Rebleeding is prevented by securing the aneurysm; vasospasm is treated with nimodipine and euvolemia, with endovascular therapy if refractory; hydrocephalus is treated with EVD; hyponatremia is treated by treating the cause and avoiding fluid restriction; and seizures are treated with antiseizure medication. By cause, trauma needs supportive care and treatment of associated injuries; vessel wall injury needs hypertension control and stopping smoking, alcohol, and sympathomimetics; AVM needs surgical resection, embolization, or radiosurgery; and inherited conditions need screening for aneurysms. The aneurysm is secured preferably within 24 hours, because rebleeding occurs in 9–17% of patients in the first 72 hours, with the highest risk in the first 6 hours. If repair is delayed, an antifibrinolytic may be given for no more than 72 hours. Until the aneurysm is secured, antithrombotics are stopped, anticoagulation is reversed, and pain is treated with paracetamol, avoiding aspirin and NSAIDs.

Relationships to other conditions

Intracerebral hemorrhage. AVM and mycotic aneurysm cause both. Blood in the subarachnoid space with meningeal signs favors SAH, and focal deficits with parenchymal blood favor ICH.

Ischemic stroke. Vasospasm and delayed cerebral ischemia on days 3–14 link the two, which is why nimodipine is given to prevent ischemia.

Stroke syndromes. Seizures appear in both complication lists, whereas rebleeding and vasospasm are listed only for SAH.

Polycystic kidney disease. PKD is grouped with the inherited conditions that need screening for aneurysms.

EPILEPSY

Seizures: lowered thresholds, the patterns they produce, and stepwise control

Excess excitation or lost inhibition lowers the seizure threshold, so cortical neurons fire in excessive synchrony. An etiology raises excitability, and triggers such as stress, sleep deprivation, and bright lights and sounds can then push the brain over the threshold. Seizures are focal or generalized, and their features follow the region involved. Management stabilizes the patient, stops prolonged seizures in timed steps, treats the cause, and selects long-term medication by seizure type. Here, we review these mechanisms, causes, patterns, and treatments, and consider how seizures relate to other neurologic conditions.

Key points

- Causes follow VITAMIN DEFG, and triggers act on top of an etiology.
- Focal seizures follow their lobe: motor in the frontal lobe, sensory in the parietal, occipital, and temporal lobes, and autonomic and psychological in the temporal lobe and insula.
- Reticular formation or thalamic involvement brings LOC, often with automatisms, and a post-ictal state.
- Absence seizures have no post-ictal state, and EEG shows a 3 Hz spike-and-wave pattern.
- A seizure lasting 5 minutes or more is status epilepticus and is treated in timed steps.

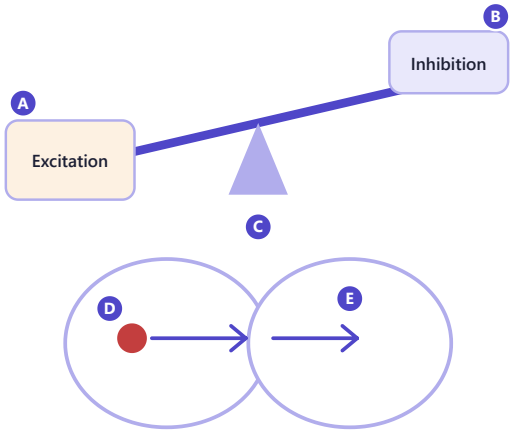


Fig. 1 | A tilted balance with excitation outweighing inhibition. Below it, firing starts at one site in one hemisphere and spreads to the other. A, Excitation; B, Inhibition; C, Synchrony; D, Focal seizure; E, Generalized seizure.

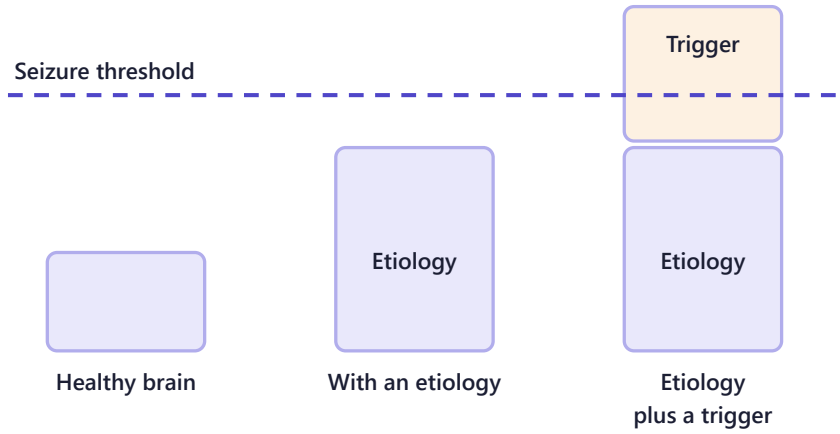


Fig. 2 | Three bars rising toward a dashed seizure threshold. A healthy brain stays far below it. An etiology raises excitability closer to it. A trigger on top of an etiology crosses it, causing a seizure.

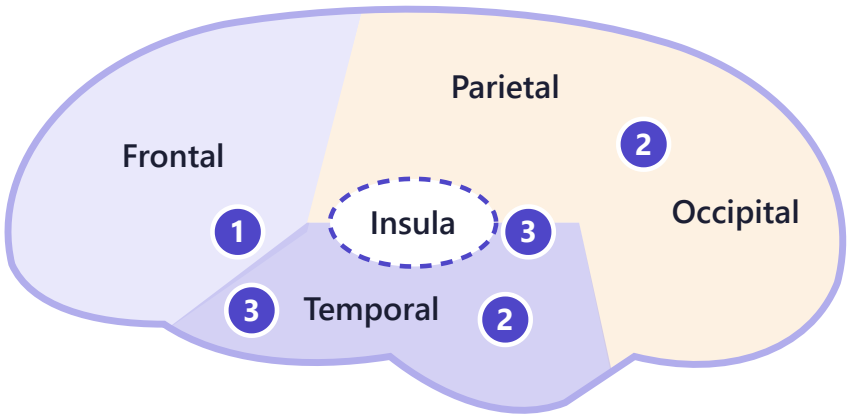


Fig. 3 | A lateral view of the brain. The frontal lobe produces motor seizures, the parietal and occipital lobes produce sensory seizures, the temporal lobe produces sensory, autonomic, and psychological seizures, and the insula produces autonomic and psychological seizures. 1, Motor (frontal); 2, Sensory (parietal, occipital, and temporal); 3, Autonomic and psychological (temporal and insular).

Pathophysiology

Glutamate opens Na+ and Ca2+ channels, causing depolarization, and GABA opens Cl– channels, causing hyperpolarization. When excitation wins, the seizure threshold falls and neurons fire together rather than independently. In a focal seizure, firing starts in one region, so symptoms match that region; a focal seizure that spreads to both hemispheres is a focal to bilateral tonic-clonic seizure. In a generalized seizure, firing rapidly engages networks in both hemispheres.

Etiology

Vascular causes are AIS, ICH, SAH, PRES from increased BP, and anoxic brain injury from global hypoxia. Infectious causes are meningitis, encephalitis, and brain abscess, and traumatic causes are epidural and subdural hematomas. Autoimmune causes are SLE and paraneoplastic syndrome, leading to autoimmune encephalitis. Metabolic causes are decreased vitamin B1, hypoglycemia or hyperglycemia, hyponatremia or hypernatremia, hypocalcemia, hypomagnesemia, hypophosphatemia, increased urea and metabolic waste, increased ammonia, and increased thyroid hormones. Iatrogenic causes are opioids, TCAs, isoniazid, salicylate, cocaine, cefepime, amphetamine, metronidazole, penicillin, bupropion, lithium, and benzodiazepine or alcohol withdrawal. Idiopathic causes are epilepsy, with non-adherence to medications, West syndrome, Lennox-Gastaut syndrome, and juvenile myoclonic epilepsy. Neoplastic causes are glioblastoma, meningioma, and metastatic cancer, and degenerative or demyelinating causes are Alzheimer disease and MS. Eclampsia, fever, and genetic causes complete the list. Triggers are stress, sleep deprivation, and bright lights and sounds.

Febrile seizures occur in children aged 6 months to 5 years with fever of 38 °C or more and no CNS infection. Most are simple, a single generalized seizure of 15 minutes or less; complex febrile seizures have focal features, last 15 minutes or more, or recur within 24 hours. Most resolve without complications, although a subset carries a higher risk of epilepsy.

Clinical presentation

A seizure can pass through an aura, an ictal event, and a post-ictal state. Focal motor seizures arise in the frontal lobe and are atonic, tonic, clonic, or myoclonic, or show a Jacksonian march. Focal sensory seizures arise in the parietal, occipital, and temporal lobes and cause pain or paresthesia, and visual, auditory, smell, or taste hallucinations. Focal autonomic and psychological seizures arise in the temporal lobe and insula and cause increased HR and BP, urinary incontinence, salivation, and sweating; fear, dread, anxiety, or pleasure; and déjà vu, depersonalization, and derealization. If the reticular formation or thalamus is involved, there is LOC, often with automatisms such as rapid blinking, chewing, and lip smacking, and a post-ictal state, and if not, there is neither. Auras include déjà vu and nausea.

Generalized seizures and the post-ictal state

A focal seizure that spreads to both hemispheres stays focal, as a focal to bilateral tonic-clonic seizure. Motor seizures are atonic, tonic, clonic, myoclonic, or tonic-clonic, with LOC and a post-ictal state, and may feature an involuntary cry or scream, upgaze, pooling of oral secretions, tongue biting, urinary or fecal incontinence, and increased HR and BP. Absence seizures, usually at ages 5–15, are non-motor, with staring, daydreaming, automatisms, and no loss of muscle tone. A post-ictal state follows focal seizures with impairment, generalized motor seizures, and focal to bilateral tonic-clonic seizures, and features confusion, lethargy, amnesia, and

Todd paralysis, which follows focal or generalized tonic-clonic seizures.

Diagnosis

Glucose and vital signs come first, so that hypoglycemia is treated. Bloods comprise FBC, electrolytes, urea, LFTs, and drug levels. Non-contrast CT head is the first imaging after a new seizure, EEG classifies the seizure, and MRI finds structural causes. A normal EEG between seizures does not exclude epilepsy. By cause, vascular disease is assessed with CT head, CT angiogram, and MRI for PRES; infection with lumbar puncture, CSF analysis, and blood cultures; trauma with non-contrast CT head; autoimmune disease with ANA and autoimmune encephalitis antibodies in blood and CSF; metabolic disease with glucose, sodium, calcium, magnesium, phosphate, urea, ammonia, and TFTs; drugs and epilepsy with a toxicology screen, antiseizure medication levels, and EEG; tumors with MRI with contrast; degenerative or demyelinating disease with MRI and cognitive assessment; eclampsia with a pregnancy test, BP, and urine protein; fever with temperature and a search for infection; and genetic causes with family history and genetic testing.

Prolactin rises 10–20 minutes after a generalized convulsive seizure but not after PNES; it also rises after convulsive syncope, and a normal level does not confirm PNES.

Seizure mimics

Syncope is favored by a trigger, a prodrome of lightheadedness, brief jerks, and rapid recovery, and is assessed with ECG and orthostatic BP. Psychogenic non-epileptic seizures show closed eyes, side-to-side head movement, asynchronous limb movement, and no post-ictal state, and are confirmed by video EEG. Cardiac arrhythmia is suggested by palpitations, exertional collapse, or a family history of sudden death. TIA causes negative symptoms such as weakness or numbness, without a march, whereas migraine with aura spreads over minutes and is followed by headache. Sleep disorders occur only during sleep, and panic attacks and movement disorders preserve awareness.

Management

During a seizure, the patient is placed in the recovery position with nothing in the mouth, the airway, oxygen, and IV access are secured, and glucose is given if low, with thiamine if alcohol use is likely. Status epilepticus is a seizure lasting 5 minutes or more. At 0–5 minutes, the patient is stabilized and glucose is checked; at 5–20 minutes, IV lorazepam or IM midazolam is given; at 20–40 minutes, IV levetiracetam, valproate, or fosphenytoin is given; and after 40 minutes, the patient needs intubation, ICU care, and an anesthetic infusion with EEG. The cause is treated: the stroke, with BP lowered in PRES; ABX or antivirals, draining any brain abscess; neurosurgical evacuation of hematomas if needed; corticosteroids and IVIG, treating any underlying tumor; correction of metabolic abnormalities, with thiamine for decreased vitamin B1; stopping a drug, benzodiazepines for withdrawal, and restarting missed medications; surgery, radiation, or chemotherapy; treatment of degenerative or demyelinating disease; magnesium sulfate and delivery in eclampsia; and treatment of fever and its cause. Long-term medication follows seizure type: levetiracetam, lamotrigine, or carbamazepine for focal seizures; valproate, levetiracetam, or lamotrigine for generalized motor seizures; ethosuximide for absence; valproate or levetiracetam for juvenile myoclonic epilepsy; ACTH or vigabatrin for West syndrome; and valproate, lamotrigine, or rufinamide for Lennox-Gastaut syndrome. Refractory epilepsy may need surgery, vagus nerve stimulation, or a ketogenic diet. By mechanism, carbamazepine, phenytoin, lamotrigine, and lacosamide block sodium channels; ethosuximide, and weakly valproate, block T-type calcium channels; benzodiazepines and phenobarbital act at the GABA-A receptor, and vigabatrin blocks GABA breakdown; levetiracetam binds SV2A; and valproate and topiramate have multiple mechanisms. All patients avoid triggers and take medications as prescribed, and valproate is avoided where possible in those who could become pregnant, because it is teratogenic.

Phenytoin causes gingival hyperplasia, nystagmus and ataxia at high levels, SJS, and fetal hydantoin syndrome, and follows zero-order kinetics at high doses. Carbamazepine causes agranulocytosis, aplastic anemia, hyponatremia, and SJS, so patients of Han Chinese ancestry are tested for HLA-B*1502. Valproate raises the risk of neural tube defects about 10-fold and causes hepatotoxicity, especially under age 2.

Relationships to other conditions

Stroke. AIS, ICH, and SAH are vascular causes of seizures, and seizures are listed among stroke complications, so both belong in the first assessment. Non-contrast CT head helps with both, and Todd paralysis after a seizure is the feature most likely to be confused with a stroke deficit.

Focal seizures with impairment vs absence seizures. Both can cause staring and automatisms. An aura and a post-ictal state favor a focal seizure, whereas no post-ictal state and a 3 Hz spike-and-wave pattern on EEG favor absence.

Focal motor vs generalized motor seizures. Both can be atonic, tonic, clonic, or myoclonic. A Jacksonian march

favors a focal seizure, whereas LOC with an involuntary cry, upgaze, or tongue biting favors a generalized one, and a focal seizure can spread as a focal to bilateral tonic-clonic seizure.

Epilepsy vs provoked seizures. Metabolic, toxic, and withdrawal causes are corrected directly, whereas epilepsy needs long-term antiseizure medication, and non-adherence is itself a cause.

Multiple sclerosis *and* ***Alzheimer disease***. Both are degenerative or demyelinating causes of seizures, so treatment targets the underlying disease as well.

Eclampsia. A seizure in pregnancy with increased BP and urine protein favors eclampsia, which is treated with magnesium sulfate and delivery rather than standard antiseizure medication.

DEMYELINATING

Multiple sclerosis: immune demyelination and lesions scattered in space and time

Autoreactive T cells release cytokines that bring macrophages across the BBB, where they destroy myelin and oligodendrocytes are injured, forming myelin plaques that slow nerve conduction. Environmental and genetic factors contribute, and the typical patient is a woman aged 20–40 years. Symptoms depend on the site of each plaque, and diagnosis rests on showing dissemination in space, usually with dissemination in time. Management treats acute relapses, modifies the disease, and controls symptoms. Here, we review these mechanisms, the symptoms they produce, diagnosis, and treatment, and consider how multiple sclerosis relates to other neurologic conditions.

Key points

- IL-1, IL-6, and TNF-α increase adhesion molecules, vasodilation, vascular permeability, and chemotaxis.
- Macrophages destroy myelin and oligodendrocytes are injured, forming myelin plaques.
- MRI comes first, and the 2024 McDonald criteria need dissemination in space, plus dissemination in time or a substitute.
- CSF oligoclonal bands support the diagnosis if MRI is not diagnostic.
- Acute relapses are treated with IV methylprednisolone, and disease-modifying therapy is long term.

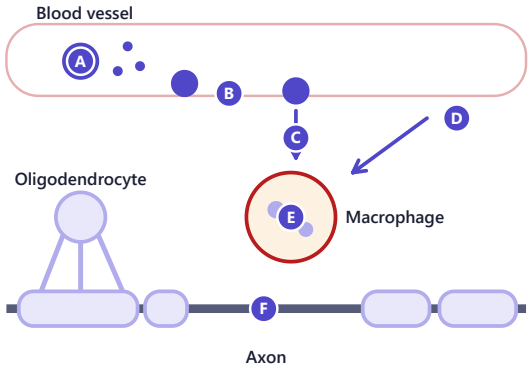


Fig. 1 | A blood vessel above an axon. T cells release cytokines in the vessel, immune cells stick to its wall and cross it, and a macrophage below destroys myelin beside an injured oligodendrocyte, leaving a bare stretch of axon. A, Cytokine release; B, Adhesion molecules; C, Vasodilation and permeability; D, Chemotaxis; E, Phagocytosis; F, Slowed conduction.

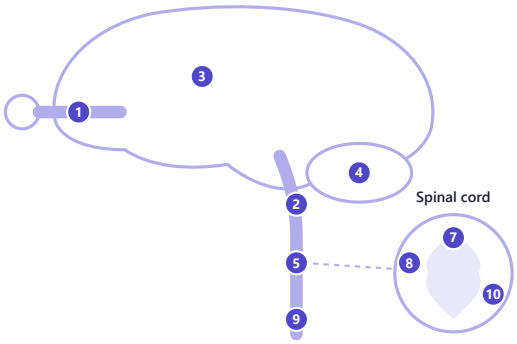


Fig. 2 | A side view of the central nervous system with numbered lesion sites: optic nerve, brainstem, corticobulbar tracts, cerebellum, and cervical cord, beside a cross-section of the spinal cord showing the dorsal columns, corticospinal tract, and spinothalamic tract. 1, Optic neuritis; 2, MLF damage; 3, Pseudobulbar palsy; 4, Cerebellar signs; 5, Lhermitte sign; 7, Dorsal column loss; 8, Upper motor neuron signs; 9, Bladder dysfunction; 10, Spinothalamic loss.

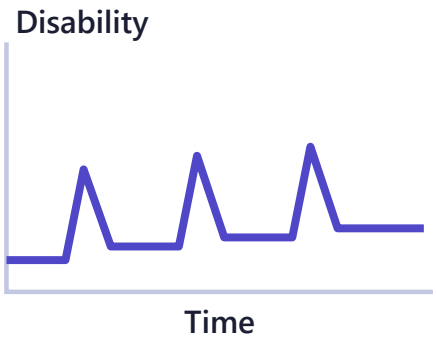


Fig. 3 | Disability over time in relapsing-remitting MS: relapses rise and fall back, leaving small steps. Relapsing-remitting: most people; relapses over days to weeks, with stability between them

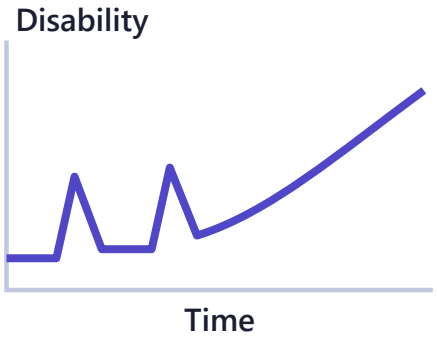


Fig. 4 | Disability over time in secondary progressive MS: early relapses, then a steady climb. Secondary progressive: gradual worsening after 10–15 years, with or without relapses

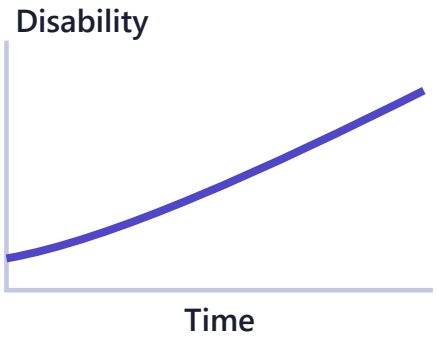


Fig. 5 | Disability over time in primary progressive MS: a steady climb from onset. Primary progressive: a small subset; worsening from onset

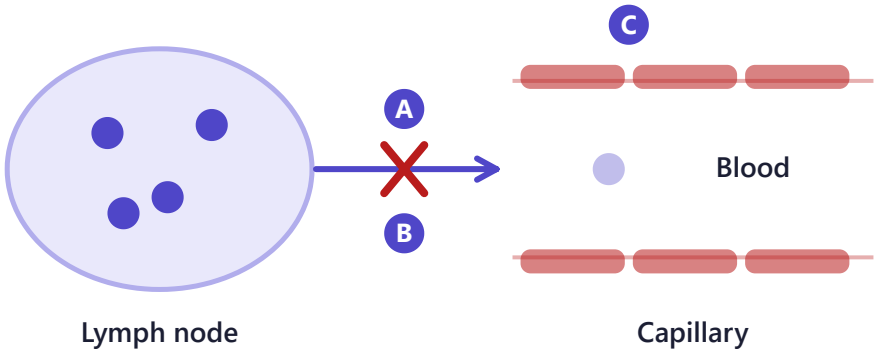


Fig. 6 | A lymph node on the left releases lymphocytes along an arrow toward a capillary on the right. Fingolimod blocks the exit, and the capillary wall is lined with endothelial cells joined tightly together. A, S1P and lymph nodes; B, Fingolimod; C, S1P and capillaries.

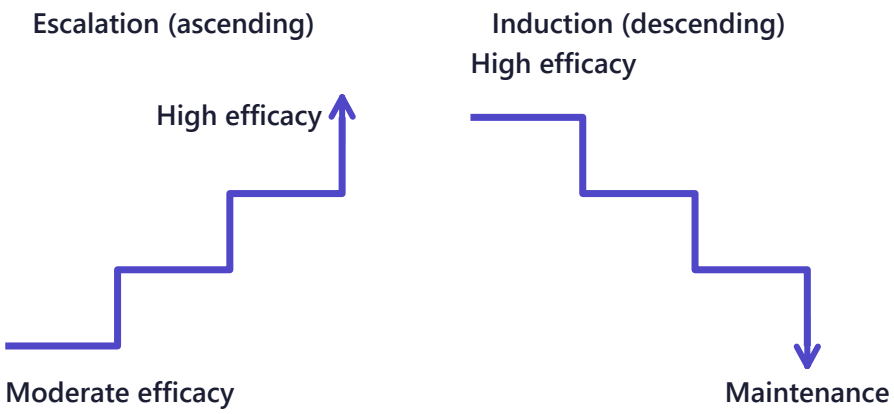


Fig. 7 | Two mirrored staircases. Escalation climbs from moderate-efficacy to high-efficacy drugs. Induction starts with high-efficacy or induction drugs and steps down to maintenance.

Pathophysiology

In a genetically susceptible person, autoreactive T cells release IL-1, IL-6, and TNF-α. These cytokines increase adhesion molecules, so immune cells stick to the vessel wall, and cause vasodilation and increased vascular permeability, so immune cells cross the BBB. Chemotaxis draws macrophages to the site, where they destroy myelin and oligodendrocytes are injured, causing myelin plaque formation. Demyelinated axons conduct poorly, so symptoms depend on the site of the plaque.

Etiology

Environmental factors are EBV, HHV-6, decreased vitamin D, smoking, obesity, and higher latitude. Genetic and demographic factors are HLA-DR2 and being female and aged 20–40 years.

Clinical presentation

Optic neuritis causes decreased visual acuity, reduced color vision, and RAPD, and MLF damage causes bilateral INO. Pseudobulbar palsy causes decreased chewing, increased jaw jerk, absent facial expressions, dysphagia, dysphonia, increased gag reflex, dysarthria, and spastic tongue. Cerebellar involvement causes ataxia and intention tremor. Lhermitte sign is an electric shock sensation down the spine on neck flexion, and in Uhthoff phenomenon, symptoms worsen with heat. Dorsal column damage causes loss of fine touch, vibration, and conscious proprioception, and spinothalamic damage causes loss of pain and temperature. Upper motor neuron signs are muscle weakness, increased tone, hyperreflexia, and positive Babinski sign. Bladder dysfunction causes incontinence with SNS damage and retention with PNS damage. Most people have a relapsing-remitting course, with relapses over days to weeks and stability between them. Gradual worsening after 10–15 years, with or without relapses, marks the secondary progressive course, and a small subset has a primary progressive course, with worsening from onset. Each course is also described as active or not active, by relapses or new MRI lesions, and progressive courses as progressing or not, and CIS is a first episode that does not yet meet dissemination in time.

The Charcot triad of intention tremor, dysarthria with scanning speech, and nystagmus reflects cerebellar and brainstem involvement, and tremor affects 58–77% of patients.

Diagnosis

First-line testing is MRI brain and spine with gadolinium, which shows T2 plaques in periventricular, cortical or juxtacortical, infratentorial, spinal cord, and optic nerve sites, with active plaques enhancing. The 2024 McDonald criteria need dissemination in space, with plaques in at least 2 of the 5 sites, and dissemination in time, with new plaques over time or enhancing and non-enhancing plaques together. CSF oligoclonal bands, kappa free light chains, 4 or more sites, or 6 plaques with a central vein sign can substitute for time. With only 1 site, the diagnosis needs dissemination in time or positive CSF, plus a central vein sign or a paramagnetic rim plaque. Incidental MRI findings qualify with 2 or more sites, plus dissemination in time, positive CSF, or a central vein sign. The 2017 criteria used 4 sites, without the optic nerve, and always needed dissemination in time or oligoclonal bands. Other causes must be excluded first. Second-line testing is lumbar puncture for CSF oligoclonal bands or kappa free light chains, if MRI is not diagnostic. Extra tests are visual evoked potentials, which show slowed conduction in the optic nerve, OCT, which shows an inter-eye difference in retinal thickness, and bloods to exclude mimics: FBC, B12, TFTs, ESR, ANA, syphilis serology, HIV, and AQP4 and MOG antibodies.

MS mimics

NMOSD causes severe optic neuritis and long spinal cord lesions, with AQP4 antibodies, and MOGAD causes

bilateral optic neuritis, often with disc swelling, with MOG antibodies. ADEM is one episode after an infection, with encephalopathy, usually in children. B12 deficiency causes dorsal column and corticospinal signs with anemia. SLE is suggested by rash, joint pain, and kidney disease, with ANA, and neurosarcoidosis by lung disease and CN palsies, with CXR and serum ACE. Neurosyphilis, HIV, and Lyme disease are found by serology. Small vessel disease occurs with older age and vascular risk factors, with no oligoclonal bands.

Management

An acute relapse is treated with IV methylprednisolone for 3–7 days, with plasma exchange if refractory. Chronic disease-modifying therapy includes the injectable drugs interferon beta and glatiramer acetate, the oral drugs dimethyl fumarate, teriflunomide, and fingolimod, and the high-efficacy drugs natalizumab, ocrelizumab, and alemtuzumab. Natalizumab can cause PML, so JC virus status is checked first. S1PR1 signaling lets lymphocytes leave lymph nodes for the blood. Fingolimod binds S1PR1, which is then degraded, so lymphocytes stay in lymph nodes and fewer reach the CNS. On endothelial cells, S1PR1 tightens junctions and lowers capillary permeability, although prolonged fingolimod can increase permeability, and S1P receptors on the SA and AV nodes explain fingolimod-induced bradycardia. Siponimod, ozanimod, and ponesimod are more selective. Two strategies guide treatment: escalation, or ascending therapy, starts with moderate-efficacy drugs and moves up if the disease stays active, whereas early high-efficacy treatment uses natalizumab, S1P receptor modulators, or anti-CD20 drugs from the start, and induction, or descending therapy, uses short-term mitoxantrone, alemtuzumab, cladribine, or stem cell transplant before maintenance. Early high-efficacy treatment suits young people with active disease, many T2 lesions, and no irreversible disability. Symptoms are controlled with baclofen for spasticity, oxybutynin for incontinence, catheterization for retention, gabapentin for neuropathic pain, and avoiding heat in Uhthoff phenomenon.

Relationships to other conditions

Stroke syndromes. Both cause focal deficits, including upper motor neuron signs and brainstem signs. Deficits that follow one arterial territory and appear suddenly favor stroke, whereas plaques separated in space and time favor multiple sclerosis.

Posterior cord syndrome. Both cause loss of vibration and proprioception, and B12 deficiency is a cause of posterior cord syndrome, which is why B12 is checked. Low B12 favors deficiency, whereas MRI plaques and CSF oligoclonal bands favor multiple sclerosis.

AQP4 antibody disease and ***MOGAD***. These are mimics that also cause demyelination, so positive antibodies point away from multiple sclerosis.

Seizures. Multiple sclerosis is a demyelinating cause of seizures, so treatment targets the underlying disease as well.

DEMYELINATING

Neuromyelitis optica spectrum disorder: an antibody-mediated astrocytopathy of the optic nerves, spinal cord, and area postrema

Neuromyelitis optica spectrum disorder is an autoimmune disease of the CNS in which AQP4-IgG binds AQP4, the water channel of astrocyte end-feet, activates complement, and destroys astrocytes, with secondary damage to oligodendrocytes and axons. It causes severe optic neuritis, LETM, area postrema syndrome, and brainstem, diencephalic, and cerebral syndromes, and follows a relapsing course in which disability builds with each attack. Diagnosis rests on the 2015 international consensus criteria, which require one core clinical characteristic with AQP4-IgG, or two with supporting MRI findings without it. Attacks are treated with IV methylprednisolone and PLEX, and relapses are prevented with long-term immunosuppression, including B-cell depletion, IL-6 receptor blockade, and complement inhibition. Here, we review its pathophysiology, risk factors, clinical features, diagnosis, and treatment.

Key points

- AQP4-IgG binds AQP4 on astrocyte end-feet and kills astrocytes through complement, so demyelination is secondary.
- Lesions favor regions dense in AQP4: the optic nerves, spinal cord, area postrema, and tissue around the third and fourth ventricles.
- The 2015 criteria require one core clinical characteristic with AQP4-IgG, or two without it, including optic neuritis, myelitis with LETM, or area postrema syndrome.
- Disability builds with each relapse, and a progressive course is very rare.
- Attacks are treated with IV methylprednisolone and often PLEX, long-term immunosuppression follows the first attack, and interferon beta and natalizumab can worsen the disease.

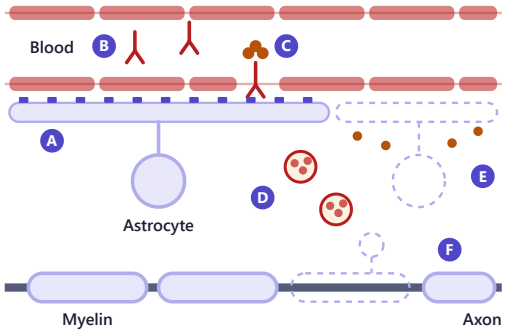


Fig. 1 | A capillary runs across the top, lined by endothelial cells, with a gap in its lower wall. Below it, an astrocyte end-foot carries AQP4 water channels, A. AQP4 antibodies, B, in the blood cross the gap and bind AQP4, where complement, C, gathers. Neutrophils and eosinophils, D, enter the tissue. On the right, the end-foot and its astrocyte are lost and glutamate, E, builds up. Along the bottom, an axon loses a stretch of myelin and its oligodendrocyte, F. A, AQP4; B, AQP4-IgG; C, Complement; D, Neutrophils and eosinophils; E, Glutamate; F, Oligodendrocytes and axons.

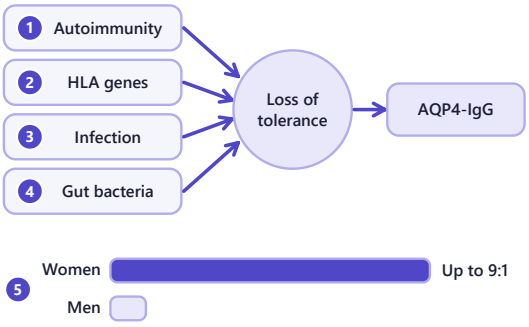


Fig. 2 | Other autoimmune disease, 1, HLA genes, 2, infection or vaccination, 3, and gut bacteria, 4, feed a loss of tolerance to AQP4 that produces AQP4 antibodies. Below, women, 5, outnumber men by up to 9 to 1. 1, Other autoimmune disease; 2, HLA genes; 3, Infection or vaccination; 4, Gut bacteria; 5, Women.

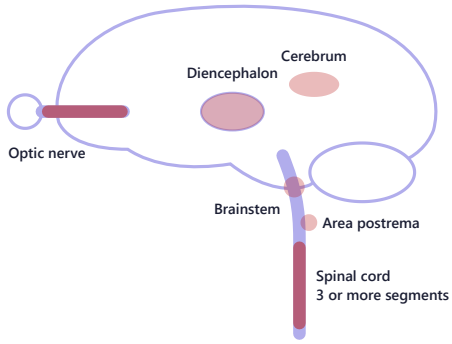


Fig. 3 | A side view of the central nervous system with the six core regions of NMOSD highlighted: the optic nerve, the diencephalon, the cerebrum, the brainstem around the fourth ventricle, the area postrema of the dorsal medulla, and a long spinal cord lesion spanning 3 or more segments.

Pathophysiology

Antibodies against AQP4, the water channel on astrocyte end-feet, activate complement and destroy astrocytes, and the inflammation that follows damages oligodendrocytes and axons in the optic nerves, spinal cord, and area postrema.

AQP4 is a water channel whose tetramers cluster into orthogonal arrays of particles on the astrocyte end-feet that surround vessels, and it is dense in the optic nerve, spinal cord, and area postrema and other regions around the ventricles. AQP4-IgG is made by plasmablasts, which IL-6 keeps alive, and reaches AQP4 where inflammation opens the BBB; in animal models, the area postrema is one route of entry. The antibody binds the extracellular loops of AQP4, favoring clustered AQP4, and activates the classical complement pathway, whose membrane attack complex kills astrocytes, so lesions lose AQP4 and GFAP. Neutrophils, eosinophils, and macrophages then infiltrate and can cause necrosis.

Binding also pulls AQP4 into the cell together with EAAT2, the glutamate transporter, so glutamate builds up. Oligodendrocytes and axons are damaged as bystanders, so demyelination is secondary and follows astrocyte loss rather than starting it, and axon damage often comes first. AQP4 is also found in the kidney, stomach, and skeletal muscle, but these are mostly spared, perhaps because CD59 limits complement outside the CNS.

Etiology

NMOSD is associated with other autoimmune diseases, including SLE, Sjögren syndrome, myasthenia gravis, autoimmune thyroid disease, and antiphospholipid syndrome, as well as celiac disease and sarcoidosis. HLA-DRB1*03:01 and complement C4 copy number confer risk, in a pattern closer to SLE than to MS. Infection or vaccination precedes 20–30% of attacks, and part of AQP4 resembles a protein of *Clostridium perfringens*, which is overabundant in the gut of some patients, suggesting molecular mimicry.

Women are affected up to 9 times as often as men in adults, and about 3 times as often in children. Onset is usually at 30–40 years, and fewer than 5% of patients are children. NMOSD is more common with Asian or African ancestry, accounting for 1–2% of demyelinating disease in the United States and Italy but 13.7% in India and over 30% in Thailand, and its prevalence is about 0.7–10 per 100000. Smoking, low vitamin D, low physical activity, and a diet low in fruit and vegetables may add risk.

Clinical presentation

Six core clinical characteristics each implicate one region. Optic neuritis is severe, often bilateral or reaching the chiasm, can cause altitudinal field defects, and can leave acuity at 20/200 or worse. Acute myelitis causes a complete cord syndrome, with paraplegia and paroxysmal tonic spasms. Area postrema syndrome, in 16–43% of patients, causes intractable hiccups, nausea, or vomiting and is often seen first by gastroenterologists and misdiagnosed as gastroparesis or reflux. Acute brainstem syndrome arises from lesions around the fourth ventricle, diencephalic syndrome causes symptomatic narcolepsy with thalamic or hypothalamic lesions, and cerebral syndrome occurs with NMOSD-typical brain lesions. Optic neuritis and myelitis can occur together or weeks apart.

Most patients, 80–90%, relapse, about 60% within a year of a first LETM, although relapses can be over a decade apart, and the 5–10% with a single attack are more often without AQP4-IgG. Disability builds with each attack, and a progressive course is very rare. Unlike in MS, relapses increase during pregnancy and after delivery, and recovery is worse with older onset, about 85% at age 20 and 60% at age 60. In relapsing disease, about 60% become blind in one or both eyes and about half develop paraplegia or monoplegia, and respiratory failure in severe attacks raises mortality. Chronic pain, including neuropathic pain, painful attacks, and headache, affects 80–100% of patients, and dysautonomia, often worse than in MS, bladder and bowel dysfunction, depression, and anxiety are common.

Diagnosis

Evaluation starts with a core clinical characteristic, then serum AQP4-IgG on a cell-based assay, MRI of the brain, orbits, and spinal cord, and exclusion of alternatives, with red flags prompting a wider search. Under the 2015 international consensus criteria, one core clinical characteristic suffices with AQP4-IgG once alternatives are excluded. Without AQP4-IgG, or when it is untested, at least 2 core characteristics in different regions are needed, from one or more attacks; one must be optic neuritis, myelitis with LETM, or area postrema syndrome with a dorsal medulla lesion, and supporting MRI findings are required. AQP4-IgG without an attack, or typical lesions without symptoms, is not NMOSD, and seronegative patients are retested after a relapse and before B-cell or antibody-targeted treatment. Area postrema syndrome requires AQP4-IgG and acute or subacute intractable nausea, vomiting, or hiccups, alone or together, for at least 48 hours, or sooner with a new area postrema lesion on MRI, with other causes excluded, and little response to antiemetics is a clue.

Cell-based assays are preferred, with a sensitivity of about 77% and 0.1% false positives, whereas ELISA and immunofluorescence have a sensitivity of 63–64% and more false positives, often at low titer, and titer does not predict severity. MRI typically shows LETM over 3 or more vertebral segments, centered on the gray matter, with swelling, T1 hypointensity, and enhancement, sometimes extending into the brainstem, and bright spotty lesions, although 7–14% of first myelitis attacks are shorter, especially early or in remission. Optic nerve lesions are bilateral, posterior, reach the chiasm, or span over half the nerve's length, and brain lesions sit where AQP4 is dense, around the third and fourth ventricles, in the dorsal medulla, thalamus, hypothalamus, periaqueductal gray, and corpus callosum. Nonspecific white matter lesions occur in up to half, and over 30% meet McDonald criteria, so they do not exclude NMOSD.

CSF OCB are usually absent, and their presence is a red flag, whereas over 50 white cells/μL, in about 35%, or neutrophils or eosinophils favor NMOSD over MS. CSF GFAP is raised for days to weeks after an attack. OCT shows more RNFL thinning than in MS, and visual fields show a central or centrocecal scotoma, with reduced color vision. Short, peripheral cord lesions, Dawson fingers, OCB, and a progressive course favor MS, confirmed with MRI and CSF; disc edema, anterior optic nerve lesions, conus lesions, ADEM, and men or children favor MOGAD, confirmed with MOG-IgG; and response to antiemetics favors gastroparesis, reflux, or food poisoning over area postrema syndrome, which brain MRI can show. LETM also follows infection, granulomatous disease, tumors, paraneoplastic syndromes, ADEM, spinal cord infarction, and dural arteriovenous fistulas.

Management

Attacks are treated with IV methylprednisolone 1 g daily for 2–7 days. Steroids alone rarely restore baseline, so PLEX is often needed, from the start or if steroids fail, and treatment within 14 days of onset more often returns patients to baseline. Antiemetics rarely help area postrema syndrome, but IV methylprednisolone or PLEX stops most attacks within 2 days.

Because each relapse adds disability, long-term immunosuppression starts after the first attack. B cells are depleted with RTX, an anti-CD20 antibody given every 6 months and a first-line drug, or inebilizumab, an anti-CD19 antibody. Satralizumab and tocilizumab block the IL-6 receptor, and eculizumab blocks complement C5 and is given after meningococcal vaccination. AZA with prednisone is also first line, and MMF, methotrexate, and tacrolimus are alternatives. Satralizumab, tocilizumab, eculizumab, and inebilizumab have proven benefit mainly in AQP4-IgG-positive disease. Interferon beta, natalizumab, and other MS disease-modifying drugs are avoided. With RTX, immunoglobulin levels are checked before treatment and yearly, because low levels raise infection risk. Relapse risk rises in pregnancy, and RTX has been continued safely in case reports.

Relationships to other conditions

Multiple sclerosis. Both are relapsing CNS diseases that affect more women and cause optic neuritis and myelitis, and NMOSD was long considered a variant of multiple sclerosis. AQP4-IgG, LETM over 3 or more segments, bilateral or chiasmal optic neuritis, area postrema syndrome, CSF pleocytosis with neutrophils or eosinophils, and severe attacks with poor recovery favor NMOSD, whereas short, peripheral cord lesions, Dawson fingers, OCB, and a progressive course favor multiple sclerosis, whose interferon beta and natalizumab can worsen NMOSD.

MOGAD. Both are antibody-mediated diseases that cause optic neuritis and LETM, and MOG-IgG was first thought to mark AQP4-negative NMO. AQP4-IgG, area postrema syndrome, posterior and chiasmal optic nerve lesions, other autoimmune disease, and more women favor NMOSD, whereas disc edema, anterior optic nerve lesions, conus lesions, ADEM, children and men, and a monophasic course favor MOGAD, which is a demyelinating disease rather than an astrocytopathy.

Vision loss and red eye. Optic neuritis causes painful vision loss in multiple sclerosis, NMOSD, and MOGAD. Severe, bilateral, or chiasmal optic neuritis with poor recovery favors NMOSD, whereas unilateral optic neuritis with good recovery favors multiple sclerosis.

DEMYELINATING

MOG antibody-associated disease: an antibody-mediated demyelinating disease distinct from MS and NMOSD

MOG antibody-associated disease is an inflammatory demyelinating disease of the CNS in which antibodies target MOG, a glycoprotein on the outermost layer of myelin and on oligodendrocytes. It most often causes optic neuritis, myelitis, or ADEM, alone or in combination, and follows a monophasic or relapsing course without a progressive phase. Diagnosis requires a demyelinating attack, serum MOG-IgG on a live cell-based assay, and exclusion of better diagnoses, including MS. Attacks are treated with corticosteroids, and relapsing disease with immunosuppression. Here, we review its pathophysiology, triggers, clinical syndromes, diagnosis, and treatment.

Key points

- MOG is a transmembrane glycoprotein on the outermost myelin lamellae, with an extracellular, N-terminal, immunoglobulin-like domain.
- MOGAD is distinct from MS and from AQP4-positive NMOSD, an astrocytopathy.
- Optic neuritis, myelitis, and ADEM are the most common syndromes.
- A clear positive titer of 1:100 or more needs no supporting features.
- MS drugs appear ineffective; relapses are prevented with AZA, MMF, IVIG, or corticosteroids.

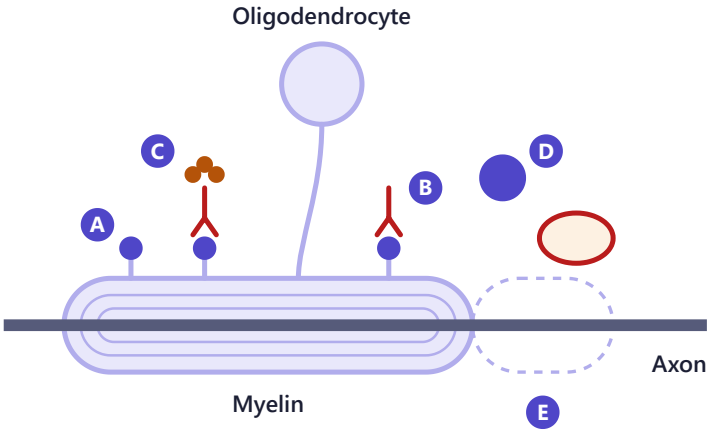


Fig. 1 | An oligodendrocyte wraps an axon in myelin. MOG, A, sits on the outermost layer of myelin with its immunoglobulin-like end facing out. MOG antibodies, B, bind it and recruit complement, C. T cells and macrophages, D, join in, and a stretch of myelin is lost, E. A, MOG; B, MOG-IgG; C, Complement; D, T cells and macrophages; E, Demyelination.

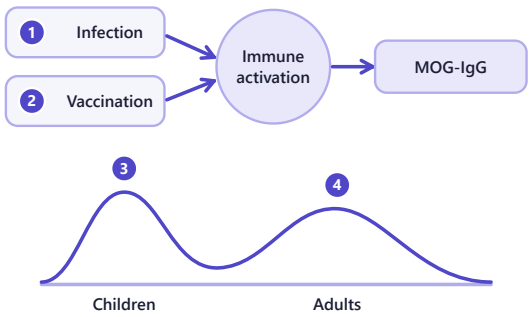


Fig. 2 | Infection, 1, or rarely vaccination, 2, drives immune activation and blood-brain barrier disruption, which lead to MOG antibodies. Below, onset age has two peaks: children, 3, and young to middle-aged adults, 4. 1, Infection; 2, Vaccination; 3, Children; 4, Young to middle-aged adults.

Pathophysiology

Antibodies against MOG on the outer surface of myelin recruit complement and immune cells, causing perivenous and confluent demyelination of the optic nerves, spinal cord, and brain.

MOG is a transmembrane glycoprotein of the outermost myelin lamellae and oligodendrocyte surface, with an extracellular, N-terminal, immunoglobulin-like domain. It is a small part of myelin, but because it faces outward, antibodies reach it easily. IgG1 MOG antibodies activate complement and cell-mediated cytotoxicity, and active lesions contain perivascular lymphocytes and myelin-laden macrophages, with perivenous and confluent white matter and cortical demyelination.

MOG-IgG was first thought to mark AQP4-negative NMO, but MOGAD is now a separate disease. AQP4-positive NMOSD targets AQP4 on astrocytes, an astrocytopathy in which axon damage precedes demyelination, whereas MOGAD targets myelin and oligodendrocytes. MOGAD is monophasic or relapsing, without a progressive phase, and affects both sexes equally, whereas NMOSD and MS affect more women, and MS can be progressive.

Etiology

MOGAD can follow infection, mainly viral, including influenza, Epstein-Barr virus, herpes simplex virus, and SARS-CoV-2, and also Mycoplasma, and, extremely rarely, vaccination. Onset has two peaks: children, who make up about a third of cases, and young to middle-aged adults. Men and women are affected equally, MOGAD is not linked to other autoimmune diseases, unlike AQP4 disease, and pregnancy may protect against relapse.

Clinical presentation

Optic neuritis is the most common syndrome and is often bilateral, with disc edema and a long optic nerve lesion. Myelitis causes weakness, a sensory level, and bladder and bowel dysfunction, with LETM or a conus lesion. ADEM, mainly in children, causes encephalopathy with multifocal deficits. Cortical encephalitis often causes seizures, and brainstem or cerebellar deficits can involve the pons and MCP. These syndromes can occur alone, in sequence, or together.

About half of patients have a single attack. Relapse is more likely in adults and with persistent MOG-IgG, and disability comes from attacks, without a progressive phase.

Diagnosis

Diagnosis requires a demyelinating attack, serum MOG-IgG on a live cell-based assay, and exclusion of better diagnoses, including MS. By the 2023 criteria, a clear positive titer of 1:100 or more needs no supporting features. A low positive titer of 1:10 to below 1:100, a positive result without a titer, or antibodies in CSF only requires negative AQP4-IgG and at least one supporting feature.

Supporting features are bilateral optic neuritis, an optic nerve lesion over half its length, disc edema, perineural optic sheath enhancement, LETM, a conus lesion, an H-sign or central cord lesion, ill-defined, fluffy brain lesions, deep gray matter lesions, and a cortical lesion with meningeal enhancement. Well-circumscribed lesions and CSF OCB favor MS, and area postrema syndrome with AQP4-IgG favors NMOSD. Because MOG-IgG is found in only 0.4% of MS, testing everyone with possible MS yields false positives.

Management

Acute attacks are treated with IV methylprednisolone 1 g daily for 5 days, or oral prednisone 1250 mg, and PLEX or immunoadsorption follows if steroids fail. Relapse prevention usually starts after a second attack, with AZA or MMF bridged by prednisolone for the first months, IVIG at 1–2 g/kg every 4 weeks, or oral corticosteroids, whose long-term risks must be weighed. RTX is less effective than in NMOSD, and MS drugs such as interferons, glatiramer acetate, and fingolimod appear ineffective.

Relationships to other conditions

Multiple sclerosis. Both are demyelinating diseases that cause optic neuritis and myelitis. Bilateral optic neuritis with disc edema, LETM, a conus lesion, fluffy lesions, and MOG-IgG favor MOGAD, whereas well-circumscribed lesions, CSF OCB, and a progressive course favor multiple sclerosis, whose drugs appear ineffective in MOGAD.

NMOSD. Both are antibody-mediated diseases that cause optic neuritis and LETM, and MOG-IgG was first thought to mark AQP4-negative NMO. Disc edema, a conus lesion, ADEM, equal numbers of men and women, and a monophasic course favor MOGAD, whereas AQP4-IgG and area postrema syndrome favor NMOSD, an astrocytopathy in which axon damage precedes demyelination.

EYE

Glaucoma: aqueous outflow, intraocular pressure, and optic nerve damage

Glaucoma is an optic neuropathy in which slow drainage of aqueous humor raises IOP and damages the optic nerve. In open-angle glaucoma, aqueous humor exits slowly through a resistant trabecular meshwork, causing painless, gradual peripheral vision loss. In angle closure, the iris blocks the angle, through pupillary block or a thickened iris base after the pupil dilates, and acute angle closure is a medical emergency. Gonioscopy distinguishes the two. Treatment lowers IOP by increasing outflow or decreasing production of aqueous humor. Here, we review the flow of aqueous humor, the mechanisms and features of each type, diagnosis, and treatment.

Key points

- Most aqueous humor drains through the trabecular meshwork, and a minority through the uveoscleral route.
- Open-angle glaucoma is slow exit of aqueous humor through the trabecular meshwork.
- Angle closure follows pupillary block or a thickened iris base after the pupil dilates, and mydriatics can trigger it.
- Gonioscopy shows whether the angle is open or closed.
- Prostaglandin analogs and cholinergic agents increase outflow, and β-blockers, α2-agonists, and carbonic anhydrase inhibitors decrease production.
- Acute angle closure is lowered medically, then treated with laser peripheral iridotomy.

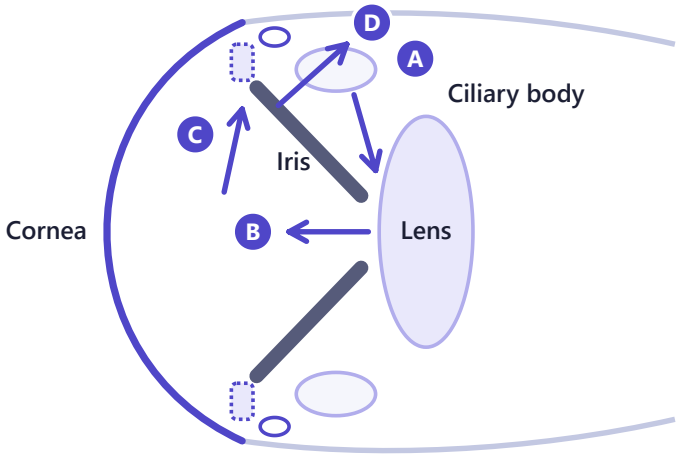


Fig. 1 | A cross-section of the front of the eye. A, the ciliary body secretes aqueous humor behind the iris. B, it flows through the pupil into the anterior chamber. C, most drains through the trabecular meshwork into Schlemm canal at the angle between iris and cornea. D, some leaves through the ciliary muscle, the uveoscleral route. A, Production; B, Through the pupil; C, Trabecular outflow; D, Uveoscleral outflow.

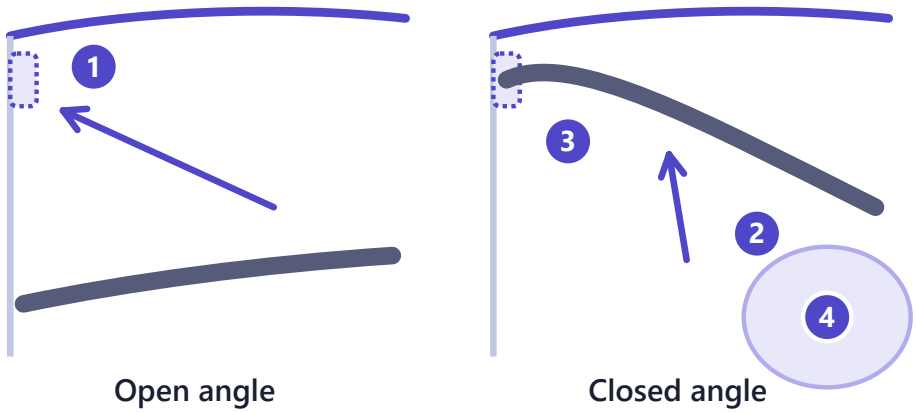


Fig. 2 | Two close-ups of the drainage angle. In open-angle glaucoma, 1, aqueous humor reaches a resistant trabecular meshwork. In angle closure, the iris covers the meshwork: by pupillary block, 2, which bows the iris forward; by a thickened iris base, 3, after the pupil dilates; or by a thick lens pushing forward, 4. 1, Trabecular resistance; 2, Pupillary block; 3, Thickened iris base; 4, Thick lens.

Pathophysiology

Aqueous humor drains too slowly, through a resistant trabecular meshwork in open-angle glaucoma or past an iris that blocks the angle in angle closure, so IOP rises above 21 mmHg and damages the optic nerve. Ganglion cell axons are lost, the cup enlarges, and vision is lost irreversibly, peripheral first. Glaucoma can occur at normal pressure, and raised pressure does not always cause it.

The ciliary body secretes aqueous humor behind the iris, with β2 receptors increasing and α2 receptors decreasing secretion. It flows through the pupil into the anterior chamber. Most drains through the trabecular meshwork into Schlemm canal, the site of resistance in open-angle glaucoma, and a minority leaves through the ciliary muscle by the uveoscleral route, which prostaglandin analogs increase.

Open-angle glaucoma is chronic and painless and makes up most glaucoma. Angle closure, about a third of cases but more often blinding, presses the iris against the trabecular meshwork and can be an acute emergency or chronic.

Etiology

In open-angle glaucoma, aqueous humor exits slowly although the angle is open. Angle closure has three main mechanisms. In pupillary block, the most common, the iris presses on the lens, aqueous humor builds behind it, and the iris bows forward onto the cornea. When the pupil dilates, mydriatics contract the radial dilator muscle, and the bunched iris base narrows the angle. A lens that thickens with age, or swells with cataract, pushes the iris forward.

Risks for open-angle glaucoma are older age, African ancestry, which raises risk 4-fold, family history, myopia, a thin cornea, and type 2 diabetes. Risks for angle closure are age 55–65, female sex, which raises risk 2–4 times, East Asian and Inuit ancestry, hyperopia, and a shallow anterior chamber. Dilating drops, such as atropine and cyclopentolate, topiramate, sulfonamides, and darkness can trigger closure. Secondary glaucoma follows steroids, pseudoexfoliation, neovascularization, trauma, or cataract.

Clinical presentation

Open-angle glaucoma has no early symptoms, and over half of cases are undetected. Vision is lost gradually in the periphery, with arcuate defects, while central vision is kept until late, and the disc shows a cup-to-disc ratio above 0.5 and a notched rim, bilaterally but asymmetrically.

Acute angle closure causes sudden severe eye pain, a red eye, blurred vision, halos, headache, nausea, and vomiting. The pupil is fixed and mid-dilated, with corneal edema, ciliary flush, a shallow anterior chamber, and a firm eye, and IOP is 30–80 mmHg. Glaucoma is the second leading cause of blindness, and acute closure can cause retinal vascular occlusion, ischemic optic neuropathy, and peripheral anterior synechiae.

Diagnosis

Tonometry measures IOP, gonioscopy shows whether the angle is open or closed, fundoscopy and OCT assess the cup and nerve fiber layer, and visual fields show peripheral and arcuate loss. In gonioscopy, a mirrored lens on the cornea shows the angle; in acute angle closure, it is closed for 360°, and the other eye is often narrow. Gonioscopy is repeated after an attack, because IOP can fall from ischemia of the ciliary body even if the angle stays closed. Disc damage at normal pressure prompts assessment of vascular risks for normal-tension glaucoma.

Management

Open-angle glaucoma is treated with drops, a prostaglandin analog first, then laser trabeculoplasty, then trabeculectomy, a tube shunt, or MIGS. Acute angle closure is treated with acetazolamide 500 mg, timolol, and apraclonidine, then IV mannitol 1–2 g/kg if needed, and pilocarpine once IOP is below 40 mmHg. Laser peripheral iridotomy, which lets aqueous humor bypass the pupil, is the treatment of choice, with lens extraction as an alternative.

Treatment lowers IOP, by increasing the outflow of aqueous humor or decreasing its production. Prostaglandin analogs, such as latanoprost, increase uveoscleral outflow and are first line; they lengthen eyelashes and darken the iris. Cholinergic agents, such as pilocarpine, contract the ciliary muscle and constrict the pupil, opening the trabecular meshwork, but are poorly tolerated. α2-agonists, brimonidine and apraclonidine, both decrease production and increase uveoscleral outflow, and brimonidine can cause apnea in children. β-blockers, such as timolol, block ciliary β receptors and are avoided in asthma, COPD, heart block, and bradycardia. Carbonic anhydrase inhibitors, dorzolamide drops or acetazolamide by mouth or IV, reduce bicarbonate and so aqueous humor. Mydriatics and cycloplegics, such as atropine and cyclopentolate, are avoided because they dilate the pupil and can trigger angle closure. Sympathomimetics, such as phenylephrine and pseudoephedrine in decongestants, also dilate the pupil and can trigger angle closure in predisposed eyes.

Relationships to other conditions

Cataract. A swollen cataract can push the iris forward and close the angle, and a hypermature cataract can leak proteins that clog the trabecular meshwork. A cataract alone causes painless blurring without raised IOP.

Vision loss and red eye. Acute angle closure is a cause of a painful red eye with reduced vision, like bacterial keratitis. A fixed, mid-dilated pupil, halos, and high IOP favor angle closure, whereas a corneal infiltrate after contact lens wear favors keratitis.

EYE

Cataract: opacification of the crystalline lens

A cataract is an opacification of the crystalline lens that scatters light and causes gradual, painless vision loss. Age-related nuclear sclerosis is the most common type, and posterior subcapsular cataract affects vision most and progresses to surgery most often. Untreated cataracts can cause phacomorphic or phacolytic glaucoma, and cataract is associated with dementia in people over 65. Diagnosis is clinical, and surgery, usually phacoemulsification with an IOL, is the only treatment. Here, we review the mechanisms, types, causes, complications, and treatment of cataract.

Key points

- Nuclear sclerosis, the most common type, is age-related and causes a myopic shift.
- Posterior subcapsular cataract sits near the nodal point, causes glare, and its growth predicts surgery most strongly.
- Steroids, diabetes, trauma, and uveitis are causes beyond age.
- Phacomorphic glaucoma closes the angle, and phacolytic glaucoma clogs the trabecular meshwork with macrophages.
- Cataract is associated with dementia over age 65, and cataract surgery with lower dementia risk.

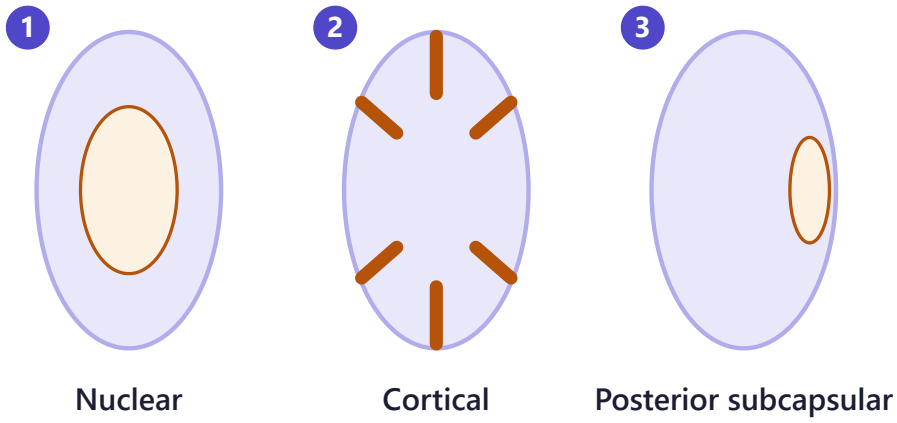


Fig. 1 | Three lenses seen from the side, front to the left. 1, nuclear sclerosis clouds the center. 2, cortical cataract forms wedge-shaped spokes from the edge. 3, posterior subcapsular cataract forms a plaque at the back of the lens. 1, Nuclear sclerosis; 2, Cortical; 3, Posterior subcapsular.

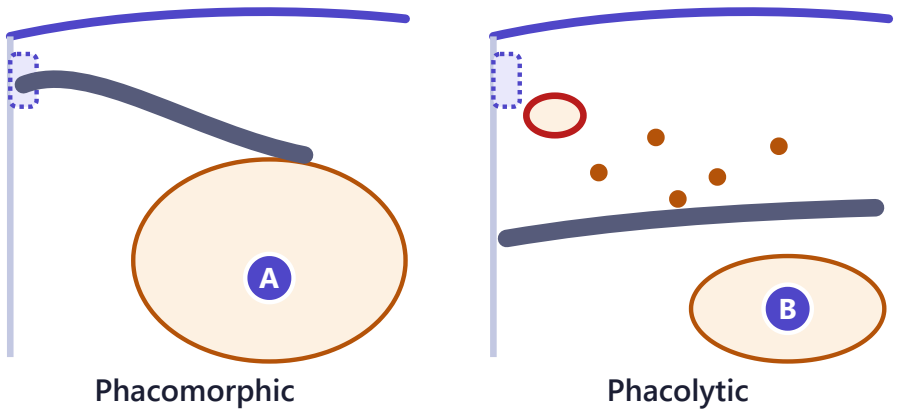


Fig. 2 | Two close-ups of the drainage angle with a cataract. A, phacomorphic glaucoma: a swollen lens pushes the iris forward against the trabecular meshwork. B, phacolytic glaucoma: a hypermature lens leaks proteins, and macrophages that take them up clog the meshwork while the angle stays open. A, Phacomorphic glaucoma; B, Phacolytic glaucoma.

Pathophysiology

Lens proteins change with age or injury, so the crystalline lens opacifies and scatters light, causing glare, halos, and gradual, painless blurring. In nuclear cataract, urochrome pigment collects in the center; in cortical cataract, water collects between lens fibers; and in subcapsular cataract, lens epithelium undergoes fibrous metaplasia. In diabetes, sorbitol accumulates and draws water into the lens.

Etiology

Nuclear sclerosis is the most common type; the center turns yellow-brown, and a myopic shift lets some patients read again without glasses. Cortical cataract forms wedge-shaped spokes from the edge of the lens. Posterior subcapsular cataract forms at the back of the lens, near the eye's nodal point, so it affects vision most, and its growth predicts progression to surgery most strongly.

Age is the most common cause, typically in the fifth and sixth decades. Topical or systemic steroids cause posterior subcapsular cataract, diabetes causes snowflake cortical opacities, trauma is a common cause of unilateral cataract in young adults, and uveitis is the most common cause of secondary cataract. Radiation, myotonic dystrophy, atopic dermatitis, smoking, and alcohol also contribute.

Congenital cataracts follow genetic causes, intrauterine infection, and metabolic disease such as galactosemia; they present with leukocoria and are treated early because they cause amblyopia.

Clinical presentation

Cataract causes gradual, painless blurring, glare and halos from headlights and sunlight, worst with posterior subcapsular cataract, faded, yellowed colors, frequent changes of glasses from the myopic shift, and monocular diplopia as light refracts through clear gaps. Untreated cataract can cause phacomorphic or phacolytic glaucoma. Cataract is associated with dementia in people over 65, and cataract surgery is linked to a lower risk of dementia, with a hazard ratio of 0.71. Cataract is a major cause of blindness worldwide, but it is reversible.

Untreated cataracts can cause lens-induced glaucoma. In phacomorphic glaucoma, a swollen, intumescent lens pushes the iris forward and closes the angle. In phacolytic glaucoma, a hypermature lens leaks proteins, and macrophages clog the trabecular meshwork while the angle stays open. Both are treated by removing the lens.

Diagnosis

Diagnosis is clinical, with visual acuity on a Snellen chart, slit lamp examination before and after dilation to grade type and density, a dimmed red reflex, and examination of the fundus and pupils to exclude retinal or optic nerve disease. An RAPD with a cataract points to retinal or optic nerve disease, because a cataract alone does not cause one. Sudden pain, a red eye, and high IOP with a cataract call for gonioscopy to look for lens-induced glaucoma, and before surgery, tamsulosin, which causes a floppy iris, and anticoagulants are noted.

Management

Surgery is the only treatment. It is offered when vision limits daily activities, such as reading or night driving, when glare is troublesome, or for lens-induced glaucoma. Phacoemulsification breaks up the lens with ultrasound through a small incision, and an IOL replaces it, whereas manual small-incision surgery suits dense, black cataracts.

Relationships to other conditions

Glaucoma. A swollen cataract can close the angle, and a hypermature cataract can clog the trabecular meshwork. Painless blurring with normal IOP favors cataract alone, whereas raised IOP, field loss, or a painful red eye favors glaucoma.

Vision loss and red eye. Cataract is a cause of gradual painless vision loss, like retinitis pigmentosa. Glare and a dimmed red reflex favor cataract, whereas night blindness and tunnel vision favor retinitis pigmentosa.

Alzheimer disease. Cataract is associated with dementia in people over 65, and cataract surgery is linked to a lower risk of dementia.

EYE

Vision loss and red eye: telling eye conditions apart by presentation

Eye conditions can be distinguished by how they present: a red eye, painful vision loss, sudden painless vision loss, or gradual painless vision loss. Viral and bacterial conjunctivitis and bacterial keratitis cause a red eye, with vision spared in conjunctivitis. Optic neuritis causes painful vision loss, retinal detachment and wet macular degeneration cause sudden painless loss, and retinitis pigmentosa and dry macular degeneration cause gradual loss. Each sits in a different part of the eye, from the conjunctiva to the optic nerve. Here, we review these eight conditions by presentation.

Key points

- Viral conjunctivitis, usually adenovirus, causes a red eye with watery discharge and normal vision and resolves within 3 weeks.
- Bacterial conjunctivitis causes mucopurulent discharge with lids stuck together, and topical antibiotics shorten it.
- Bacterial keratitis follows contact lens wear and needs scraping and urgent topical antibiotics.
- Optic neuritis causes painful monocular vision loss with an RAPD and is often the first sign of MS.
- Retinal detachment causes flashes, floaters, and a curtain and needs urgent surgery.
- Wet macular degeneration causes central distortion and is treated with anti-VEGF injections.
- Retinitis pigmentosa causes night blindness, then tunnel vision, with bone spicule pigment.
- Dry macular degeneration, most AMD, causes gradual central loss with drusen, and AREDS supplements slow it.

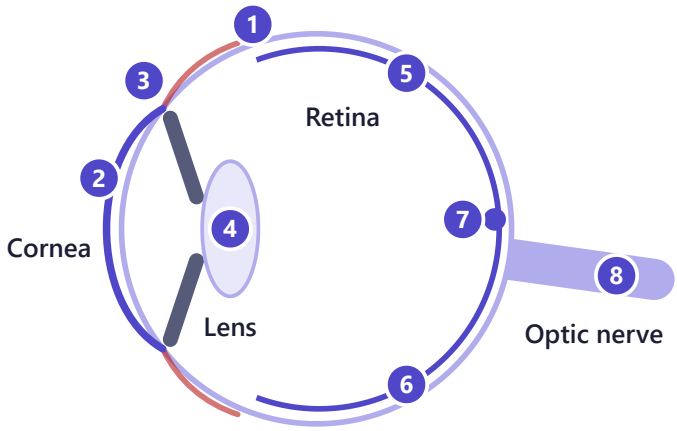


Fig. 1 | A side view of the eye, front to the left, numbering where each condition sits: 1, conjunctiva; 2, cornea; 3, drainage angle; 4, lens; 5, retina; 6, peripheral photoreceptors; 7, macula; 8, optic nerve. 1, Conjunctivitis; 2, Bacterial keratitis; 3, Glaucoma; 4, Cataract; 5, Retinal detachment; 6, Retinitis pigmentosa; 7, Macular degeneration; 8, Optic neuritis.

Red eye

Viral conjunctivitis, caused by adenovirus in up to 90% of cases, causes a sudden red eye with watery discharge, itching, and foreign body sensation, and vision is normal. Follicles and preauricular lymph nodes are typical, often after a respiratory infection. It is highly contagious for 10–14 days and resolves within 3 weeks with artificial tears, cool compresses, and hand hygiene, and antibiotics do not help.

Bacterial conjunctivitis, caused mainly by Staphylococcus aureus in adults and by Haemophilus influenzae, Streptococcus pneumoniae, and Moraxella in children, accounts for most childhood cases. It causes a red eye with mucopurulent discharge and lids stuck together on waking, often bilaterally, with normal vision. It is often self-limited, and topical antibiotics, such as chloramphenicol, polymyxin B with trimethoprim, or a fluoroquinolone, 4 times daily for 5–7 days, shorten it. Hyperacute conjunctivitis from Neisseria gonorrhoeae produces copious purulent discharge, can perforate the cornea, and needs Gram stain, culture, and urgent treatment.

Bacterial keratitis is a sight-threatening corneal infection, most often after contact lens wear, caused by

organisms such as Staphylococcus aureus and Pseudomonas. It causes pain, redness, photophobia, discharge, and reduced vision, with an epithelial defect, a stromal infiltrate, and sometimes hypopyon. Corneal scraping for Gram stain and culture precedes urgent topical antibiotics, moxifloxacin or fortified cefazolin and tobramycin. A painful red eye with halos and a fixed, mid-dilated pupil is acute angle closure instead.

Allergic conjunctivitis follows IgE-mediated mast cell degranulation and causes itching, redness, tearing, and chemosis, often seasonally with atopy, and is treated with allergen avoidance, lubricating drops, and topical antihistamines and mast cell stabilizers. Anterior uveitis causes a painful red eye with blurred vision, photophobia, and ciliary flush, is associated with HLA-B27 disease, juvenile idiopathic arthritis, IBD, sarcoidosis, and Behçet disease, and is treated with topical corticosteroids and cycloplegics.

Painful vision loss

Optic neuritis causes sudden vision loss in one eye, with pain on eye movement that often comes first and reduced color vision. An RAPD is present, and the disc is normal or, in a third, swollen. It affects women more than men, at ages 20–50, and is often the first sign of MS. MRI of the brain and orbits with gadolinium is obtained, and IV methylprednisolone followed by an oral taper speeds recovery without changing the outcome. It is also the most common syndrome of MOGAD, in which it is often bilateral.

Sudden painless vision loss

Retinal detachment is rhegmatogenous, from a tear, in most cases, or tractional, as in diabetic retinopathy, or exudative. Myopia, lattice degeneration, and trauma raise risk. New floaters and flashes precede field loss that spreads from the periphery like a curtain. Urgent surgery, by pneumatic retinopexy, scleral buckle, or vitrectomy, seals the break, and the outcome is best if the macula is still attached.

In wet macular degeneration, VEGF drives new vessels from the choroid through Bruch membrane, and they leak and bleed. It affects 10–15% of patients with AMD, with age, smoking, family history, and hypertension as risks. It causes central distortion and blurring, with a central scotoma on an Amsler grid. OCT shows fluid, fluorescein angiography confirms the vessels, and intravitreal anti-VEGF agents, ranibizumab or aflibercept, are the treatment; without them, 79–90% of eyes become legally blind.

CRAO, usually from carotid or cardiac emboli and occasionally from GCA, causes sudden, severe, painless vision loss with an RAPD and a pale retina with a cherry-red spot, and is treated as a stroke equivalent. CRVO, mostly after age 50 with hypertension, glaucoma, diabetes, or hyperlipidemia, gives a blood and thunder fundus of dilated, tortuous veins and hemorrhages in all quadrants. GCA, after age 50, causes new headache, jaw claudication, and polymyalgia rheumatica, and sudden vision loss from anterior ischemic optic neuropathy; ESR and CRP are followed by temporal artery biopsy, and glucocorticoids are started at once to prevent blindness.

Gradual painless vision loss

In retinitis pigmentosa, gene mutations, inherited in an autosomal recessive, autosomal dominant, or X-linked pattern, kill rods first, then cones; Usher syndrome adds hearing loss. Night blindness comes first, then tunnel vision, and central vision is lost late. The fundus shows bone spicule pigment, narrowed arterioles, and waxy disc pallor, and the ERG is reduced. There is no standard treatment beyond glasses and low-vision support, and gene therapy is emerging.

Dry macular degeneration makes up most AMD. Lipofuscin and lipid deposits, drusen, build up beneath the RPE, and the RPE and photoreceptors then atrophy, ending in geographic atrophy. Age over 60, smoking, family history and complement gene variants, and hypertension raise risk. It causes gradual loss of central vision, with yellow drusen and RPE pigment changes. Antioxidant and zinc supplements, as tested in AREDS, slow progression, smoking is stopped, and an Amsler grid detects conversion to wet AMD. Cataract and open-angle glaucoma also cause gradual painless vision loss.

Papilledema is bilateral optic disc swelling from raised ICP, with headache and transient visual obscurations, from tumors, CSF obstruction, venous sinus thrombosis, meningitis, or idiopathic intracranial hypertension, and calls for urgent imaging. Hypertensive retinopathy shows arteriolar narrowing, AV nicking, flame hemorrhages, hard exudates, and cotton wool spots, with papilledema in malignant hypertension. Horner syndrome, ptosis, miosis, and anhidrosis, follows lesions of the brainstem, lung apex, or internal carotid artery, and Argyll Robertson pupils, small pupils that constrict to near but not light, mark neurosyphilis.

Relationships to other conditions

Glaucoma. Acute angle closure causes a painful red eye like bacterial keratitis, and open-angle glaucoma causes gradual peripheral loss like retinitis pigmentosa. Raised IOP and a large cup favor glaucoma.

Cataract. Cataract causes gradual painless vision loss, like retinitis pigmentosa. Glare and a dimmed red reflex favor cataract, whereas night blindness and bone spicule pigment favor retinitis pigmentosa.

Multiple sclerosis. Optic neuritis is often the first sign of multiple sclerosis, so MRI of the brain is obtained to look for other lesions.

MOGAD. Optic neuritis is the most common syndrome of MOGAD. Bilateral optic neuritis with disc edema favors MOGAD.

EAR

Hearing loss: conductive, sensorineural, and mixed

Hearing loss is conductive when sound is blocked before the cochlea, sensorineural when the cochlea or cochlear nerve fails, and mixed when both occur. Conductive causes include cerumen, effusion, perforation, cholesteatoma, and otosclerosis; sensorineural causes include presbycusis, noise, ototoxic drugs, Ménière disease, and vestibular schwannoma. The Weber and Rinne tests and audiometry separate the types, and tinnitus, vertigo, and asymmetry point to particular causes. Treatment addresses the cause, with hearing aids and cochlear implants for permanent loss. Here, we review the types, causes, clinical clues, diagnosis, and management of hearing loss.

Key points

- Conductive loss makes sound quieter, and sensorineural loss makes it quieter and distorted.
- In conductive loss, the Weber test lateralizes to the affected ear and bone conduction beats air on the Rinne test.
- In sensorineural loss, the Weber test lateralizes to the better ear and air conduction beats bone.
- Presbycusis is bilateral and high-frequency, noise causes a 4 kHz notch, and Ménière disease causes fluctuating low-frequency loss.
- Asymmetric SNHL or unilateral tinnitus calls for MRI to look for vestibular schwannoma.
- Sudden SNHL is an emergency treated with oral steroids.
- GJB2 and GJB6 mutations at DFNB1 are the most common genetic cause, and PAX3 mutations cause Waardenburg syndrome types 1 and 3.

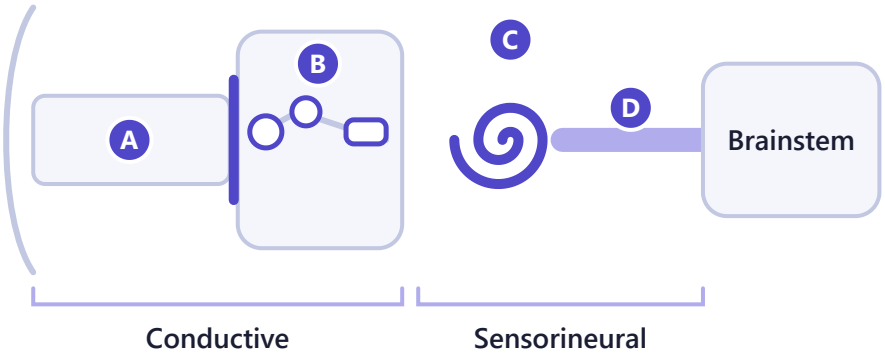


Fig. 1 | Sound passes from the ear canal, A, to the eardrum and ossicles of the middle ear, B, to the hair cells of the cochlea, C, and along the cochlear nerve to the brainstem, D. Loss at A or B is conductive; loss at C or D is sensorineural. A, Ear canal; B, Middle ear; C, Cochlea; D, Cochlear nerve.

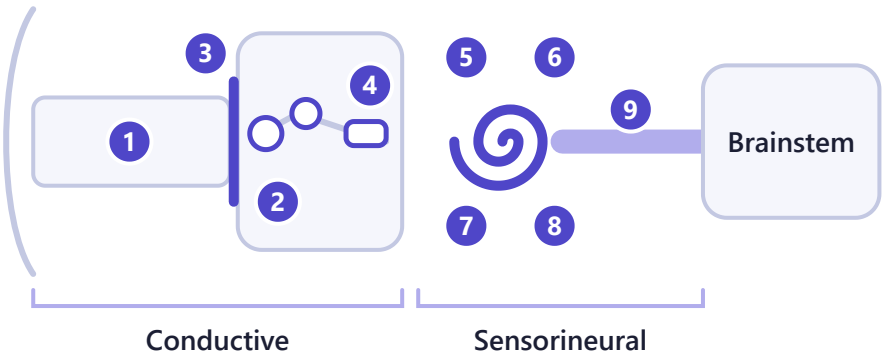


Fig. 2 | Causes of hearing loss placed on the ear: 1, cerumen in the canal; 2, effusion in the middle ear; 3, eardrum perforation or cholesteatoma; 4, otosclerosis fixing the stapes; 5 to 8, presbycusis, noise, ototoxic drugs, and Ménière disease in the cochlea; 9, vestibular schwannoma on the nerve. 1, Cerumen impaction; 2, Otitis media with effusion; 3, Perforation or cholesteatoma; 4, Otosclerosis; 5, Presbycusis; 6, Noise; 7, Ototoxic drugs; 8, Ménière disease; 9, Vestibular schwannoma.

Pathophysiology

Sound is collected by the outer ear, amplified by the eardrum and ossicles, transduced by cochlear hair cells, and carried by the cochlear nerve, part of CN VIII, to the brainstem. Loss before the cochlea is conductive and makes sound quieter. Loss at or after the cochlea is sensorineural and makes sound quieter and distorted. Mixed loss involves both, as otosclerosis can.

In conductive loss, bone conduction is normal, leaving an air-bone gap on the audiogram. In sensorineural loss, air and bone conduction are both reduced without a gap. Sensorineural loss is the most common type, and conductive loss is common in children.

Etiology

Conductive loss follows cerumen impaction, foreign bodies, and otitis externa; otitis media with effusion, the most common cause of acquired hearing loss in children; eardrum perforation; cholesteatoma, trapped squamous debris that erodes the middle ear; and otosclerosis, in which spongy bone fixes the stapes. Otosclerosis affects women twice as often, runs in families in over half of cases, and worsens in pregnancy.

Sensorineural loss follows presbycusis, the most common cause worldwide; noise, which damages hair cell stereocilia; ototoxic drugs, such as aminoglycosides, loop diuretics, and cisplatin; Ménière disease, from endolymphatic hydrops; and vestibular schwannoma, a benign Schwann cell tumor of CN VIII. In children, genetic causes account for over half of cases, and congenital CMV is another cause. Meningitis and diabetes can also cause it, and sudden SNHL is usually idiopathic.

Genetic causes account for about half of childhood hearing loss, 70% of it nonsyndromic and 30% syndromic, and nonsyndromic loss is autosomal recessive in 75–80% of cases, prelingual, severe, and stable. The DFNB1 locus on chromosome 13q12 holds GJB2 and GJB6, which encode connexins 26 and 30, gap junction proteins whose loss lowers the endocochlear potential; DFNB1 mutations are the most common genetic cause of severe to profound congenital deafness, and a GJB6 deletion with one GJB2 mutation also causes it. Waardenburg syndrome follows failed migration of neural crest cells, so melanocytes are missing from the hair, skin, iris, and stria vascularis. Type 1, from PAX3 mutations, is autosomal dominant, with a white forelock, heterochromia, patchy depigmentation, dystopia canthorum, and SNHL in about 20%; type 2, from MITF mutations, lacks dystopia canthorum and causes SNHL in up to 50%; type 3, also from PAX3, adds upper limb anomalies; and type 4, from EDNRB or EDN3 mutations and usually autosomal recessive, adds Hirschsprung disease.

NF2, an autosomal dominant disorder of the merlin gene on chromosome 22, causes bilateral vestibular schwannomas with meningiomas and ependymomas.

Clinical presentation

Presbycusis causes difficulty in noise and loss of high-pitched consonants. Noise causes tinnitus and hyperacusis. Otosclerosis causes progressive conductive loss with a normal eardrum, sometimes with a red promontory, the Schwartze sign. Ménière disease causes fluctuating loss with episodic vertigo, tinnitus, and aural fullness, and vestibular schwannoma causes unilateral loss, tinnitus, and imbalance, rarely with facial palsy. In children, hearing loss is the most common cause of speech delay.

Most tinnitus is subjective, heard only by the patient, and noise trauma is its most common cause, at the frequency lost. Objective tinnitus, which the examiner can hear, can

arise from an aneurysm. Unilateral tinnitus with asymmetric hearing suggests vestibular schwannoma, and tinnitus with vertigo and fluctuating loss suggests Ménière disease. High-dose aspirin causes reversible tinnitus, whereas cisplatin and loop diuretics can cause permanent tinnitus. Tinnitus usually accompanies hearing loss, but 20% of people in tinnitus clinics hear normally.

Diagnosis

Otoscopy looks for wax, perforation, effusion, or cholesteatoma. With a 512 Hz tuning fork, the Weber test lateralizes to the affected ear in conductive loss and to the better ear in sensorineural loss, and the Rinne test shows bone louder than air in an ear with conductive loss, whereas air is louder than bone normally and in sensorineural loss.

Audiometry measures air and bone conduction. Presbycusis gives a bilateral, high-frequency, downsloping curve, noise gives a notch at 3–6 kHz, often 4 kHz, with recovery at 8 kHz, Ménière disease gives low- to mid-frequency SNHL, and conductive loss gives an air-bone gap. Sudden SNHL, a loss of over 30 dB at 3 consecutive frequencies within 72 hours, calls for audiometry and MRI of the internal auditory canals, and asymmetric SNHL or unilateral tinnitus calls for MRI with gadolinium to look for vestibular schwannoma. Speech delay in a child always calls for a formal hearing test, and congenital or childhood SNHL calls for GJB2 testing once acquired causes are excluded, with gene-specific testing if a syndrome is suspected.

Management

Conductive loss is treated by its cause: removal or microsuction of cerumen, ventilation tubes for persistent effusion, and hearing aids or stapes surgery for otosclerosis. Sudden SNHL is an emergency treated with oral prednisone 1 mg/kg daily, up to 60 mg, for 7 days, then tapered, because spontaneous recovery is most likely in the first 2 weeks. Chronic SNHL is treated with hearing aids, since lost hearing does not return, and cochlear implants help severe bilateral loss, working best when placed before age 2 in children born deaf. Families with genetic hearing loss receive genetic counseling, since parents of recessive cases are usually unaffected carriers. Ear protection and avoiding or monitoring ototoxic drugs prevent loss, and a small vestibular schwannoma may be observed, whereas surgery or radiation does not restore hearing.

Tinnitus with hearing loss is treated with hearing aids, distress with cognitive behavior therapy and sound therapy, and aspirin-induced tinnitus resolves when aspirin is stopped.

Relationships to other conditions

Vertigo. Ménière disease, labyrinthitis, and vestibular schwannoma cause both hearing loss and vestibular symptoms. Hearing loss with vertigo points to the inner ear or CN VIII, whereas vestibular neuritis and BPPV spare hearing.

Stroke symptoms and complications. Lateral pontine syndromes from the anterior inferior cerebellar artery can cause ipsilateral hearing loss with vertigo. Other brainstem signs favor stroke over an ear cause.

EAR

Vertigo: timing, nystagmus, and hearing as guides to the cause

Vertigo is a false sense of motion arising from asymmetric vestibular input, and the VOR turns that asymmetry into nystagmus. Peripheral causes account for over 90% of cases and include BPPV, vestibular neuritis, labyrinthitis, Ménière disease, and vestibular schwannoma; central causes include vestibular migraine and posterior circulation stroke. Timing, triggers, the pattern of nystagmus, and whether hearing loss and tinnitus are present point to the cause, and the HINTS examination separates peripheral from central acute vertigo. Treatment targets the cause. Here, we review the vestibular system, causes of vertigo by timing, nystagmus, diagnosis, and treatment.

Key points

- Peripheral nystagmus is horizontal-torsional, unidirectional, and suppressed by fixation; central nystagmus can change direction or be vertical.
- In acute vertigo, an abnormal head impulse, unidirectional nystagmus, and no skew suggest a peripheral cause.
- BPPV causes episodes under 1 minute with upbeat, torsional nystagmus on the Dix-Hallpike maneuver and is treated with the Epley maneuver.
- Vestibular neuritis spares hearing, whereas labyrinthitis adds hearing loss.
- Ménière disease causes attacks of 20 minutes to 12 hours with fluctuating low-frequency hearing loss, tinnitus, and fullness.
- Vestibular suppressants should be limited to a few days.

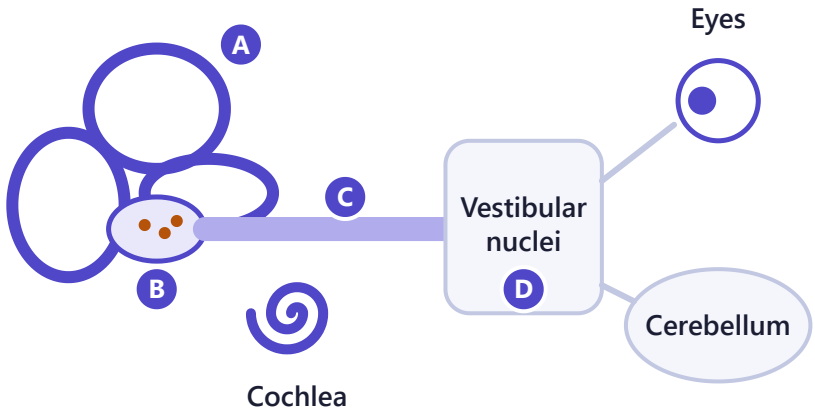


Fig. 1 | The semicircular canals sense rotation, A, and the utricle senses gravity with its otoconia, B. Signals travel along the vestibular nerve, C, to the vestibular nuclei and cerebellum, D, which drive the eyes through the vestibulo-ocular reflex. A, Semicircular canals; B, Utricle; C, Vestibular nerve; D, Vestibular nuclei and cerebellum.

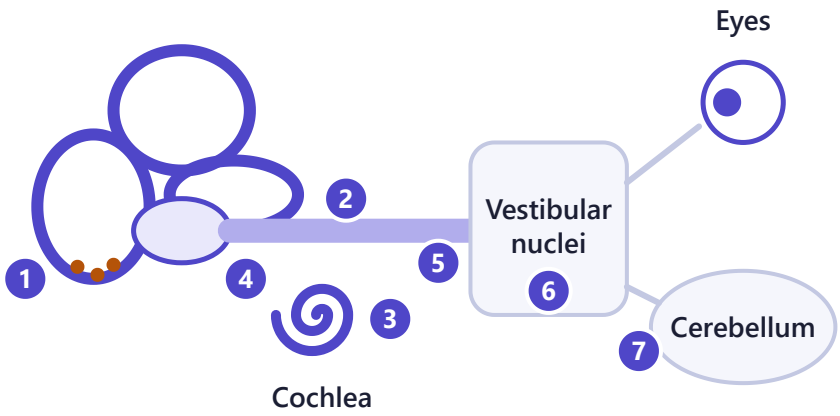


Fig. 2 | Causes of vertigo placed on the vestibular system: 1, BPPV, with otoconia loose in the posterior canal; 2, vestibular neuritis of the vestibular nerve; 3, labyrinthitis of the whole labyrinth, including the cochlea; 4, Ménière disease, with endolymphatic hydrops; 5, vestibular schwannoma near the brainstem; 6, vestibular migraine; 7, stroke of the cerebellum or brainstem. 1, BPPV; 2, Vestibular neuritis; 3, Labyrinthitis; 4, Ménière disease; 5, Vestibular schwannoma; 6, Vestibular migraine; 7, Posterior circulation stroke.

Pathophysiology

Unequal signals from the two vestibular systems are read as rotation. The VOR drives the eyes in a slow drift toward the weaker side, and corrective fast phases beat away from it, producing nystagmus, which is named for its fast phase and strengthens with gaze toward that phase. The semicircular canals sense rotation, the utricle senses gravity with its otoconia, and the vestibular nerve, part of CN VIII, carries signals to the vestibular nuclei and cerebellum.

Peripheral causes account for over 90% of vertigo. Peripheral nystagmus is horizontal-torsional and unidirectional, never purely vertical or torsional, and is suppressed by fixation, with an abnormal head impulse and no skew. Central vertigo, mostly from ischemia of the cerebellum or brainstem, can cause direction-changing, vertical, or purely torsional nystagmus that fixation does not suppress, with a normal head impulse and sometimes skew deviation. Hearing loss points to the ear.

Etiology

BPPV, the most common peripheral cause, follows loose otoconia in a canal, usually the posterior, and age and head injury raise its risk. Vestibular neuritis is presumed viral inflammation of the vestibular nerve, and labyrinthitis inflames the whole labyrinth, usually after a viral infection. Ménière disease follows endolymphatic hydrops, and vestibular schwannoma is a slow-growing tumor of CN VIII. Vestibular migraine is a common central cause, and posterior circulation stroke, most often lateral medullary syndrome, is the cause not to miss. Anticonvulsants, salicylates, and antibiotics can affect vestibular function.

Clinical presentation

Acute, continuous vertigo lasting days with nausea and unsteadiness suggests vestibular neuritis, labyrinthitis, or posterior circulation stroke. Recurrent episodes under 1 minute triggered by rolling over, looking up, or bending suggest BPPV. Spontaneous attacks of 20 minutes to 12 hours with ear symptoms suggest Ménière disease, attacks of 5 minutes to 72 hours with migraine features suggest vestibular migraine, and attacks of minutes suggest TIA. Progressive imbalance with unilateral hearing loss and tinnitus suggests vestibular schwannoma.

Posterior canal BPPV causes upbeat, torsional nystagmus on the Dix-Hallpike maneuver that fatigues with repetition, and lateral canal BPPV causes geotropic or apogeotropic horizontal nystagmus on the head-roll test. Among central patterns, downbeat nystagmus points to the cervicomedullary junction or midline cerebellum, upbeat nystagmus to the pons and midbrain, gaze-evoked nystagmus to sedatives and other drugs, and periodic alternating nystagmus, which reverses every 90–120 seconds, to the brainstem and cerebellum. Complications include falls, chronic dizziness when suppressants are used too long, and missed stroke.

Diagnosis

Vertigo is first separated from presyncope and imbalance, then classified by timing and triggers. Hearing, tinnitus, and nystagmus then help point to the cause. Vestibular neuritis causes spontaneous, unidirectional, horizontal-torsional nystagmus with normal hearing, and labyrinthitis adds unilateral hearing loss and tinnitus. Ménière disease causes fluctuating low-frequency SNHL, tinnitus, and fullness, with peripheral nystagmus during attacks, and vestibular schwannoma causes progressive unilateral SNHL and tinnitus, with

imbalance more than vertigo. Stroke usually spares hearing and causes direction-changing or vertical nystagmus with a normal head impulse and skew deviation.

In acute, continuous vertigo, the HINTS examination combines the head impulse, nystagmus, and a test of skew: a normal head impulse, direction-changing nystagmus, or skew suggests a central cause, and HINTS is more sensitive for stroke than MRI in the first 48 hours. In positional vertigo, the Dix-Hallpike maneuver, with the head turned 45° to the tested ear and then hanging back, provokes upbeat, torsional nystagmus in posterior canal BPPV.

A central HINTS result, focal signs, a new headache, or stroke risk calls for MRI and MRA, since CT is less sensitive. Unilateral hearing loss or tinnitus calls for audiometry and MRI with gadolinium, and Ménière disease is diagnosed by 2 or more attacks of 20 minutes to 12 hours with documented low-frequency loss. The absence of focal signs does not exclude a central cause.

Management

Treatment targets the cause, and vestibular suppressants only relieve acute symptoms. BPPV is treated with the Epley maneuver, which repositions otoconia out of the posterior canal. Vestibular neuritis and labyrinthitis are treated with antiemetics, antihistamines such as meclizine, or benzodiazepines for about 3 days, then vestibular rehabilitation, and steroids lack evidence. Ménière disease is treated with less salt, caffeine, and alcohol, thiazides or betahistine, intratympanic steroids, and intratympanic gentamicin if refractory. Vestibular migraine is treated as migraine, stroke with stroke care, and vestibular schwannoma with observation, surgery, or radiation. Vestibular rehabilitation trains the brain to balance with visual and proprioceptive cues, and suppressants beyond a few days delay central compensation.

Relationships to other conditions

Hearing loss. Ménière disease, labyrinthitis, and vestibular schwannoma cause hearing loss and tinnitus with vestibular symptoms. Normal hearing favors vestibular neuritis or BPPV.

Stroke symptoms and complications. Lateral medullary and cerebellar strokes can present with acute vertigo. A normal head impulse, direction-changing nystagmus, skew, or other brainstem signs favor stroke.

Multiple sclerosis. Brainstem demyelination is a central cause of vertigo. Other focal signs and internuclear ophthalmoplegia favor a central lesion.

DEMENTIA

Alzheimer disease: amyloid plaques, tau tangles, and progressive neuronal loss

Amyloid plaques and neurofibrillary tangles cause gliosis and neuronal loss, leading to progressive dementia. Plaques arise when β-secretase and γ-secretase cleave APP to release Aβ, and tangles arise when hyperphosphorylated tau leaves microtubules and aggregates inside neurons. Age is the main risk factor, and mutations in APP, PSEN1, and PSEN2 cause early-onset disease. Loss of neurons in the entorhinal cortex, hippocampus, and cholinergic and monoaminergic nuclei produces memory loss and later global decline. Diagnosis in life is clinical, with biomarkers, and the ABC score needs brain tissue, usually at autopsy. Here, we review these mechanisms, the symptoms they produce, diagnosis, and treatment, and consider how Alzheimer disease relates to other neurologic conditions.

Key points

- α-secretase cuts APP inside the Aβ region, so no Aβ forms, whereas β-secretase followed by γ-secretase releases Aβ.
- PSEN1 and PSEN2 encode the catalytic component of γ-secretase, and their mutations cause early-onset familial disease.
- Aβ begins in the neocortex and moves inward, whereas NFTs begin in the entorhinal cortex and move outward.
- Acetylcholinesterase inhibitors offset the loss of cholinergic neurons from the nucleus basalis of Meynert.
- The ABC score grades Aβ plaques, NFTs, and neuritic plaques, and needs brain tissue, usually at autopsy.

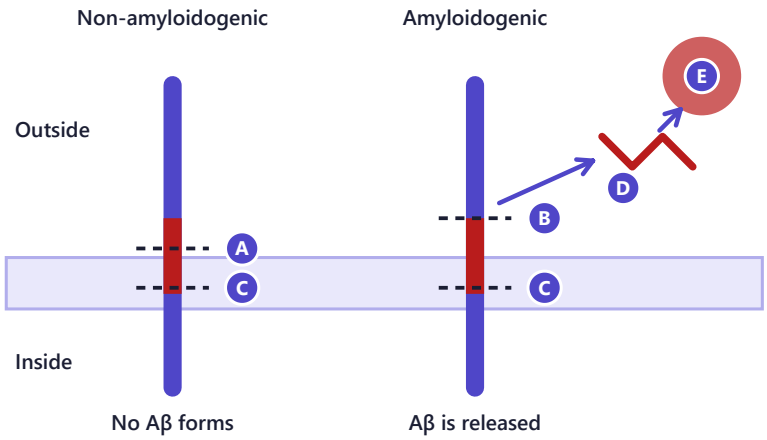


Fig. 1 | Amyloid precursor protein crossing a membrane in two pathways. In the non-amyloidogenic pathway, alpha secretase cuts inside the amyloid beta region, then gamma secretase cuts, and no amyloid beta forms. In the amyloidogenic pathway, beta secretase cuts at the start of the region, then gamma secretase cuts, releasing amyloid beta, which forms fibrils and plaques. A, α-secretase; B, β-secretase; C, γ-secretase; D, Fibrillization; E, Plaque formation.

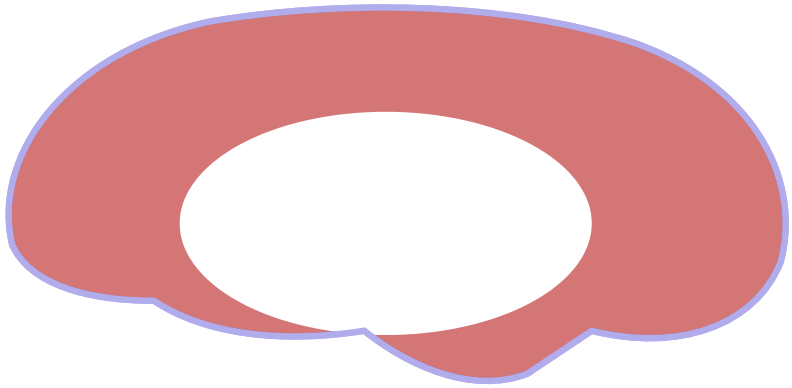


Fig. 2 | A brain with amyloid beta across the neocortex, sparing the deep structures. Aβ: neocortex first, moving inward

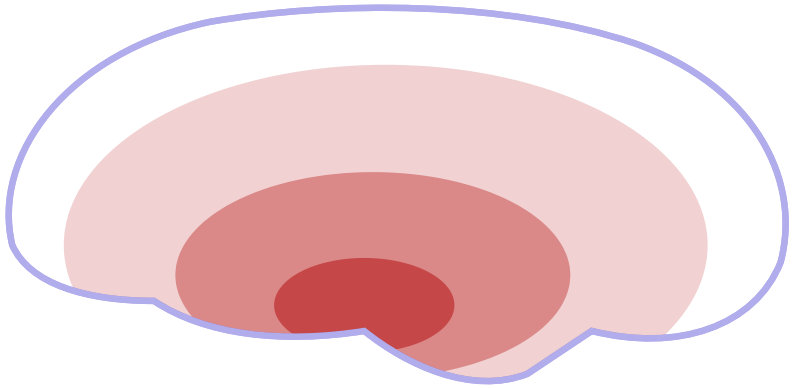


Fig. 3 | A brain with neurofibrillary tangles densest in the entorhinal cortex, fading outward through limbic areas to the neocortex. NFTs: entorhinal cortex first, moving outward

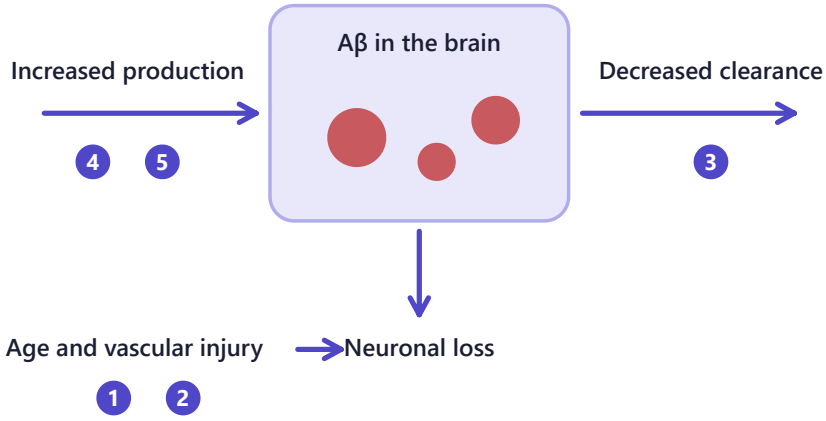


Fig. 4 | Amyloid beta builds up in the brain when production increases or clearance decreases. The build-up leads to neuronal loss, and age and vascular injury add to that loss. 1, Age; 2, Vascular risk factors; 3, APOE ε4; 4, Early-onset mutations; 5, Down syndrome.

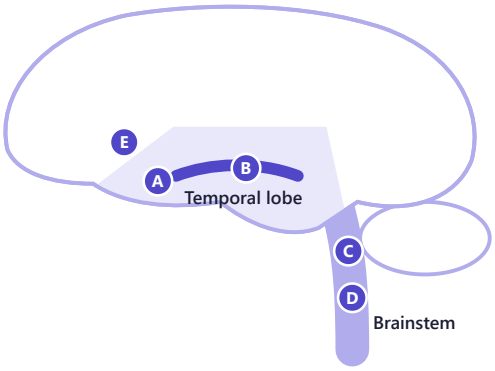


Fig. 5 | A side view of the brain. The entorhinal cortex and hippocampus lie in the medial temporal lobe, the nucleus basalis of Meynert lies in the basal forebrain, and the locus coeruleus and raphe nuclei lie in the brainstem. A, Entorhinal cortex; B, Hippocampus CA1; C, Locus coeruleus; D, Raphe nuclei; E, Nucleus basalis of Meynert.

Pathophysiology

Aβ is produced from APP, forms plaques outside neurons alongside NFTs inside them, and activates microglia and astrocytes, causing gliosis. Neuronal loss then causes atrophy and dementia.

APP is processed by two pathways. In the non-amyloidogenic pathway, α-secretase cuts inside the Aβ region, so no Aβ forms when γ-secretase then cuts within the membrane. In the amyloidogenic pathway, β-secretase cuts at the start of the Aβ region, and the γ-secretase cut releases Aβ. Aβ aggregates into oligomers and then fibrils, which deposit outside neurons as plaques.

Tau is a microtubule-associated protein that stabilizes axons. Hyperphosphorylation detaches tau from microtubules, and tau then aggregates inside neurons as NFTs. Axonal transport fails, and the neurons die.

The two lesions spread differently. Aβ follows the Thal phases, from the neocortex to the allocortex, including the hippocampus, then the diencephalon and striatum, the brainstem, and the cerebellum. NFTs follow the Braak stages, from the entorhinal cortex (I–II) to limbic areas (III–IV) and the neocortex (V–VI). NFTs track symptoms better than plaques do, because they follow the circuits that fail.

Etiology

Age is the main risk factor, and vascular injury from hypertension, diabetes, obesity, and smoking adds to neuronal loss. Decreased Aβ clearance is represented by APOE ε4, which slows clearance and raises the risk of late-onset disease. Increased Aβ production is represented by early-onset mutations in APP, PSEN1, and PSEN2, and by Down syndrome, in which chromosome 21 carries an extra APP gene. The presenilins PSEN1 and PSEN2 are the catalytic component of γ-secretase, and PEN2, APH1, and nicastrin are the other components of the complex.

Clinical presentation

Symptoms follow the structures affected. The entorhinal cortex is the first cortical site of NFTs, causing early memory loss and getting lost, and early atrophy of hippocampus CA1 accompanies short-term memory loss. Noradrenergic loss in the locus coeruleus changes attention and behavior, and serotonergic loss in the raphe nuclei causes depression and sleep changes. Cholinergic neuron loss in the nucleus basalis of Meynert causes memory and attention decline.

Early disease causes short-term memory loss, then executive dysfunction. Middle disease adds language and visuospatial decline, apathy, agitation, and wandering, and late disease causes apraxia, mutism, incontinence, and total dependence. Complications are aspiration pneumonia from dysphagia; infection, with pneumonia the most common cause of death; falls, dehydration, and malnutrition; and agitation, delirium, wandering, and caregiver burden. Life expectancy is about 4–8 years after diagnosis at age 65 or older. Aβ in vessel walls causes cerebral amyloid angiopathy, with lobar ICH.

Diagnosis

Assessment begins with a history that includes a collateral source, followed by cognitive testing with the MMSE or MoCA. Bloods are FBC, TFTs, B12, and electrolytes, and MRI shows medial temporal atrophy. Biomarkers are added if the diagnosis is unclear: CSF shows decreased Aβ42 with increased tau and phosphorylated tau, and amyloid PET shows plaques in life. Genetic testing is considered with early onset and a strong family history.

The ABC score is a neuropathologic score that needs brain tissue, usually at autopsy. A grades Aβ plaques by Thal phase, with A0 for none, A1 for phases 1–2, A2 for phase 3, and A3 for phases 4–5. B grades NFTs by Braak stage, with B0 for none, B1 for stages I–II, B2 for III–IV, and B3 for V–VI. C grades neuritic plaques by CERAD score, with C0 for none, C1 sparse, C2 moderate, and C3 frequent. The three grades combine into a level of

Alzheimer disease neuropathologic change: not, low, intermediate, or high.

Dementia mimics

Delirium is favored by sudden onset, fluctuating attention, and an acute illness, and prompts a search for the cause. Depression shows low mood, poor effort, and awareness of memory problems. Hypothyroidism causes fatigue, cold intolerance, and weight gain, and is found with TFTs, and B12 deficiency causes anemia and neuropathy. Normal pressure hydrocephalus causes gait disturbance and urinary incontinence early, and is assessed with MRI. Dementia with Lewy bodies is favored by visual hallucinations, parkinsonism, and fluctuations. Vascular dementia shows stepwise decline with vascular risk factors, and frontotemporal dementia occurs at a younger age, with early behavior or language change; both are assessed with MRI.

Management

Acetylcholinesterase inhibitors slow ACh breakdown, offsetting cholinergic neuron loss, and give a modest benefit in mild to moderate disease. The drugs are donepezil, rivastigmine, and galantamine, and they can cause bradycardia and GI upset. Memantine is an NMDA receptor antagonist for moderate to severe disease. The anti-amyloid antibodies lecanemab and donanemab clear plaques in early disease, with monitoring for ARIA.

Supportive care covers safety, routine, and caregiver support. Behavior is managed with non-drug approaches first, vascular risk factors are treated, and advance care planning begins early.

Relationships to other conditions

Dementia with Lewy bodies. Both cause progressive dementia. Visual hallucinations, parkinsonism, and fluctuations favor dementia with Lewy bodies, whereas early short-term memory loss with medial temporal atrophy favors Alzheimer disease.

Frontotemporal dementia. Both cause progressive dementia. Early behavior or language change with spared memory favors frontotemporal dementia, whereas early episodic memory loss favors Alzheimer disease.

Intracerebral hemorrhage. Aβ links the two conditions, since Aβ in vessel walls causes cerebral amyloid angiopathy and lobar ICH. Sudden focal deficits with blood on imaging favor hemorrhage, whereas gradual cognitive decline favors Alzheimer disease.

Seizures. Alzheimer disease is a degenerative cause of seizures, so treatment targets the underlying disease as well.

DEMENTIA

Frontotemporal dementia: frontal and temporal degeneration of behavior and language

Frontotemporal dementia is a group of syndromes caused by frontotemporal lobar degeneration, in which tau, TDP-43, or FUS inclusions kill neurons in the frontal and anterior temporal lobes. The behavioral variant causes disinhibition, apathy, loss of empathy, compulsive behavior, hyperorality, and executive dysfunction with anosognosia, and the language variants cause semantic or nonfluent primary progressive aphasia. About 40% of cases are familial. Diagnosis rests on history, cognitive testing, and frontotemporal atrophy on imaging, and treatment is supportive. Here, we review its pathology, variants, diagnosis, and management.

Key points

- FTLD inclusions contain tau, TDP-43, or, rarely, FUS, and the variant does not reliably predict the protein.
- MAPT causes FTLD-tau, and GRN and C9orf72 cause FTLD-TDP; C9orf72 also causes ALS.
- bvFTD causes disinhibition, apathy, loss of empathy, compulsive behavior, hyperorality, and executive dysfunction, with anosognosia.
- Left anterior temporal atrophy causes the semantic variant, and right anterior temporal atrophy causes prosopagnosia and getting lost.
- SSRIs help behavior, but cholinesterase inhibitors and memantine have no proven benefit.

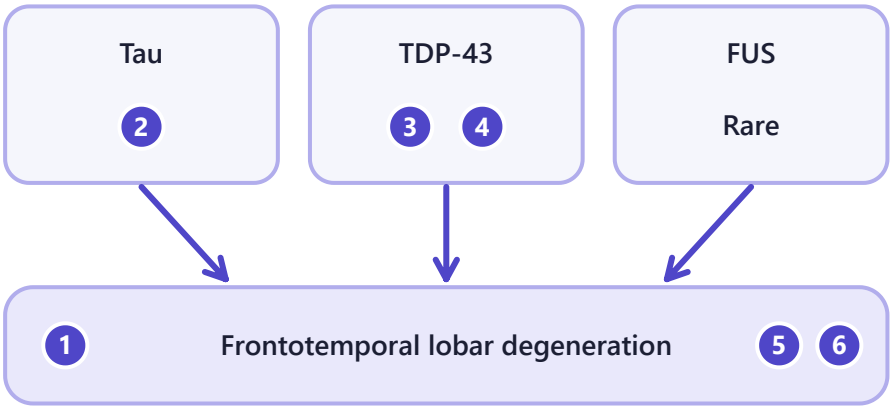


Fig. 1 | Three protein boxes: tau, linked to MAPT, 2; TDP-43, linked to GRN, 3, and C9orf72, 4; and FUS, which is rare. All feed into frontotemporal lobar degeneration, which is usually sporadic, 1, with head trauma, 5, and thyroid disease, 6, as risk factors. 1, Sporadic; 2, MAPT; 3, GRN; 4, C9orf72; 5, Head trauma; 6, Thyroid disease.

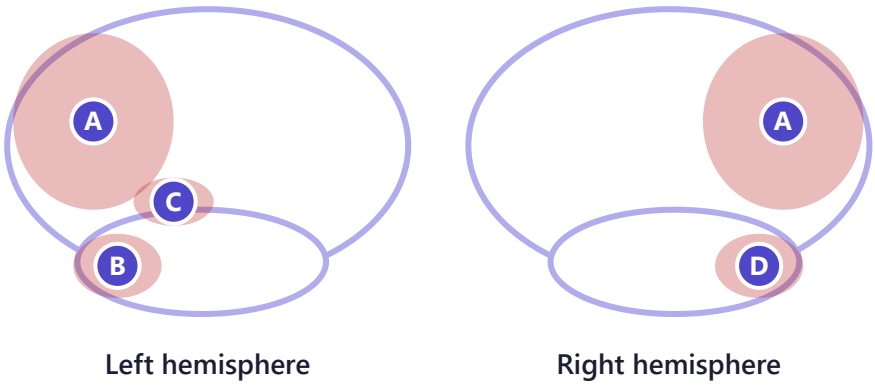


Fig. 2 | Side views of the left and right hemispheres, front toward the outside. A, frontal lobes on both sides: behavioral variant. B, left anterior temporal lobe: semantic variant. C, left inferior frontal and insular cortex: nonfluent variant. D, right anterior temporal lobe: prosopagnosia and getting lost. A, Behavioral variant (bvFTD); B, Semantic variant (svPPA); C, Nonfluent variant (nfvPPA); D, Right temporal variant.

Pathophysiology

Tau, TDP-43, or FUS inclusions kill neurons in the frontal and anterior temporal lobes, producing knife-edge atrophy, so behavior, personality, and language fail while memory is spared early. FTLD-tau includes tau inclusions such as Pick bodies, with ballooned neurons. FTLD-TDP accounts for about half of FTLD, and TDP-43 is also the protein of ALS, whereas FTLD-FUS is rare. The clinical variant does not reliably predict which protein is present.

Etiology

Most cases are sporadic, but about 40% are familial. MAPT, the tau gene, causes FTLD-tau, and GRN and C9orf72 cause FTLD-TDP; the C9orf72 repeat expansion is the most common cause of familial FTD and ALS. Head trauma and thyroid disease raise risk by 3.3-fold and 2.5-fold. Mean onset is at 58 years, making FTD the second most common early-onset dementia, and bvFTD and svPPA are more common in men, nfvPPA in women.

Clinical presentation

The behavioral variant, the most common form, follows frontal involvement. It causes disinhibition, with socially inappropriate and impulsive behavior; apathy, often mistaken for depression; loss of empathy; compulsive behavior, including repetitive movements, rituals, and stereotyped speech; hyperorality, including a sweet tooth, binge eating, and more alcohol; and executive dysfunction, with memory and visuospatial skills spared. Personality changes early, and patients develop anosognosia, losing insight into their changed behavior, and cannot explain why they act as they do.

The semantic variant follows left, dominant, anterior temporal atrophy and causes fluent speech that has lost meaning, with anomia and poor single-word comprehension. The nonfluent variant follows left inferior frontal and insular atrophy and causes effortful, halting, agrammatic speech with apraxia of speech, while word meaning is kept. Right, nondominant, anterior temporal atrophy causes prosopagnosia, getting lost, behavior change, and memory loss.

FTD overlaps with MND, CBD, and PSP. Complications include falls, hypersexuality, unsafe driving and finances, and insomnia and daytime sleepiness, and average survival is 7.5 years.

Diagnosis

Diagnosis begins with an informant history of behavior and personality change and cognitive testing, such as go-no-go, letter fluency, and similarities. MRI or CT shows frontal and anterior temporal atrophy, and SPECT or FDG-PET shows frontotemporal hypoperfusion or hypometabolism, though these findings are not specific. Genetic testing is offered when there is a family history.

Possible bvFTD requires 3 of the 6 behavioral features. The semantic variant requires anomia and impaired single-word comprehension, and the nonfluent variant requires agrammatism or effortful speech with apraxia of speech. Early episodic memory and visuospatial loss favor Alzheimer disease, word-finding pauses with poor repetition favor logopenic PPA, which is usually caused by AD, low mood favors depression, and visual hallucinations, parkinsonism, and fluctuations favor DLB.

Management

No treatment slows FTD, so care targets behavior, safety, and support. SSRIs can improve behavior but not cognition. Cholinesterase inhibitors and memantine have no proven benefit, and donepezil can worsen behavior. Antipsychotics have mixed results, and patients are prone to extrapyramidal effects. Speech therapy helps the PPA variants, physical and occupational therapy

support function and safety, caregivers need education and support, families are offered genetic counseling, and finances, driving, and eating are monitored.

Relationships to other conditions

Alzheimer disease. Both cause progressive dementia, and logopenic PPA is usually caused by AD. Early episodic memory and visuospatial loss favor Alzheimer disease, whereas early behavior or language change with spared memory favors frontotemporal dementia, which does not respond to cholinesterase inhibitors.

Atypical parkinsonism. PSP and CBD are 4R tauopathies that overlap with FTLD. Early parkinsonism, gaze palsy, falls, or asymmetric apraxia favors atypical parkinsonism.

Dementia with Lewy bodies. Both can cause early behavioral and executive change. Visual hallucinations, parkinsonism, and fluctuations favor dementia with Lewy bodies.

DEMENTIA

Dementia with Lewy bodies: α-synuclein, fluctuating cognition, and antipsychotic sensitivity

Dementia with Lewy bodies is a progressive dementia in which α-synuclein aggregates into Lewy bodies across the brainstem, limbic areas, and cortex, causing cholinergic and dopaminergic deficits. Its core features are fluctuating cognition, recurrent visual hallucinations, RBD, and parkinsonism, and attention, executive function, and visuoperceptual ability fail before memory. Diagnosis is clinical, supported by indicative biomarkers, and treatment targets each symptom, because no treatment slows neuronal loss. Antipsychotics carry a risk of serious reactions. Here, we review these mechanisms, the symptoms they produce, diagnosis, and treatment, and consider how dementia with Lewy bodies relates to other neurologic conditions.

Key points

- The four core features are fluctuating cognition, visual hallucinations, RBD, and parkinsonism.
- Probable DLB needs 2 or more core features, or 1 core feature with an indicative biomarker.
- Indicative biomarkers are reduced DAT uptake, low MIBG uptake, and REM sleep without atonia.
- DLB is diagnosed when dementia begins before, with, or within 1 year of parkinsonism.
- Acetylcholinesterase inhibitors come first, and antipsychotics are avoided whenever possible.

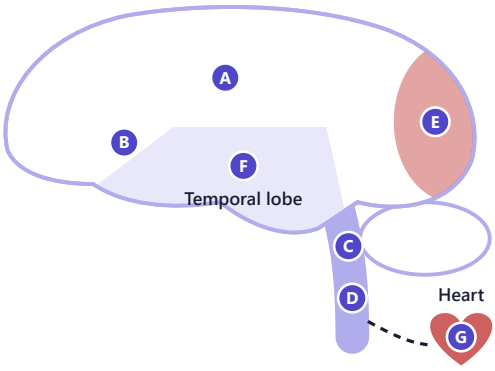


Fig. 1 | A side view of the brain and heart. Lewy bodies affect the cortex, the nucleus basalis of Meynert, the substantia nigra, and the brainstem. The occipital cortex shows reduced metabolism, the medial temporal lobe is relatively preserved, and the heart shows low MIBG uptake. A, Cortex; B, Nucleus basalis of Meynert; C, Substantia nigra; D, Brainstem nuclei; E, Occipital cortex; F, Medial temporal lobe; G, Autonomic nerves.

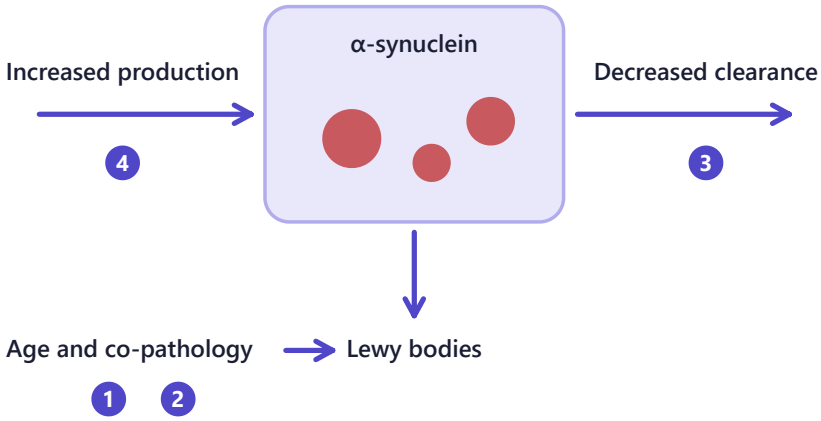


Fig. 2 | Alpha-synuclein builds up in the brain when its production increases or its clearance decreases. The build-up forms Lewy bodies, and age and Alzheimer co-pathology add to the damage. 1, Age; 2, APOE ε4; 3, GBA1 variants; 4, SNCA variants.

Pathophysiology

α-synuclein misfolds and aggregates, and Lewy bodies form as inclusions of α-synuclein and ubiquitin inside neurons across the brainstem, limbic areas, and cortex. Oligomers may be the main toxic form, because Lewy body numbers track neuronal loss poorly.

Each site produces a different effect. Cortical involvement causes attention, executive, and visuospatial decline. Cholinergic loss from the nucleus basalis of Meynert contributes to attention and cognitive decline, and dopaminergic loss from the substantia nigra causes parkinsonism. Brainstem nuclei are also affected, including the locus coeruleus, dorsal raphe, and dorsal motor nucleus of the vagus, with REM sleep without atonia and RBD. The occipital cortex shows reduced metabolism, with visual hallucinations, whereas the medial temporal lobe is relatively preserved, so memory is spared early. Autonomic nerve involvement causes low cardiac MIBG uptake and autonomic dysfunction.

Aβ plaques and tau tangles commonly coexist, and up to 30% of DLB meets the threshold for Alzheimer disease. This co-pathology brings more memory loss, faster progression, and higher mortality.

Etiology

Many studies find DLB more often in men, most people have no family history, and no single cause is known. Incidence rises with age, and APOE ε4 is a risk gene shared with Alzheimer disease. Decreased clearance is represented by GBA1 variants, which affect a lysosomal enzyme and are shared with Parkinson disease. SNCA, the α-synuclein gene, represents increased production, with rare family multiplications.

Clinical presentation

Attention, executive function, and visuoperceptual ability are affected early, and memory impairment becomes evident later. Fluctuating cognition causes pronounced variations in attention and alertness. Visual hallucinations are recurrent, well formed, and detailed. RBD causes acting out of dreams and may precede cognitive decline, and parkinsonism causes bradykinesia, rest tremor, or rigidity.

Supportive features are autonomic dysfunction, with orthostatic hypotension, which is often early, constipation, and urinary incontinence; postural instability, repeated falls, syncope, and transient unresponsiveness; hypersomnia and hyposmia; and delusions, misidentification, other hallucinations, apathy, anxiety, and depression. Complications are antipsychotic reactions, with sudden change in consciousness, impaired swallowing, acute confusion, and worse parkinsonism, as well as traumatic fractures from falls. Life expectancy is 5–8 years after diagnosis, and causes of death include falls, pneumonia, swallowing problems, cardiac complications, and suicide.

Diagnosis

Assessment begins with a history that includes a collateral source, cognitive testing with the MMSE or MoCA, bloods with a chemistry panel, TFTs, B12, syphilis, and HIV, and MRI, which shows relatively preserved medial temporal lobes. Biomarkers are added if the diagnosis is unclear.

Dementia is essential, defined as progressive cognitive decline that interferes with daily activities. Probable DLB needs 2 or more core features, or 1 core feature with 1 or more indicative biomarkers, and biomarkers alone cannot make the diagnosis. Possible DLB is 1 core feature with no indicative biomarker, or an indicative biomarker with no core feature. DLB is less likely if another illness explains the picture, or if parkinsonism is the only core feature and first appears in severe dementia.

Indicative biomarkers are reduced basal ganglia uptake on DAT SPECT or PET, low uptake on MIBG myocardial scintigraphy, and REM sleep without atonia on polysomnography. Supportive biomarkers are relative preservation of the medial temporal lobes on CT or MRI, low occipital activity on SPECT or PET with the cingulate island sign on FDG-PET, and prominent posterior slow-wave activity on EEG. CSF α-synuclein seed amplification assays have over 90% sensitivity and specificity, and only autopsy gives a definite diagnosis. DLB is diagnosed when dementia begins before, with, or within 1 year of parkinsonism, and PDD when dementia begins 1 year or more after established Parkinson disease.

DLB mimics

Alzheimer disease is favored by prominent early memory loss and medial temporal atrophy on MRI. Delirium has a sudden onset, an acute illness, and a reversible cause. Vascular dementia shows stepwise decline with vascular risk factors. Frontotemporal dementia occurs at a younger age, with early behavior or language change, and prion disease causes rapid progression with myoclonus. Drug-induced parkinsonism follows recent dopamine-blocking drug use and is assessed with a medication review. Progressive supranuclear palsy causes early falls and vertical gaze palsy.

Management

No treatment slows neuronal loss, so treatment targets each symptom. Acetylcholinesterase inhibitors, rivastigmine, donepezil, and galantamine, slow ACh breakdown, offsetting cholinergic loss, and improve cognition, function, and hallucinations. Memantine may

add benefit for cognition. Parkinsonism is less responsive than in Parkinson disease and is treated with low-dose levodopa, increased slowly. RBD is treated with melatonin, or cautious bedtime clonazepam, and depression with SSRIs, SNRIs, or mirtazapine. Autonomic dysfunction is treated as in other diseases, with little DLB trial data. Falls call for a safety assessment, bone density, and vitamin D.

Antipsychotics are avoided whenever possible, because serious reactions occur in up to 50% of people. If one is essential, low-dose quetiapine is used, though a small trial was negative, and haloperidol, risperidone, olanzapine, and aripiprazole are generally avoided. Supportive care covers caregiver education, home modifications, physical, occupational, and speech therapy, and early advance care planning.

Relationships to other conditions

Alzheimer disease. Both cause progressive dementia, the two pathologies often coexist, and both are treated with acetylcholinesterase inhibitors. Prominent early memory loss and medial temporal atrophy favor Alzheimer disease, whereas fluctuations, visual hallucinations, RBD, and parkinsonism favor dementia with Lewy bodies.

Typical Parkinson disease. Both involve α-synuclein and parkinsonism. Dementia before, with, or within 1 year of parkinsonism favors dementia with Lewy bodies, whereas dementia 1 year or more after established Parkinson disease is PDD.

MOVEMENT DISORDERS

Motor signs and syndromes: localizing weakness, abnormal movements, and nerve lesions

Motor signs localize lesions along a chain from the cortex through the brainstem and spinal cord to the peripheral nerve, with the basal ganglia, cerebellum, and dorsal columns tuning movement alongside. UMN lesions cause spastic weakness with brisk reflexes, whereas LMN lesions cause flaccid weakness with atrophy and fasciculations. Facial palsy that spares the forehead is supranuclear, gait patterns point to their lesion, ataxia separates cerebellar from sensory causes, and tremor type, dystonia, chorea, ballismus, myoclonus, and apraxia each have characteristic sites. Peripheral nerve lesions produce wrist drop, claw hand, ape hand, and foot drop. Here, we review these signs and what they localize.

Key points

- UMN lesions cause spasticity, brisk reflexes, and a Babinski sign; LMN lesions cause flaccid weakness, atrophy, and fasciculations.
- Supranuclear facial palsy spares the forehead, whereas nuclear and peripheral palsy weakens the whole half of the face.
- Midline cerebellar lesions cause gait ataxia, hemispheric lesions cause limb ataxia on the same side, and dorsal column lesions cause sensory ataxia with a positive Romberg test.
- Resting tremor suggests PD, postural tremor essential tremor, intention tremor the cerebellum, and rubral tremor a midbrain lesion.
- Hemiballismus classically follows a lesion of the contralateral subthalamic nucleus and tends to resolve.
- Radial nerve injury causes wrist drop, ulnar claw hand, median ape hand, and common peroneal foot drop.

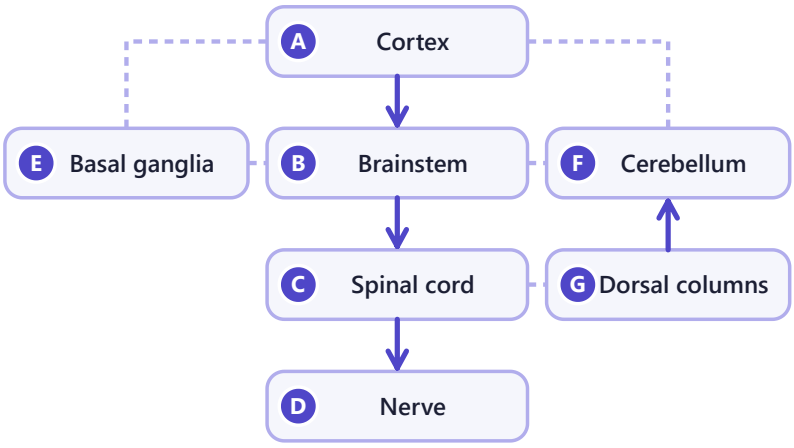


Fig. 1 | The motor system as a chain: cortex, A, through the internal capsule and brainstem, B, to the spinal cord, C, and the peripheral nerve, D. Beside the chain, the basal ganglia, E, and cerebellum, F, tune movement, and the dorsal columns, G, return position sense. A, Cortex and internal capsule; B, Brainstem; C, Spinal cord; D, Peripheral nerve; E, Basal ganglia; F, Cerebellum; G, Dorsal columns.

Where motor signs arise

Voluntary movement passes from the cortex through the internal capsule and brainstem to the spinal cord and peripheral nerve, while the basal ganglia and cerebellum tune it and the dorsal columns return position sense. UMN lesions, from cortex to cord, cause spastic, velocity-dependent tone, brisk reflexes, a Babinski sign, and weakness worst in arm extensors and leg flexors, as in stroke, MS, and cord injury. LMN lesions, from anterior horn cell to nerve, cause flaccid weakness, reduced reflexes, atrophy, and fasciculations in the pattern of a nerve or root, as in polio, SMA, and nerve injury. A UMN lesion above the pyramidal decussation causes contralateral signs, and one below it, in the cord, causes ipsilateral signs.

Upper motor neuron weakness

Hemiparesis and hemiplegia follow lesions of the contralateral cortex, internal capsule, or brainstem, or of the ipsilateral cord, and cause one-sided weakness with spasticity, brisk reflexes, and a Babinski sign; the face is

involved if the lesion lies above the facial nucleus. Stroke, tumor, trauma, infection, and MS are causes, and the gait is hemiparetic. Quadriplegia follows cervical cord or bilateral brainstem lesions, and injury above C5 also paralyzes the diaphragm. Central cord syndrome, the most common incomplete injury, weakens the arms more than the legs after neck hyperextension with stenosis.

Supranuclear facial palsy, from a contralateral cortical or corticobulbar lesion, weakens the lower face and spares the forehead, which receives input from both hemispheres; bilateral lesions cause pseudobulbar palsy, with slurred speech, dysphagia, a brisk jaw jerk, and pseudobulbar affect.

Lower motor neuron weakness

Nuclear and peripheral facial palsy, from the ipsilateral facial nucleus or CN VII, weakens the whole half of the face, with hyperacusis, taste loss, and dry eye. Bell palsy causes 38–83% of cases and peaks within 72 hours after ear pain; it is treated with steroids, antivirals, and eye protection, and over 95% recover with prompt treatment. A pontine stroke also weakens the forehead but brings other brainstem signs.

Radial nerve injury at the spiral groove of the humerus, from Saturday night palsy, a humeral shaft fracture, or crutches, causes wrist drop with loss of wrist, finger, and thumb extension and a numb dorsoradial hand; posterior interosseous nerve lesions at the elbow are purely motor. Splinting and rehabilitation are the treatment, with surgery if it does not recover. Ulnar nerve compression, most often at the cubital tunnel and also at the Guyon canal, causes tingling of the ring and little fingers, weak grip and pinch with a Froment sign, and, when severe, intrinsic wasting and clawing.

Median nerve compression, most often in the carpal tunnel, causes night pain and tingling of the thumb, index, middle, and half of the ring finger, with Phalen and Tinel signs; thenar sensation is spared because its branch leaves above the wrist, and severe cases cause thenar wasting and ape hand. Repetitive or vibrating work, pregnancy, diabetes, hypothyroidism, rheumatoid arthritis, and obesity raise risk, and surgical release treats severe or refractory cases. Common peroneal nerve compression at the fibular head, from leg crossing, casts, kneeling, squatting, surgery positioning, or trauma, causes foot drop with weak dorsiflexion and eversion, a numb dorsum of the foot and lateral leg, and a steppage gait. Inversion, supplied by the tibial nerve, is spared, which separates it from L5 radiculopathy, the most common cause of foot drop. An AFO and conservative care usually suffice.

Erb palsy, a birth traction injury of C5–C6 associated with shoulder dystocia, holds the arm in the waiter's tip position, with the shoulder adducted and internally rotated, elbow extended, forearm pronated, and wrist flexed, and abolishes the biceps reflex. Klumpke palsy, a C8–T1 injury from hyperabduction traction or an apical lung tumor, causes intrinsic hand weakness and clawing with medial forearm sensory loss, and Horner syndrome if sympathetic fibers are involved. Long thoracic nerve injury weakens the serratus anterior and wings the scapula; axillary nerve injury, from anterior dislocation or a proximal humerus fracture, weakens the deltoid and teres minor and numbs the regimental badge area; and musculocutaneous nerve injury weakens elbow flexion and numbs the lateral forearm.

Ataxia

Midline vermis lesions cause gait ataxia, with a broad-based gait and truncal sway, from alcohol, drugs such as phenytoin, lithium, and carbamazepine, or degeneration. Hemispheric lesions cause limb ataxia on the same side, with dysmetria, past-pointing, intention tremor, dysdiadochokinesia, rebound, pendular reflexes, scanning speech, and gaze-evoked nystagmus, from posterior circulation stroke, tumor, abscess, or MS. Sensory ataxia follows dorsal column or large-fiber loss, with lost position sense, a stomping gait, pseudoathetosis, and a positive Romberg test, as in B12 deficiency and tabes dorsalis. The Romberg test detects dorsal column disease, because cerebellar ataxia persists with the eyes open.

Tremor

Resting tremor, easing with movement and unilateral at onset, suggests PD. Postural tremor of both hands at 6–12 Hz, sometimes with head and voice tremor, is essential tremor, which is often familial and treated with propranolol or primidone. Intention tremor grows as the hand nears a target and follows cerebellar lesions. Rubral, or Holmes, tremor is slow, under 4.5 Hz, and coarse, and occurs at rest, in posture, and with intention, because midbrain and cerebellothalamic lesions, usually from stroke, disrupt both cerebellar and dopamine pathways; it begins weeks to 2 years after the lesion.

Dystonia, chorea, and ballismus

Dystonia is sustained co-contraction of agonists and antagonists that twists the body into abnormal postures, worsens with action, and eases with a sensory trick. Cervical dystonia is the most common in adults, with blepharospasm and writer's cramp as other focal forms, and early-onset generalized dystonia is linked to DYT1 and DYT6. Antipsychotics and dopa-responsive dystonia are other causes. Botulinum toxin treats focal dystonia, trihexyphenidyl childhood generalized dystonia, and

levodopa dopa-responsive dystonia, with baclofen and benzodiazepines as alternatives.

Hemichorea and hemiballismus classically follow a lesion of the contralateral subthalamic nucleus. Chorea is irregular, flowing, and dance-like, and ballismus is violent, flinging, and high in amplitude. Ischemic stroke is the most common cause over 65, and nonketotic hyperglycemia in type 2 diabetes is another. The movements tend to resolve, and treatment addresses the cause, with haloperidol or risperidone, tetrabenazine, clonazepam, or topiramate. Bilateral chorea with dementia and a family history suggests Huntington disease.

Myoclonus and apraxia

Myoclonus is a sign of sudden, brief, lightning-like jerks. Asterixis is negative myoclonus, brief lapses in tone of the outstretched wrists, as in liver failure. Physiologic myoclonus, such as hiccups, startle, and jerks on falling asleep, needs only reassurance, JME causes morning jerks from puberty, and secondary myoclonus follows hypoxic brain injury, drugs including anticonvulsants, and kidney or liver failure.

Apraxia is loss of learned, skilled movement despite normal strength, sensation, and coordination. Ideational apraxia loses the concept, ideomotor apraxia disconnects concept from output, limb-kinetic apraxia loses fine output, and constructional apraxia impairs drawing and copying. It follows lesions of the dominant parietal and premotor praxis network, occurring in 50–80% of left hemisphere strokes, and is a feature of Alzheimer disease and corticobasal syndrome; physical and occupational therapy are the treatment.

Gait

A spastic hemiparetic gait swings the extended leg in an arc, and a scissor gait, crouched with crossing legs, follows cerebral palsy. A neuropathic steppage gait lifts the foot high and lands on the toes because of foot drop from peroneal, L5, or neuropathic lesions. A cerebellar gait is broad-based and unsteady, a sensory ataxic gait stomps and worsens in the dark, as in B12 deficiency, a festinating gait shuffles with reduced arm swing and freezing in PD, and a waddling gait follows proximal weakness. Most gait disorders are not neurologic, and pain causes an antalgic gait.

Relationships to other conditions

Stroke symptoms and complications. Stroke is the most common cause of hemiparesis, supranuclear facial palsy, hemiballismus, apraxia, and rubral tremor. A sudden onset in an arterial territory favors stroke.

Typical Parkinson disease. Resting tremor and a festinating gait point to Parkinson disease, whereas postural tremor without bradykinesia favors essential tremor.

Huntington disease. Chorea is the hallmark of Huntington disease. Bilateral chorea with dementia and a family history favors it over hemichorea from a focal lesion.

Atypical parkinsonism. Corticobasal degeneration causes apraxia, dystonia, and myoclonus, and multiple system atrophy causes cerebellar ataxia. Their combination with parkinsonism favors these disorders.

MOVEMENT DISORDERS

Parkinson disease: α-synuclein, dopamine neuron loss, and dopaminergic treatment

Parkinson disease is a progressive neurodegenerative disorder in which α-synuclein aggregates into Lewy bodies and dopamine neurons of the SNc die, depleting dopamine in the striatum. Genetic factors, such as SNCA and LRRK2, interact with environmental exposures, such as pesticides and MPTP, and with age. Mitochondrial dysfunction, oxidative stress, failure of the ubiquitin-proteasome system, and glutamate excitotoxicity drive neuron loss. Diagnosis is clinical, by the MDS criteria, with DAT SPECT when unclear, and treatment restores dopamine signaling, because no treatment slows the disease. Here, we review these mechanisms, the symptoms they produce, diagnosis, and treatment, and consider how Parkinson disease relates to other neurologic conditions.

Key points

- Lewy bodies are α-synuclein-stained inclusions, and the SNc loses its pigment as dopamine neurons die.
- Motor signs appear once 50–60% of nigrostriatal neurons are lost.
- Parkinsonism means bradykinesia plus rest tremor or rigidity.
- The MDS criteria weigh absolute exclusions, red flags, and supportive criteria.
- Levodopa with carbidopa is the most effective treatment, and dopamine agonists cause impulse control disorders.

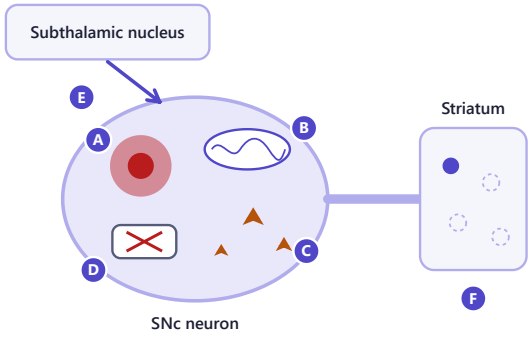


Fig. 1 | A dopamine neuron in the substantia nigra pars compacta projecting to the striatum. Inside the cell body are a Lewy body, a failing mitochondrion, reactive oxygen species, and a blocked proteasome. Glutamate from the subthalamic nucleus floods the neuron, and dopamine release in the striatum falls. A, α-synuclein aggregation; B, Mitochondrial dysfunction; C, Oxidative stress; D, Ubiquitin-proteasome failure; E, Glutamate excitotoxicity; F, Dopamine loss.



Fig. 2 | A midbrain cross-section with a dark band of pigmented substantia nigra on each side. Normal: pigmented substantia nigra



Fig. 3 | A midbrain cross-section in Parkinson disease with pale, depigmented substantia nigra. Parkinson disease: depigmented SNc

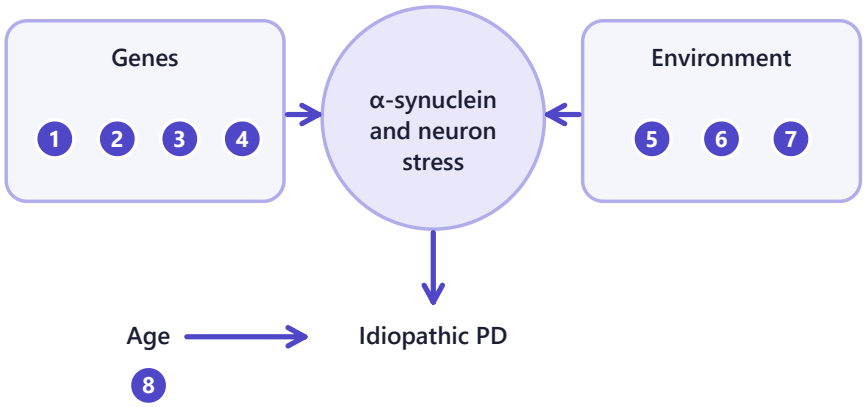


Fig. 4 | Genes on the left and environmental exposures on the right both feed into neuron stress in the center, which leads to idiopathic Parkinson disease; age adds to the risk. 1, SNCA; 2, LRRK2; 3, PRKN, PINK1, and DJ-1; 4, GBA1; 5, Pesticides and herbicides; 6, MPTP; 7, Head injury; 8, Age.

Pathophysiology

Genetic and environmental factors make α-synuclein aggregate into Lewy bodies, and dopamine neurons of the SNc die, leaving too little dopamine in the striatum to drive movement. Motor signs appear once 50–60% of nigrostriatal neurons are lost.

Several mechanisms kill dopamine neurons. Misfolded α-synuclein forms Lewy bodies and Lewy neurites. MPTP and rotenone accumulate in mitochondria and impair them. Reactive oxygen species cause oxidative stress, which DJ-1 normally counters by protecting mitochondria. Failure of the ubiquitin-proteasome system leaves misfolded proteins uncleared, and parkin is a ubiquitin ligase in this system. An overactive subthalamic nucleus floods the SNc with glutamate, causing excitotoxicity.

Loss of neuromelanin-containing dopamine neurons makes the SNc pale. Lewy bodies are α-synuclein-stained inclusions in neuron cell bodies, and Lewy neurites are inclusions in neuron processes. By Braak staging, pathology spreads from the gut, the dorsal motor nucleus of the vagus, and the olfactory bulb to the locus coeruleus, the SNc, and finally the cortex, so constipation, RBD, and anosmia can precede motor signs by years.

Etiology

SNCA, the α-synuclein gene, causes familial PD when extra copies are present. LRRK2 is the most common dominant gene, accounting for 40% of familial cases, and PRKN, PINK1, and DJ-1 serve mitochondrial quality control, while GBA1 encodes a lysosomal enzyme. Environmental factors include pesticides and herbicides, such as rotenone, MPTP, which is taken up by DAT into dopamine neurons, and mild to moderate head injury. Age is a major factor, and PD affects 1% of people over 60.

Most PD is idiopathic: incompletely penetrant genes, such as LRRK2, raise risk most with toxin exposure, and known genes explain only about 3–5% of cases. Smoking, caffeine, and physical activity are linked to lower risk.

Clinical presentation

Bradykinesia, slow movement with decreasing amplitude, is the core feature. Rest tremor is unilateral at onset and reemerges at rest but is not required. Rigidity is lead pipe or cogwheel, with reduced arm swing, and postural instability, with festination, freezing, and falls, appears late.

Non-motor features include prodromal constipation, RBD, and anosmia; autonomic orthostatic hypotension, urinary retention, dysphagia, and erectile dysfunction; and depression and anxiety, with dementia and psychosis later. Complications are motor fluctuations, with 'off' periods as levodopa wears off, levodopa-induced dyskinesia, impulse control disorders, mainly with dopamine agonists, and falls.

Diagnosis

Diagnosis is clinical. Parkinsonism means bradykinesia plus rest tremor or rigidity. By the 2015 MDS criteria, clinically probable PD needs parkinsonism, no absolute exclusion criteria, and any red flags balanced by supportive criteria, and clinically established PD needs at least 2 supportive criteria and no red flags.

Absolute exclusion criteria are cerebellar abnormalities, downward vertical supranuclear gaze palsy, bvFTD or primary progressive aphasia, parkinsonism limited to the legs for over 3 years, no response to high-dose levodopa, cortical sensory loss, limb apraxia, or progressive aphasia, and normal DAT imaging. Red flags are a wheelchair within 5 years, no progression over 5 years, early severe bulbar dysfunction, inspiratory stridor, severe autonomic failure within 5 years, recurrent falls within 3 years, disproportionate anterocollis, no non-motor features after 5 years, and unexplained pyramidal signs. Supportive criteria are a clear, dramatic response to dopaminergic therapy, levodopa-induced dyskinesia, rest tremor of a limb, and olfactory loss or cardiac MIBG denervation.

^{123}I-ioflupane SPECT shows presynaptic dopamine transporters in the striatum. Uptake is reduced with dopamine loss, in PD, MSA, PSP, CBD, and DLB, and normal in essential tremor and drug-induced and vascular parkinsonism. A clear response to levodopa supports PD, and a poor response suggests another diagnosis.

Atypical parkinsonism and mimics

Progressive supranuclear palsy causes vertical gaze palsy and early falls. Multiple system atrophy causes early autonomic failure, with cerebellar and pyramidal signs, and corticobasal degeneration causes asymmetric apraxia, cortical sensory loss, and alien limb. Dementia with Lewy bodies brings early dementia and visual hallucinations. Drug-induced parkinsonism follows dopamine-blocking drugs, and essential tremor causes a bilateral postural or action tremor; both have normal DAT imaging.

Management

No treatment slows PD, so treatment restores dopamine signaling and manages symptoms. Levodopa crosses the BBB and becomes dopamine, and carbidopa blocks its peripheral conversion; together they are the most effective treatment. Pramipexole is a dopamine agonist, selective for D3 receptors, rasagiline is a selective MAO-B inhibitor, and entacapone is a COMT inhibitor that prolongs each levodopa dose.

Amantadine, a weak NMDA antagonist, increases dopamine release and blocks its reuptake, easing bradykinesia, rigidity, and tremor, often only briefly. Anticholinergics such as trihexyphenidyl block muscarinic receptors and mainly help tremor, at the cost of dry mouth and other anticholinergic effects.

As neurons are lost, 'on' time shortens, and entacapone or extended-release levodopa is added. Dopamine agonists such as pramipexole are the main trigger of impulse control disorders, including gambling, hypersexuality, compulsive shopping, and binge eating, which call for reducing or stopping the agonist. DBS treats refractory motor fluctuations. Non-motor symptoms are treated with PEG for constipation, SSRIs for depression and anxiety, cholinesterase inhibitors for dementia, pimavanserin or other antipsychotics for psychosis, and cognitive behavior therapy for impulsive behavior.

Relationships to other conditions

Dementia with Lewy bodies. Both involve α-synuclein, Lewy bodies, and parkinsonism. Dementia before, with, or within 1 year of parkinsonism favors dementia with Lewy bodies, whereas dementia 1 year or more after established Parkinson disease is Parkinson disease dementia.

Atypical parkinsonism. Progressive supranuclear palsy, multiple system atrophy, and corticobasal degeneration also cause parkinsonism with reduced DAT uptake, but they respond poorly to levodopa and add features such as vertical gaze palsy, early autonomic failure, or asymmetric apraxia.

Atypical parkinsonism: multiple system atrophy, progressive supranuclear palsy, and corticobasal degeneration

Atypical parkinsonism describes neurodegenerative disorders that cause parkinsonism with a poor response to levodopa and additional signs. Multiple system atrophy is an α-synucleinopathy with GCIs in oligodendrocytes, whereas progressive supranuclear palsy and corticobasal degeneration are 4R tauopathies. Each damages a characteristic pattern of structures, producing early autonomic failure and cerebellar signs, vertical gaze palsy and early falls, or asymmetric cortical signs. Diagnosis is clinical, supported by MRI patterns, and treatment is symptomatic. Here, we review the proteinopathies, clinical features, imaging, and management of these disorders.

Key points

- α-synuclein collects in PD, DLB, and MSA, 4R tau in PSP and CBD, Aβ with 3R and 4R tau in AD, and TDP-43 in MND and LATE.
- MSA presents with parkinsonism, cerebellar signs, or pyramidal signs, with early autonomic failure.
- PSP causes vertical supranuclear gaze palsy, early falls, akinesia, and cognitive dysfunction.
- CBD causes asymmetric rigidity, dystonia, and myoclonus with apraxia, cortical sensory loss, and alien limb.
- The hot cross bun sign suggests MSA, and midbrain atrophy suggests PSP.
- No treatment slows these disorders, so all three are treated by symptom.

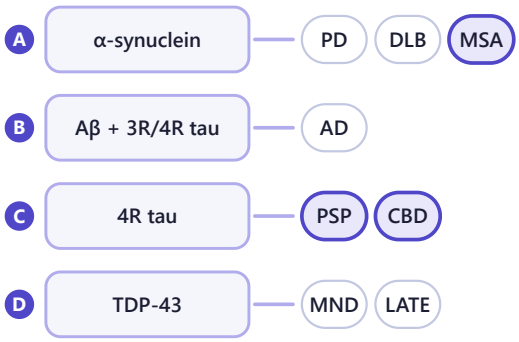


Fig. 1 | Four rows group neurodegenerative diseases by the protein that collects. A, alpha-synuclein: Parkinson disease, dementia with Lewy bodies, and multiple system atrophy. B, amyloid beta with three-repeat and four-repeat tau: Alzheimer disease. C, four-repeat tau: progressive supranuclear palsy and corticobasal degeneration. D, TDP-43: motor neuron disease and LATE. The three atypical parkinsonian disorders are shaded. Shaded: atypical parkinsonism A, α-synuclein; B, Aβ plus 3R and 4R tau; C, 4R tau; D, TDP-43.

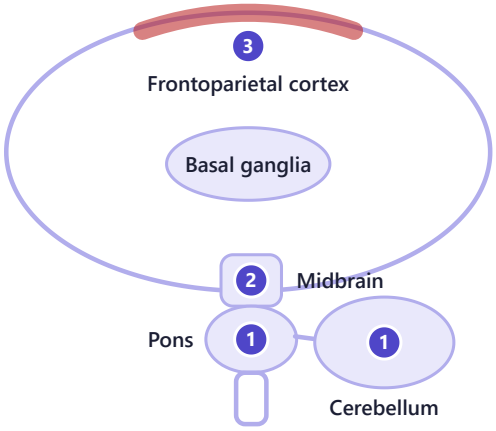


Fig. 2 | A side view of the brain and brainstem. Multiple system atrophy, 1, affects the pons and cerebellum; progressive supranuclear palsy, 2, affects the midbrain; corticobasal degeneration, 3, affects the frontoparietal cortex. All three involve the basal ganglia. 1, Multiple system atrophy; 2, Progressive supranuclear palsy; 3, Corticobasal degeneration.

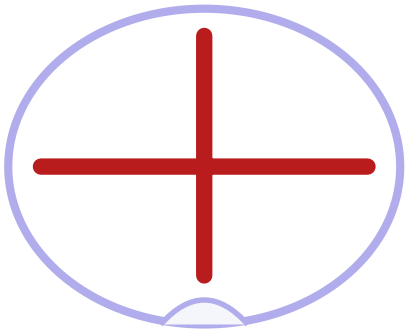


Fig. 3 | An axial slice of the pons with a bright cross through it, the hot cross bun sign. MSA: hot cross bun sign

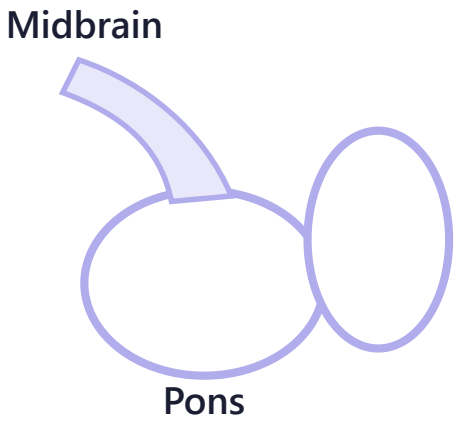


Fig. 4 | A midline side view of the brainstem with a thin, beak-shaped midbrain above a normal pons, the hummingbird sign. PSP: midbrain atrophy

Pathophysiology

An abnormal protein collects in neurons and glia, and cells of the basal ganglia, brainstem, cerebellum, or cortex die, causing parkinsonism that responds poorly to levodopa because striatal targets are lost too.

Neurodegenerative diseases can be grouped by the protein that collects. α-synuclein collects in PD and DLB in neurons and in MSA in oligodendrocytes. AD combines Aβ with tangles that mix 3R and 4R tau. PSP and CBD are 4R tauopathies affecting neurons and astrocytes. TDP-43 collects in MND, LATE, and about half of FTLD.

In MSA, GCIs of α-synuclein in oligodendrocytes accompany widespread neuron loss in the putamen, pons, MCP, and cerebellum, with increased putaminal iron. In PSP, neuronal and glial 4R tau, including tufted astrocytes, affects the basal ganglia and brainstem, with variable cortical involvement. In CBD, neuronal and glial 4R tau, astrocytic plaques, and ballooned neurons affect the posterior frontal and anterior parietal cortex and basal ganglia asymmetrically.

Etiology

All three disorders are sporadic, and their causes are unknown. A family history of parkinsonism or ataxia argues against MSA. In CBD, misfolded tau is thought to spread from cell to cell in a prion-like manner, seeding further pathology.

Clinical presentation

MSA presents with parkinsonism (MSA-P), with bradykinesia, rigidity, tremor, and postural instability, from striatonigral degeneration; with cerebellar signs (MSA-C), including gait and limb ataxia, dysarthria, and nystagmus, from olivopontocerebellar atrophy; or with both, and pyramidal signs, such as hyperreflexia and a Babinski sign, are common. MSA-P accounts for 70–80% of cases in Western populations, whereas MSA-C accounts for 67–84% in Asian populations and begins earlier. Its features include early autonomic failure, with orthostatic hypotension, urinary dysfunction, and erectile dysfunction, a poor levodopa response, dysphagia, hypometric saccades, falls, myoclonus, RBD, which often comes first, and stridor and sleep apnea.

The hallmark of PSP is vertical supranuclear gaze palsy, beginning with slow vertical saccades, with downgaze

affected before upgaze. Postural instability causes early backward falls with a stiff, upright gait, and akinesia is symmetric and axial. Cognitive dysfunction includes apathy, disinhibition, and executive dysfunction. The 2017 MDS criteria describe subtypes: PSP-RS, the classic form, with gaze palsy, early falls, and cognitive change, leads to death about 7 years from onset; PSP-P, which may respond to levodopa, and PSP-PAGF, with gait freezing, allow survival of a decade or more; and cortical variants include PSP-CBS, PSP-bvFTD, and PSP-PNFA.

CBD causes asymmetric limb rigidity or akinesia, dystonia, and myoclonus, with cortical signs: limb apraxia, cortical sensory loss, and alien limb. Nonfluent aphasia and executive dysfunction can occur.

Complications include falls with head injury, dysphagia with aspiration pneumonia, and, in MSA, stridor and sleep apnea from vocal cord paralysis. Patients with PSP become dependent within 3–4 years, and median survival in CBD is about 8 years.

Diagnosis

Diagnosis is clinical: parkinsonism with red flags and a poor or short-lived response to a levodopa trial. MRI shows the pattern of atrophy, and DAT SPECT is reduced in all three, as in PD, so it does not separate them from PD. Only autopsy confirms the protein.

Early autonomic failure, cerebellar or pyramidal signs, and stridor favor MSA, in which MRI shows the hot cross bun sign in the pons, atrophy of the putamen, MCP, pons, and cerebellum, and putaminal iron. Vertical gaze palsy and early falls favor PSP, in which MRI shows midbrain atrophy, the hummingbird sign. Asymmetric apraxia, cortical sensory loss, and alien limb favor CBD, in which MRI shows asymmetric frontoparietal atrophy and FDG-PET shows asymmetric hypometabolism. Probable CBS is asymmetric, with 2 of limb rigidity or akinesia, limb dystonia, and limb myoclonus, plus 2 of orobuccal or limb apraxia, cortical sensory deficit, and alien limb; possible CBS may be symmetric, with 1 of each. CBS is not specific, since PSP, AD, and other pathologies can cause it.

Management

No treatment slows MSA, PSP, or CBD, so all three are treated the same way, by symptom. Parkinsonism is treated with a levodopa trial of up to 900–1200 mg daily for at least a month, tapered if there is no benefit. Focal dystonia is treated with botulinum toxin, with baclofen or clonazepam as alternatives, and falls with physical and occupational therapy and balance training. DBS is not recommended in PSP.

Orthostatic hypotension is treated with midodrine or fludrocortisone, urinary dysfunction with α-blockers, mirabegron, or selective antimuscarinics, and constipation with fiber, fluids, and osmotic laxatives. RBD is treated with melatonin, then low-dose clonazepam, and sialorrhea with botulinum toxin. Early speech pathology referral addresses dysphagia and speech, and cholinesterase inhibitors may help cognition but can worsen motor function in PSP. Care involves neurology and physical, occupational, and speech therapy, with early advance care planning and later hospice.

Relationships to other conditions

Typical Parkinson disease. Both cause parkinsonism with reduced DAT uptake. A clear, lasting levodopa response and rest tremor favor typical Parkinson disease, whereas a poor response with early autonomic failure, gaze palsy, early falls, or asymmetric cortical signs favors atypical parkinsonism.

Dementia with Lewy bodies. Both are α-synucleinopathies when the atypical disorder is MSA. Early dementia and visual hallucinations favor dementia with Lewy bodies.

Frontotemporal dementia. PSP and CBD are tauopathies that overlap with FTLD and can cause behavior and language change. Behavior or language change first, without early parkinsonism, favors frontotemporal dementia.

MOVEMENT DISORDERS

Huntington disease: CAG repeat expansion, striatal degeneration, and chorea

Huntington disease is an autosomal dominant neurodegenerative disorder caused by an expanded CAG repeat in HTT, which adds a polyglutamine stretch to huntingtin. Medium spiny neurons of the caudate and putamen degenerate, indirect-pathway neurons first, causing chorea, followed by hypokinesia, psychiatric change, and a subcortical dementia. Repeat length predicts penetrance and age of onset, and anticipation follows paternal transmission. Diagnosis rests on genetic testing, and treatment is symptomatic, with VMAT2 inhibitors such as tetrabenazine for chorea. Here, we review the genetics, pathology, clinical course, diagnosis, and treatment of Huntington disease.

Key points

- HD is autosomal dominant: 36–39 CAG repeats have reduced penetrance, and 40 or more almost always cause disease.
- Anticipation occurs mainly with paternal transmission, because repeats lengthen in spermatogenesis.
- Indirect-pathway medium spiny neurons are lost first, causing chorea; direct-pathway loss later causes rigidity.
- Degeneration starts in the caudate and dorsal putamen and spreads dorsomedial to ventrolateral; the Vonsattel grade tracks disability.
- Tetrabenazine inhibits VMAT2, reducing dopamine release and chorea, but can worsen depression.
- Aspiration pneumonia is the most common cause of death, followed by suicide.

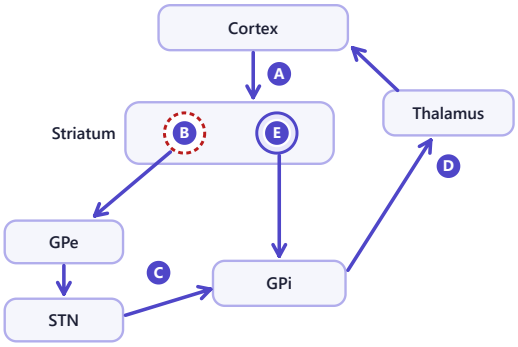


Fig. 1 | The basal ganglia circuit in Huntington disease. A, the cortex drives the striatum. B, indirect-pathway medium spiny neurons, which project to the external globus pallidus, are lost first. C, through the external pallidum and subthalamic nucleus, the internal pallidum then inhibits the thalamus less. D, the released thalamus overdrives the cortex, causing chorea. E, direct-pathway neurons, which project to the internal pallidum, are lost later, causing rigidity and bradykinesia. A, Cortex; B, Indirect-pathway neurons; C, GPe, STN, and GPi; D, Thalamus; E, Direct-pathway neurons.



Fig. 2 | The head of the caudate bulging into the lateral ventricle at each Vonsattel grade: convex at grades 0 to 2, flat at grade 3, and concave at grade 4, as caudate neurons are lost. Caudate head beside the lateral ventricle

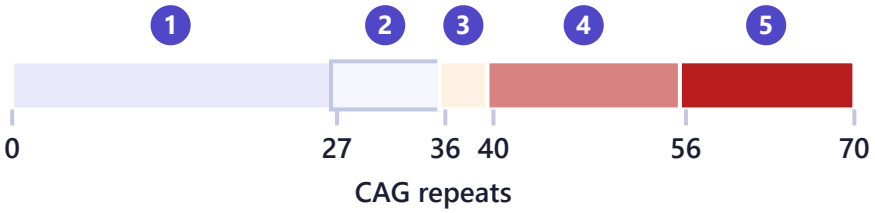


Fig. 3 | A bar of CAG repeat lengths from 0 to 70: 1, 26 or fewer is normal; 2, 27 to 35 is intermediate; 3, 36 to 39 has reduced penetrance; 4, 40 or more is fully penetrant; 5, over 55 causes juvenile disease. 1, 26 or fewer; 2, 27–35, intermediate; 3, 36–39, reduced penetrance; 4, 40 or more; 5, Over 55.

Pathophysiology

An expanded CAG repeat in HTT, on chromosome 4p16.3, adds a polyglutamine stretch to huntingtin, which causes both a toxic gain and a loss of function and forms nuclear inclusions. Over 90% of the GABA medium spiny neurons of the striatum degenerate. Indirect-pathway neurons, which contain enkephalin and project to the GPe, are lost first, so the GPi inhibits the thalamus less and the released thalamus overdrives the cortex, causing chorea. Direct-pathway neurons, projecting to the GPi, are lost later, causing rigidity and bradykinesia.

Degeneration starts in the tail and head of the caudate and the dorsal putamen and proceeds dorsomedial to ventrolateral and caudal to rostral. By Vonsattel grade, grade 0 shows no visible change, grades 1–2 show 50% loss of caudate head neurons with the caudate still convex, and grades 3–4 show a flat, then concave caudate, with 95% of its neurons lost. The cortex, globus pallidus, cerebellum, amygdala, and hippocampus are affected later. The grade tracks clinical disability, and caudate atrophy tracks repeat length and motor score.

Etiology

HD is autosomal dominant, so each child of an affected parent has a 50% chance of inheriting it. Repeats of 26 or fewer are normal, and 27–35 cause no disease but are unstable and can expand in children. Repeats of 36–39 have reduced penetrance, 40 or more almost always cause disease, with longer repeats bringing earlier onset, and over 55, usually over 60, cause juvenile disease. Repeats lengthen during spermatogenesis, so anticipation occurs mainly with paternal transmission. The family history can be negative after early parental death, late onset in the parent, an intermediate allele, or an unrecognized diagnosis, and homozygotes have a similar onset but faster progression.

Prevalence is 2.7 per 100000 worldwide and 10.6–13.7 per 100000 in people of European descent, and mean onset is at 45 years.

Clinical presentation

Chorea starts distally and spreads to the face, trunk, and proximal limbs, and later gives way to bradykinesia, dystonia, and rigidity, with dysarthria, dysphagia, and falls. Psychiatric change often comes first, with irritability, impulsivity, and aggression; apathy is the most common feature, depression is frequent, psychosis appears late, and insight is poor. Cognitive change begins with executive dysfunction and progresses to a subcortical dementia with slow retrieval that spares aphasia and apraxia.

The disease passes from presymptomatic, gene positive without signs, through prodromal, with subtle motor and cognitive signs, apathy, and depression, to manifest, when signs impair daily life. Juvenile disease, before age 20, causes rigidity and bradykinesia (the Westphal variant), dystonia, learning difficulty, and seizures in 30–50% before age 10. Late-onset disease, after 50 in about 25%, causes chorea and gait problems with a milder, longer course. Aspiration pneumonia is the most common cause of death, followed by suicide, which about 25% attempt, and median survival is 15–18 years from onset.

Diagnosis

A family history with chorea and behavioral or cognitive change suggests HD, and genetic testing of CAG repeat length is the gold standard. MRI shows early caudate atrophy, even before symptoms, with cortical and cerebellar atrophy later. Predictive testing is offered to at-risk adults with genetic counseling, and prenatal testing uses chorionic villus sampling at 10–12 weeks or amniocentesis at 15–17 weeks.

Chorea-acanthocytosis, autosomal recessive, causes orofacial chorea worse with eating, tongue biting, myopathy, and onset around 30, with acanthocytes on the blood film. McLeod syndrome, X-linked, causes acanthocytes, hemolysis, and myopathy, with absent Kell antigen. Wilson disease causes orofacial dystonia with parkinsonism, and spinocerebellar ataxias cause prominent ataxia; marked ataxia points away from HD.

Management

No treatment slows HD, so treatment targets symptoms. Tetrabenazine inhibits VMAT2 on presynaptic vesicles, blocking the loading of dopamine and reducing its release, which lessens chorea. Tetrabenazine, deutetrabenazine, and valbenazine are first line but can cause depression, suicidality, fatigue, and akathisia, so mood and suicidal thoughts are monitored. Antipsychotics are used off label, with aripiprazole matching tetrabenazine, at the cost of extrapyramidal effects and tardive dyskinesia.

Rigidity in the Westphal variant is treated with levodopa, dopamine agonists, or amantadine, and focal dystonia with botulinum toxin. Depression, anxiety, and OCD are treated with SSRIs or TCAs, with limited evidence, and no drug helps cognition. Supportive care covers diet, nursing, and equipment, smoking and alcohol are avoided, and patients and families receive counseling and genetic counseling.

Relationships to other conditions

Typical Parkinson disease. Both disturb the basal ganglia, in opposite directions. Loss of indirect-pathway neurons in HD causes excess movement, whereas dopamine loss in Parkinson disease causes too little; the Westphal variant of HD, with rigidity, responds to the same antiparkinsonian drugs.

Alzheimer disease. Both cause dementia. HD causes a subcortical dementia with slow retrieval that spares aphasia and apraxia, whereas Alzheimer disease causes a cortical dementia with early memory loss.

Abbreviations

ABC Amyloid, Braak, CERAD

ABX Antibiotics

ACA Anterior cerebral artery

ACE Angiotensin-converting enzyme

ACh Acetylcholine

AChE Acetylcholinesterase

AChR Acetylcholine receptor

AComm Anterior communicating artery

ACTH Adrenocorticotropic hormone

AD Alzheimer disease

ADEM Acute disseminated encephalomyelitis

AFO Ankle-foot orthosis

AICA Anterior inferior cerebellar artery

AIS Acute ischemic stroke

ALS Amyotrophic lateral sclerosis

AMD Age-related macular degeneration

ANA Antinuclear antibody

APH1 Anterior pharynx-defective 1

APOE Apolipoprotein E

APP Amyloid precursor protein

APS Antiphospholipid syndrome

aPTT Activated partial thromboplastin time

AQP4 Aquaporin-4

AQP4-IgG Aquaporin-4 immunoglobulin G antibody

AREDS Age-Related Eye Disease Study

ARIA Amyloid-related imaging abnormalities

ASD Atrial septal defect

AT-III Antithrombin III

AV Atrioventricular

AVM Arteriovenous malformation

AZA Azathioprine

Aβ42 Amyloid beta 42, the 42-amino-acid form

B12 Vitamin B12 (cobalamin)

BBB Blood-brain barrier

BP Blood pressure

BPPV Benign paroxysmal positional vertigo

bvFTD Behavioral variant frontotemporal dementia

C9orf72 Chromosome 9 open reading frame 72 gene

CAG Cytosine-adenine-guanine trinucleotide

CBD Common bile duct

CBS Corticobasal syndrome

CD59 A complement regulatory protein

CERAD Consortium to Establish a Registry for Alzheimer Disease

CIS Clinically isolated syndrome

CMV Cytomegalovirus

CN Cranial nerve

CN III Cranial nerve III (oculomotor)

CN VI Cranial nerve VI (abducens)

CN VII Cranial nerve VII (facial)

CN VIII Cranial nerve VIII (vestibulocochlear)

CN X Cranial nerve X (the vagus nerve)

CNS Central nervous system

COMT Catechol-O-methyltransferase

COPD Chronic obstructive pulmonary disease

CRAO Central retinal artery occlusion

CRP C-reactive protein

CRVO Central retinal vein occlusion

CSF Cerebrospinal fluid

CT Computed tomography

CXR Chest x-ray

D1 Dopamine receptor 1

D2 Dopamine receptor 2

DAT Dopamine transporter

DBS Deep brain stimulation

DFNB1 Deafness, autosomal recessive, locus 1

DJ-1 Parkinson disease protein 7 gene

DLB Dementia with Lewy bodies

DOAC Direct oral anticoagulant

DOACs Direct oral anticoagulants

DRG Dorsal root ganglion

DVT Deep vein thrombosis

DYT1 Dystonia 1 gene (TOR1A)

DYT6 Dystonia 6 gene (THAP1)

EAAT2 Excitatory amino acid transporter 2

EBV Epstein-Barr virus

ECG Electrocardiogram

EDN3 Endothelin 3 gene

EDNRB Endothelin receptor type B gene

EEG Electroencephalogram

ELISA Enzyme-linked immunosorbent assay

ERG Electroretinogram

ESR Erythrocyte sedimentation rate

EVD External ventricular drain

FBC Full blood count

FDG Fluorodeoxyglucose

FDG-PET Fluorodeoxyglucose positron emission tomography

FFP Fresh frozen plasma

FTD Frontotemporal dementia

FTLD Frontotemporal lobar degeneration

FUS Fused in sarcoma protein

GABA Gamma-aminobutyric acid

GBA1 Glucosylceramidase beta 1 gene

GCA Giant cell arteritis

GCIs Glial cytoplasmic inclusions

GCS Glasgow Coma Scale

GFAP Glial fibrillary acidic protein

GI Gastrointestinal

GJB2 Gap junction beta 2 gene (connexin 26)

GJB6 Gap junction beta 6 gene (connexin 30)

GPe Globus pallidus externus

GPi Globus pallidus internus

GRN Granulin gene

HbA1c Glycated hemoglobin

HD Huntington disease

HFrEF Heart failure with reduced ejection fraction

HHV-6 Human herpesvirus 6

HINTS Head impulse, nystagmus, and test of skew

HIV Human immunodeficiency virus

HLA Human leukocyte antigen

HLA-B*1502 Human leukocyte antigen B*15:02 allele

HLA-B27 Human leukocyte antigen B27

HR Heart rate

HTT Huntingtin gene

IBD Inflammatory bowel disease

ICA Internal carotid artery

ICH Intracerebral hemorrhage

ICP Intracranial pressure

ICU Intensive care unit

IgE Immunoglobulin E

IgG1 Immunoglobulin G subclass 1

IL-1 Interleukin 1

IL-6 Interleukin 6

IM Intramuscular

INO Internuclear ophthalmoplegia

INR International normalized ratio

IOL Intraocular lens

IOP Intraocular pressure

IV Intravenous

IVH Intraventricular hemorrhage

IVIG Intravenous immunoglobulin

JME Juvenile myoclonic epilepsy

LATE Limbic-predominant age-related TDP-43 encephalopathy

LETM Longitudinally extensive transverse myelitis

LFTs Liver function tests

LGN Lateral geniculate nucleus

LMN Lower motor neuron

LMWH Low-molecular-weight heparin

LOC Loss of consciousness

LRRK2 Leucine-rich repeat kinase 2 gene

LV Left ventricular

M1 Primary motor cortex

MAO-B Monoamine oxidase B

MAPT Microtubule-associated protein tau gene

MCA Middle cerebral artery

MCP Middle cerebellar peduncle

MDS International Parkinson and Movement Disorder Society

MGN Medial geniculate nucleus

MI Myocardial infarction

MIBG Metaiodobenzylguanidine

MIGS Minimally invasive glaucoma surgery

MITF Melanocyte-inducing transcription factor gene

MLF Medial longitudinal fasciculus

MMF Mycophenolate mofetil

MMSE Mini-Mental State Examination

MND Motor neuron disease

MoCA Montreal Cognitive Assessment

MOG Myelin oligodendrocyte glycoprotein

MOG-IgG Myelin oligodendrocyte glycoprotein immunoglobulin G antibody

MOGAD MOG antibody-associated disease

MPTP 1-methyl-4-phenyl-1,2,3,6-tetrahydropyridine

MR Magnetic resonance

MRA Magnetic resonance angiography

MRI Magnetic resonance imaging

MS Multiple sclerosis

MSA Multiple system atrophy

MSA-C Multiple system atrophy, cerebellar type

MSA-P Multiple system atrophy, parkinsonian type

MuSK Muscle-specific kinase

NF2 Neurofibromatosis type 2

NFTs Neurofibrillary tangles

nfvPPA Nonfluent variant primary progressive aphasia

NIHSS National Institutes of Health Stroke Scale

NMDA N-methyl-D-aspartate

NMO Neuromyelitis optica

NMOSD Neuromyelitis optica spectrum disorder

NSAIDs Nonsteroidal anti-inflammatory drugs

OCB Oligoclonal bands

OCD Obsessive-compulsive disorder

OCP Oral contraceptive pill

OCT Optical coherence tomography

PAX3 Paired box 3 gene

PCA Posterior cerebral artery

PCC Prothrombin complex concentrate

PComm Posterior communicating artery

PD Parkinson disease

PDD Parkinson disease dementia

PE Pulmonary embolism

PEG Polyethylene glycol

PEN2 Presenilin enhancer 2

PET Positron emission tomography

PFO Patent foramen ovale

PICA Posterior inferior cerebellar artery

PINK1 PTEN-induced kinase 1 gene

PKD Polycystic kidney disease

PLEX Plasma exchange

PML Progressive multifocal leukoencephalopathy

PNES Psychogenic nonepileptic seizures

PNS Parasympathetic nervous system

PPA Primary progressive aphasia

PRES Posterior reversible encephalopathy syndrome

PRKN Parkin gene

PSEN1 Presenilin 1

PSEN2 Presenilin 2

PSP Progressive supranuclear palsy

PSP-bvFTD Progressive supranuclear palsy, behavioral variant frontotemporal dementia

PSP-CBS Progressive supranuclear palsy, corticobasal syndrome

PSP-P Progressive supranuclear palsy, parkinsonism predominant

PSP-PAGF Progressive supranuclear palsy, pure akinesia with gait freezing

PSP-PNFA Progressive supranuclear palsy, progressive nonfluent aphasia

PSP-RS Progressive supranuclear palsy, Richardson syndrome

PT Prothrombin time

RAPD Relative afferent pupillary defect

RBD REM sleep behavior disorder

RCC Renal cell carcinoma

REM Rapid eye movement

RHD Rheumatic heart disease

RNFL Retinal nerve fiber layer

RPE Retinal pigment epithelium

RTX Rituximab

S1 Primary somatosensory cortex

S1P Sphingosine 1-phosphate

S1PR1 Sphingosine 1-phosphate receptor 1

SA Sinoatrial

SAH Subarachnoid hemorrhage

SARS-CoV-2 Severe acute respiratory syndrome coronavirus 2

SIADH Syndrome of inappropriate antidiuretic hormone secretion

SJS Stevens-Johnson syndrome

SLE Systemic lupus erythematosus

SMA Superior mesenteric artery

SNARE Soluble NSF attachment protein receptor

SNc Substantia nigra pars compacta

SNCA Synuclein alpha gene

SNHL Sensorineural hearing loss

SNr Substantia nigra pars reticulata

SNRIs Serotonin-norepinephrine reuptake inhibitors

SNS Sympathetic nervous system

SPECT Single-photon emission computed tomography

SSRIs Selective serotonin reuptake inhibitors

STN Subthalamic nucleus

SV2A Synaptic vesicle protein 2A

svPPA Semantic variant primary progressive aphasia

SWI Susceptibility-weighted imaging

TCAs Tricyclic antidepressants

TDP-43 TAR DNA-binding protein 43

TFTs Thyroid function tests

TIA Transient ischemic attack

TNF Tumor necrosis factor

tPA Tissue plasminogen activator

UMN Upper motor neuron

V1 Primary visual cortex

VA Ventral anterior nucleus of the thalamus

VEGF Vascular endothelial growth factor

VL Ventral lateral nucleus of the thalamus

VMAT2 Vesicular monoamine transporter 2

VOR Vestibulo-ocular reflex

VPL Ventral posterolateral nucleus of the thalamus